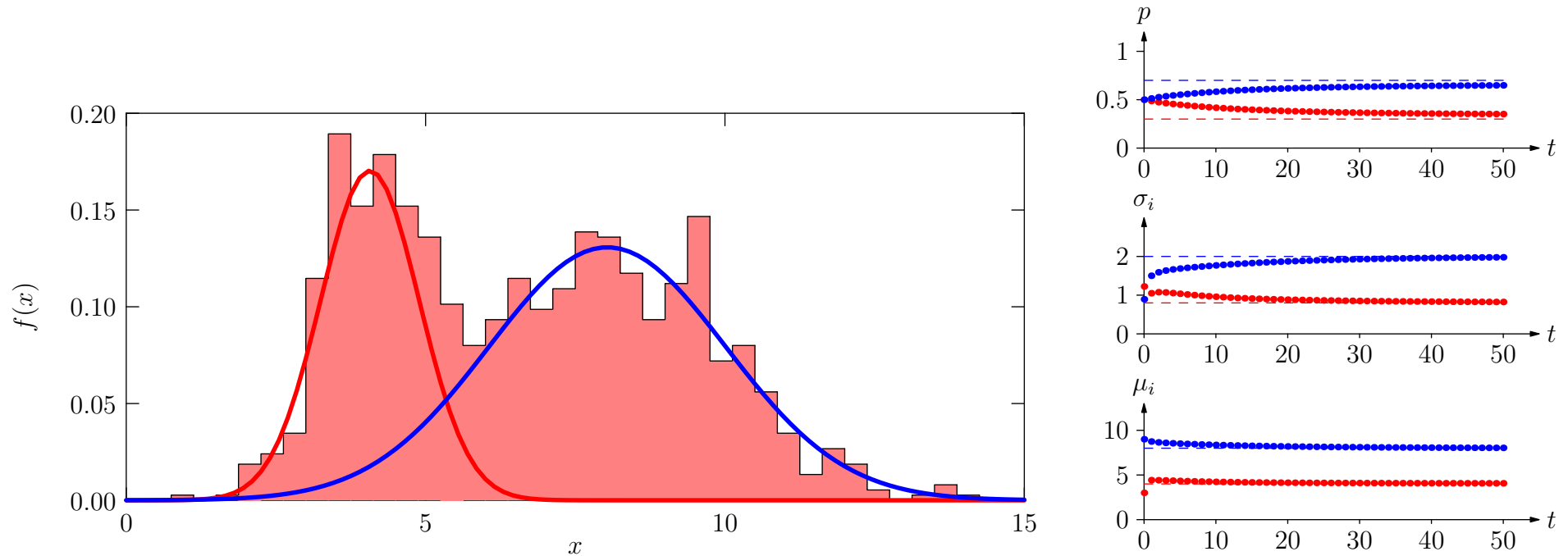


Advanced Machine Learning

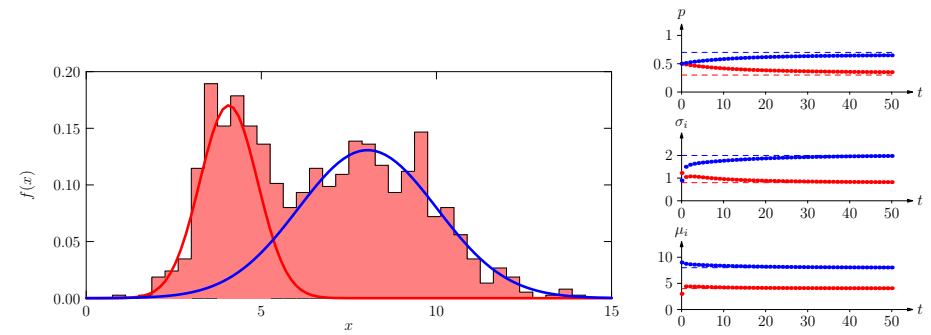
Probabilistic Inference



*Hierarchical Models, Mixture of Gaussians, Expectation
Maximisation*

Outline

1. **Building Probabilistic Models**
2. Mixture of Gaussians
3. Expectation Maximisation



Building Probabilistic Models

- To describe a system with uncertainty we use random variables, X, Y, Z , etc.
- We use the convention of writing random variables in capitals (this is sometimes confusing as when you observe a random variables it is no longer random)
- The variables are described by probability mass function $\mathbb{P}(X, Y, Z)$ or if our variables are continuous, but probability densities $f_{X, Y, Z}(x, y, z)$
- A major rule of probability is

$$\sum_X \mathbb{P}(X, Y, Z) = \mathbb{P}(Y, Z)$$

Building Probabilistic Models

- To describe a system with uncertainty we use random variables, X , Y , Z , etc.
- We use the convention of writing random variables in capitals (this is sometimes confusing as when you observe a random variables it is no longer random)
- The variables are described by probability mass function $\mathbb{P}(X, Y, Z)$ or if our variables are continuous, but probability densities $f_{X, Y, Z}(x, y, z)$
- A major rule of probability is

$$\sum_X \mathbb{P}(X, Y, Z) = \mathbb{P}(Y, Z)$$

Building Probabilistic Models

- To describe a system with uncertainty we use random variables, X , Y , Z , etc.
- We use the convention of writing random variables in capitals (this is sometimes confusing as when you observe a random variables it is no longer random)
- The variables are described by probability mass function $\mathbb{P}(X,Y,Z)$ or if our variables are continuous, but probability densities $f_{X,Y,Z}(x,y,z)$
- A major rule of probability is

$$\sum_X \mathbb{P}(X,Y,Z) = \mathbb{P}(Y,Z)$$

Building Probabilistic Models

- To describe a system with uncertainty we use random variables, X , Y , Z , etc.
- We use the convention of writing random variables in capitals (this is sometimes confusing as when you observe a random variables it is no longer random)
- The variables are described by probability mass function $\mathbb{P}(X, Y, Z)$ or if our variables are continuous, but probability densities $f_{X, Y, Z}(x, y, z)$
- A major rule of probability is

$$\sum_X \mathbb{P}(X, Y, Z) = \mathbb{P}(Y, Z)$$

Conditional Probabilities

- When developing models it is often useful to consider conditional probabilities e.g. $\mathbb{P}(X, Y|Z)$ or $f_{X|Y, Z}(x|y, z)$
- A second major rule in probabilistic modelling is

$$\mathbb{P}(X, Y) = \mathbb{P}(X|Y) \mathbb{P}(Y) = \mathbb{P}(Y|X) \mathbb{P}(X)$$

- This is a mathematical identity that does not imply causality (it defines conditional probability)
- It is the origins of Bayes' rule: $\mathbb{P}(X|Y) = \frac{\mathbb{P}(Y|X) \mathbb{P}(X)}{\mathbb{P}(Y)}$

Conditional Probabilities

- When developing models it is often useful to consider conditional probabilities e.g. $\mathbb{P}(X, Y|Z)$ or $f_{X|Y, Z}(x|y, z)$

- A second major rule in probabilistic modelling is

$$\mathbb{P}(X, Y) = \mathbb{P}(X|Y) \mathbb{P}(Y) = \mathbb{P}(Y|X) \mathbb{P}(X)$$

- This is a mathematical identity that does not imply causality (it defines conditional probability)
- It is the origins of Bayes' rule: $\mathbb{P}(X|Y) = \frac{\mathbb{P}(Y|X) \mathbb{P}(X)}{\mathbb{P}(Y)}$

Conditional Probabilities

- When developing models it is often useful to consider conditional probabilities e.g. $\mathbb{P}(X, Y|Z)$ or $f_{X|Y, Z}(x|y, z)$
- A second major rule in probabilistic modelling is

$$\mathbb{P}(X, Y) = \mathbb{P}(X|Y) \mathbb{P}(Y) = \mathbb{P}(Y|X) \mathbb{P}(X)$$

- This is a mathematical identity that does not imply causality (it defines conditional probability)
- It is the origins of Bayes' rule: $\mathbb{P}(X|Y) = \frac{\mathbb{P}(Y|X) \mathbb{P}(X)}{\mathbb{P}(Y)}$

Conditional Probabilities

- When developing models it is often useful to consider conditional probabilities e.g. $\mathbb{P}(X, Y|Z)$ or $f_{X|Y, Z}(x|y, z)$
- A second major rule in probabilistic modelling is

$$\mathbb{P}(X, Y) = \mathbb{P}(X|Y) \mathbb{P}(Y) = \mathbb{P}(Y|X) \mathbb{P}(X)$$

- This is a mathematical identity that does not imply causality (it defines conditional probability)
- It is the origins of Bayes' rule: $\mathbb{P}(X|Y) = \frac{\mathbb{P}(Y|X) \mathbb{P}(X)}{\mathbb{P}(Y)}$

Discriminative Models

- We often think of our observations as given and the predictions as random variables
- For example we might be given some features x and we wish to predict a class $C \in \mathcal{C}$
- Our objective is then to find the probability $\mathbb{P}(C|x)$
- This is known as a **discriminative model**
- E.g. in *foundations of machine learning* you learnt how to find the Bayes' optimal discrimination surface

Discriminative Models

- We often think of our observations as given and the predictions as random variables
- For example we might be given some features x and we wish to predict a class $C \in \mathcal{C}$
- Our objective is then to find the probability $\mathbb{P}(C|x)$
- This is known as a **discriminative model**
- E.g. in *foundations of machine learning* you learnt how to find the Bayes' optimal discrimination surface

Discriminative Models

- We often think of our observations as given and the predictions as random variables
- For example we might be given some features x and we wish to predict a class $C \in \mathcal{C}$
- Our objective is then to find the probability $\mathbb{P}(C|x)$
- This is known as a **discriminative model**
- E.g. in *foundations of machine learning* you learnt how to find the Bayes' optimal discrimination surface

Discriminative Models

- We often think of our observations as given and the predictions as random variables
- For example we might be given some features x and we wish to predict a class $C \in \mathcal{C}$
- Our objective is then to find the probability $\mathbb{P}(C|x)$
- This is known as a **discriminative model**
- E.g. in *foundations of machine learning* you learnt how to find the Bayes' optimal discrimination surface

Discriminative Models

- We often think of our observations as given and the predictions as random variables
- For example we might be given some features x and we wish to predict a class $C \in \mathcal{C}$
- Our objective is then to find the probability $\mathbb{P}(C|x)$
- This is known as a **discriminative model**
- E.g. in *foundations of machine learning* you learnt how to find the Bayes' optimal discrimination surface

Generative Models

- Sometimes it is easy to think about the joint process of generating the features and outputs together
- This leads to a joint distribution $\mathbb{P}(\mathbf{X}, Y)$ where $\mathbf{X}$ are your features and Y is your output you are trying to predict
- This is known as a **generative model**
- Generative models are often more natural to think about
- We can use them to do discrimination using

$$\mathbb{P}(Y|\mathbf{X}) = \frac{\mathbb{P}(\mathbf{X}, Y)}{\mathbb{P}(\mathbf{X})} = \frac{\mathbb{P}(\mathbf{X}, Y)}{\sum_Y \mathbb{P}(\mathbf{X}, Y)}$$

Generative Models

- Sometimes it is easy to think about the joint process of generating the features and outputs together
- This leads to a joint distribution $\mathbb{P}(\mathbf{X}, Y)$ where $\mathbf{X}$ are your features and Y is your output you are trying to predict
- This is known as a **generative model**
- Generative models are often more natural to think about
- We can use them to do discrimination using

$$\mathbb{P}(Y|\mathbf{X}) = \frac{\mathbb{P}(\mathbf{X}, Y)}{\mathbb{P}(\mathbf{X})} = \frac{\mathbb{P}(\mathbf{X}, Y)}{\sum_Y \mathbb{P}(\mathbf{X}, Y)}$$

Generative Models

- Sometimes it is easy to think about the joint process of generating the features and outputs together
- This leads to a joint distribution $\mathbb{P}(\mathbf{X}, Y)$ where $\mathbf{X}$ are your features and Y is your output you are trying to predict
- This is known as a **generative model**
- Generative models are often more natural to think about
- We can use them to do discrimination using

$$\mathbb{P}(Y|\mathbf{X}) = \frac{\mathbb{P}(\mathbf{X}, Y)}{\mathbb{P}(\mathbf{X})} = \frac{\mathbb{P}(\mathbf{X}, Y)}{\sum_Y \mathbb{P}(\mathbf{X}, Y)}$$

Generative Models

- Sometimes it is easy to think about the joint process of generating the features and outputs together
- This leads to a joint distribution $\mathbb{P}(\mathbf{X}, Y)$ where $\mathbf{X}$ are your features and Y is your output you are trying to predict
- This is known as a **generative model**
- Generative models are often more natural to think about
- We can use them to do discrimination using

$$\mathbb{P}(Y|\mathbf{X}) = \frac{\mathbb{P}(\mathbf{X}, Y)}{\mathbb{P}(\mathbf{X})} = \frac{\mathbb{P}(\mathbf{X}, Y)}{\sum_Y \mathbb{P}(\mathbf{X}, Y)}$$

Generative Models

- Sometimes it is easy to think about the joint process of generating the features and outputs together
- This leads to a joint distribution $\mathbb{P}(\mathbf{X}, Y)$ where $\mathbf{X}$ are your features and Y is your output you are trying to predict
- This is known as a **generative model**
- Generative models are often more natural to think about
- We can use them to do discrimination using

$$\mathbb{P}(Y|\mathbf{X}) = \frac{\mathbb{P}(\mathbf{X}, Y)}{\mathbb{P}(\mathbf{X})} = \frac{\mathbb{P}(\mathbf{X}, Y)}{\sum_Y \mathbb{P}(\mathbf{X}, Y)}$$

Latent Variables

- Sometimes we have models that involve random variables that we don't observe and we don't care about
- These are called **latent variables**
- If we have a latent variable Z and observed variable $\mathbf{X}$ and we are predicting a variable Y then we would **marginalise** over the latent variable

$$\mathbb{P}(\mathbf{X}, Y) = \sum_Z \mathbb{P}(\mathbf{X}, Y, Z)$$

Latent Variables

- Sometimes we have models that involve random variables that we don't observe and we don't care about
- These are called **latent variables**
- If we have a latent variable Z and observed variable $\mathbf{X}$ and we are predicting a variable Y then we would **marginalise** over the latent variable

$$\mathbb{P}(\mathbf{X}, Y) = \sum_Z \mathbb{P}(\mathbf{X}, Y, Z)$$

Latent Variables

- Sometimes we have models that involve random variables that we don't observe and we don't care about
- These are called **latent variables**
- If we have a latent variable Z and observed variable X and we are predicting a variable Y then we would **marginalise** over the latent variable

$$\mathbb{P}(X, Y) = \sum_Z \mathbb{P}(X, Y, Z)$$

Modelling Virus

- Suppose we want to estimate the number of hospitalisation from Corona virus in the next month
- Our observable is the number of reported cases
- In our model we might want to estimate the number of actual cases
- This would be a latent variable (it is not an observable or our final target, but it is very useful intermediate in our model)
- This will be a random variable (it is uncertain, but we can build a probabilistic model giving a distribution of number of actual cases)

Modelling Virus

- Suppose we want to estimate the number of hospitalisation from Corona virus in the next month
- Our observable is the number of reported cases
- In our model we might want to estimate the number of actual cases
- This would be a latent variable (it is not an observable or our final target, but it is very useful intermediate in our model)
- This will be a random variable (it is uncertain, but we can build a probabilistic model giving a distribution of number of actual cases)

Modelling Virus

- Suppose we want to estimate the number of hospitalisation from Corona virus in the next month
- Our observable is the number of reported cases
- In our model we might want to estimate the number of actual cases
- This would be a latent variable (it is not an observable or our final target, but it is very useful intermediate in our model)
- This will be a random variable (it is uncertain, but we can build a probabilistic model giving a distribution of number of actual cases)

Modelling Virus

- Suppose we want to estimate the number of hospitalisation from Corona virus in the next month
- Our observable is the number of reported cases
- In our model we might want to estimate the number of actual cases
- This would be a latent variable (it is not an observable or our final target, but it is very useful intermediate in our model)
- This will be a random variable (it is uncertain, but we can build a probabilistic model giving a distribution of number of actual cases)

Modelling Virus

- Suppose we want to estimate the number of hospitalisation from Corona virus in the next month
- Our observable is the number of reported cases
- In our model we might want to estimate the number of actual cases
- This would be a latent variable (it is not an observable or our final target, but it is very useful intermediate in our model)
- This will be a random variable (it is uncertain, but we can build a probabilistic model giving a distribution of number of actual cases)

Hierarchical Models

- Of course, if I was really modelling the spread of a disease I would care about the probability, $f(C|A,V)$, of catching the disease, C , given the persons age A and the variant of the disease V
- I would want to know the distribution of ages $f(A)$ and try to infer the probability of different variants $\mathbb{P}(V)$
- I would care about the probability, $f(R|A,V)$, of cases being reported given age and variant
- And the probability, $f(H|A,V)$, of hospitalisation given A and V
- This would involve an elaborate (hierarchical) model with a large number of latent variables

Hierarchical Models

- Of course, if I was really modelling the spread of a disease I would care about the probability, $f(C|A,V)$, of catching the disease, C , given the persons age A and the variant of the disease V
- I would want to know the distribution of ages $f(A)$ and try to infer the probability of different variants $\mathbb{P}(V)$
- I would care about the probability, $f(R|A,V)$, of cases being reported given age and variant
- And the probability, $f(H|A,V)$, of hospitalisation given A and V
- This would involve an elaborate (hierarchical) model with a large number of latent variables

Hierarchical Models

- Of course, if I was really modelling the spread of a disease I would care about the probability, $f(C|A,V)$, of catching the disease, C , given the persons age A and the variant of the disease V
- I would want to know the distribution of ages $f(A)$ and try to infer the probability of different variants $\mathbb{P}(V)$
- I would care about the probability, $f(R|A,V)$, of cases being reported given age and variant
- And the probability, $f(H|A,V)$, of hospitalisation given A and V
- This would involve an elaborate (hierarchical) model with a large number of latent variables

Hierarchical Models

- Of course, if I was really modelling the spread of a disease I would care about the probability, $f(C|A,V)$, of catching the disease, C , given the persons age A and the variant of the disease V
- I would want to know the distribution of ages $f(A)$ and try to infer the probability of different variants $\mathbb{P}(V)$
- I would care about the probability, $f(R|A,V)$, of cases being reported given age and variant
- And the probability, $f(H|A,V)$, of hospitalisation given A and V
- This would involve an elaborate (hierarchical) model with a large number of latent variables

Hierarchical Models

- Of course, if I was really modelling the spread of a disease I would care about the probability, $f(C|A,V)$, of catching the disease, C , given the persons age A and the variant of the disease V
- I would want to know the distribution of ages $f(A)$ and try to infer the probability of different variants $\mathbb{P}(V)$
- I would care about the probability, $f(R|A,V)$, of cases being reported given age and variant
- And the probability, $f(H|A,V)$, of hospitalisation given A and V
- This would involve an elaborate (hierarchical) model with a large number of latent variables

Probabilistic Inference

- We can use Bayes' rules to learn a set of parameter Θ that occur in our likelihood function

$$\mathbb{P}(\Theta|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)}{\mathbb{P}(\mathcal{D})}$$

- This provides us a full probabilistic description of the parameters
- It doesn't overfit (we are not choosing the best)
- Bayesian inference provides a description of its own uncertainty
- We need to specify a likelihood and prior, but this is usually not difficult

Probabilistic Inference

- We can use Bayes' rules to learn a set of parameter Θ that occur in our likelihood function

$$\mathbb{P}(\Theta|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)}{\mathbb{P}(\mathcal{D})}$$

- This provides us a full probabilistic description of the parameters
- It doesn't overfit (we are not choosing the best)
- Bayesian inference provides a description of its own uncertainty
- We need to specify a likelihood and prior, but this is usually not difficult

Probabilistic Inference

- We can use Bayes' rules to learn a set of parameter Θ that occur in our likelihood function

$$\mathbb{P}(\Theta|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)}{\mathbb{P}(\mathcal{D})}$$

- This provides us a full probabilistic description of the parameters
- It doesn't overfit (we are not choosing the best)
- Bayesian inference provides a description of its own uncertainty
- We need to specify a likelihood and prior, but this is usually not difficult

Probabilistic Inference

- We can use Bayes' rules to learn a set of parameter Θ that occur in our likelihood function

$$\mathbb{P}(\Theta|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)}{\mathbb{P}(\mathcal{D})}$$

- This provides us a full probabilistic description of the parameters
- It doesn't overfit (we are not choosing the best)
- Bayesian inference provides a description of its own uncertainty
- We need to specify a likelihood and prior, but this is usually not difficult

Probabilistic Inference

- We can use Bayes' rules to learn a set of parameter Θ that occur in our likelihood function

$$\mathbb{P}(\Theta|\mathcal{D}) = \frac{\mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)}{\mathbb{P}(\mathcal{D})}$$

- This provides us a full probabilistic description of the parameters
- It doesn't overfit (we are not choosing the best)
- Bayesian inference provides a description of its own uncertainty
- We need to specify a likelihood and prior, but this is usually not difficult

Problem with Bayes

- Bayes is problematic because it is often hard
- The posterior is often not expressible as a nice probability function
- We need to compute the *evidence* or *margin likelihood* we use

$$\mathbb{P}(\mathcal{D}) = \sum_{\Theta} \mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)$$

- But sometimes the number of values that Θ can take are so large that we cannot easily compute this
- Nevertheless we can usually do this using Monte Carlo techniques

Problem with Bayes

- Bayes is problematic because it is often hard
- The posterior is often not expressible as a nice probability function
- We need to compute the *evidence* or *margin likelihood* we use

$$\mathbb{P}(\mathcal{D}) = \sum_{\Theta} \mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)$$

- But sometimes the number of values that Θ can take are so large that we cannot easily compute this
- Nevertheless we can usually do this using Monte Carlo techniques

Problem with Bayes

- Bayes is problematic because it is often hard
- The posterior is often not expressible as a nice probability function
- We need to compute the *evidence or margin likelihood* we use

$$\mathbb{P}(\mathcal{D}) = \sum_{\Theta} \mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)$$

- But sometimes the number of values that Θ can take are so large that we cannot easily compute this
- Nevertheless we can usually do this using Monte Carlo techniques

Problem with Bayes

- Bayes is problematic because it is often hard
- The posterior is often not expressible as a nice probability function
- We need to compute the *evidence* or *margin likelihood* we use

$$\mathbb{P}(\mathcal{D}) = \sum_{\Theta} \mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)$$

- But sometimes the number of values that Θ can take are so large that we cannot easily compute this
- Nevertheless we can usually do this using Monte Carlo techniques

Problem with Bayes

- Bayes is problematic because it is often hard
- The posterior is often not expressible as a nice probability function
- We need to compute the *evidence* or *margin likelihood* we use

$$\mathbb{P}(\mathcal{D}) = \sum_{\Theta} \mathbb{P}(\mathcal{D}|\Theta) \mathbb{P}(\Theta)$$

- But sometimes the number of values that Θ can take are so large that we cannot easily compute this
- Nevertheless we can usually do this using Monte Carlo techniques

Maximum A Posteriori (MAP) Solution

- One work around is to compute the mode of the posterior

$$\Theta_{\text{MAP}} = \underset{\Theta}{\operatorname{argmax}} f(\mathcal{D}|\Theta) f(\Theta) = \underset{\Theta}{\operatorname{argmax}} \log(f(\mathcal{D}|\Theta)) + \log(f(\Theta))$$

- We don't need to calculate $f(\mathcal{D})$ or explicitly calculate the posterior distribution
- But it is not Bayesian
- You can overfit and you don't get an estimate of the error in your inference

Maximum A Posteriori (MAP) Solution

- One work around is to compute the mode of the posterior

$$\Theta_{\text{MAP}} = \underset{\Theta}{\operatorname{argmax}} f(\mathcal{D}|\Theta) f(\Theta) = \underset{\Theta}{\operatorname{argmax}} \log(f(\mathcal{D}|\Theta)) + \log(f(\Theta))$$

- We don't need to calculate $f(\mathcal{D})$ or explicitly calculate the posterior distribution
- But it is not Bayesian
- You can overfit and you don't get an estimate of the error in your inference

Maximum A Posteriori (MAP) Solution

- One work around is to compute the mode of the posterior

$$\Theta_{\text{MAP}} = \underset{\Theta}{\operatorname{argmax}} f(\mathcal{D}|\Theta) f(\Theta) = \underset{\Theta}{\operatorname{argmax}} \log(f(\mathcal{D}|\Theta)) + \log(f(\Theta))$$

- We don't need to calculate $f(\mathcal{D})$ or explicitly calculate the posterior distribution
- But it is not Bayesian
- You can overfit and you don't get an estimate of the error in your inference

Maximum A Posteriori (MAP) Solution

- One work around is to compute the mode of the posterior

$$\Theta_{\text{MAP}} = \underset{\Theta}{\operatorname{argmax}} f(\mathcal{D}|\Theta) f(\Theta) = \underset{\Theta}{\operatorname{argmax}} \log(f(\mathcal{D}|\Theta)) + \log(f(\Theta))$$

- We don't need to calculate $f(\mathcal{D})$ or explicitly calculate the posterior distribution
- But it is not Bayesian (despite what you are sometime told)
- You can overfit and you don't get an estimate of the error in your inference

Maximum A Posteriori (MAP) Solution

- One work around is to compute the mode of the posterior

$$\Theta_{\text{MAP}} = \underset{\Theta}{\operatorname{argmax}} f(\mathcal{D}|\Theta) f(\Theta) = \underset{\Theta}{\operatorname{argmax}} \log(f(\mathcal{D}|\Theta)) + \log(f(\Theta))$$

- We don't need to calculate $f(\mathcal{D})$ or explicitly calculate the posterior distribution
- But it is not Bayesian (despite what you are sometime told)—its not properly probabilistic
- You can overfit and you don't get an estimate of the error in your inference

Maximum A Posteriori (MAP) Solution

- One work around is to compute the mode of the posterior

$$\Theta_{\text{MAP}} = \underset{\Theta}{\operatorname{argmax}} f(\mathcal{D}|\Theta) f(\Theta) = \underset{\Theta}{\operatorname{argmax}} \log(f(\mathcal{D}|\Theta)) + \log(f(\Theta))$$

- We don't need to calculate $f(\mathcal{D})$ or explicitly calculate the posterior distribution
- But it is not Bayesian (despite what you are sometime told)—its not properly probabilistic
- You can overfit and you don't get an estimate of the error in your inference

Maximum Likelihood

- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
- Maximum likelihood is often sufficient for *government work*, but it isn't the best you can do
- In high-dimensional problems using a non-uniform prior can make a big difference
- And, of course, doing a full probabilistic calculation has real advantages

Maximum Likelihood

- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
- Maximum likelihood is often sufficient for *government work*, but it isn't the best you can do
- In high-dimensional problems using a non-uniform prior can make a big difference
- And, of course, doing a full probabilistic calculation has real advantages

Maximum Likelihood

- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
- Maximum likelihood is often sufficient for *government work*, but it isn't the best you can do
- In high-dimensional problems using a non-uniform prior can make a big difference
- And, of course, doing a full probabilistic calculation has real advantages

Maximum Likelihood

- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
- Maximum likelihood is often sufficient for *government work*, but it isn't the best you can do
- In high-dimensional problems using a non-uniform prior can make a big difference
- And, of course, doing a full probabilistic calculation has real advantages

Maximum Likelihood

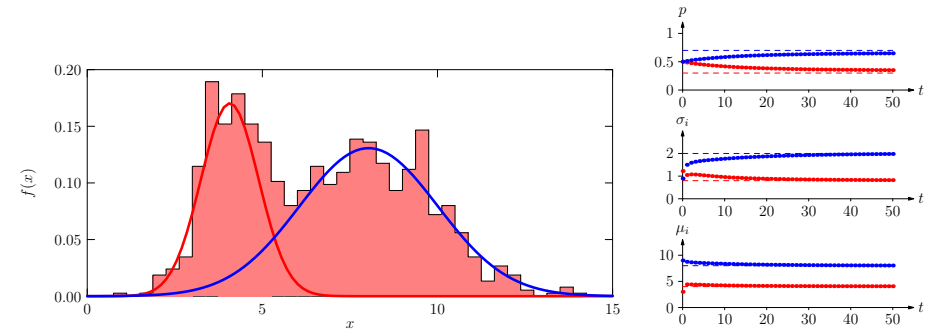
- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
- Maximum likelihood is often sufficient for *government work*, but it isn't the best you can do
- In high-dimensional problems using a non-uniform prior can make a big difference
- And, of course, doing a full probabilistic calculation has real advantages

Maximum Likelihood

- When we assume a uniform prior then the MAP solution is just maximising the likelihood
- Weirdly this hack was accepted as part of mainstream statistics even when Bayesian statistics was considered unscientific
- Maximum likelihood is often sufficient for *government work*, but it isn't the best you can do
- In high-dimensional problems using a non-uniform prior can make a big difference
- And, of course, doing a full probabilistic calculation has real advantages (no overfitting, estimate of error, etc.)

Outline

1. Building Probabilistic Models
2. **Mixture of Gaussians**
3. Expectation Maximisation



Mixture of Gaussians

- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
- Let $Z_i \in \{0,1\}$ be an indicator that particle i is of type A
- The probability of X_i is given by

$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

Mixture of Gaussians

- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
- Let $Z_i \in \{0,1\}$ be an indicator that particle i is of type A
- The probability of X_i is given by

$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

Mixture of Gaussians

- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
- Let $Z_i \in \{0,1\}$ be an indicator that particle i is of type A
- The probability of X_i is given by

$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

Mixture of Gaussians

- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
- Let $Z_i \in \{0,1\}$ be an indicator that particle i is of type A
- The probability of X_i is given by

$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

Mixture of Gaussians

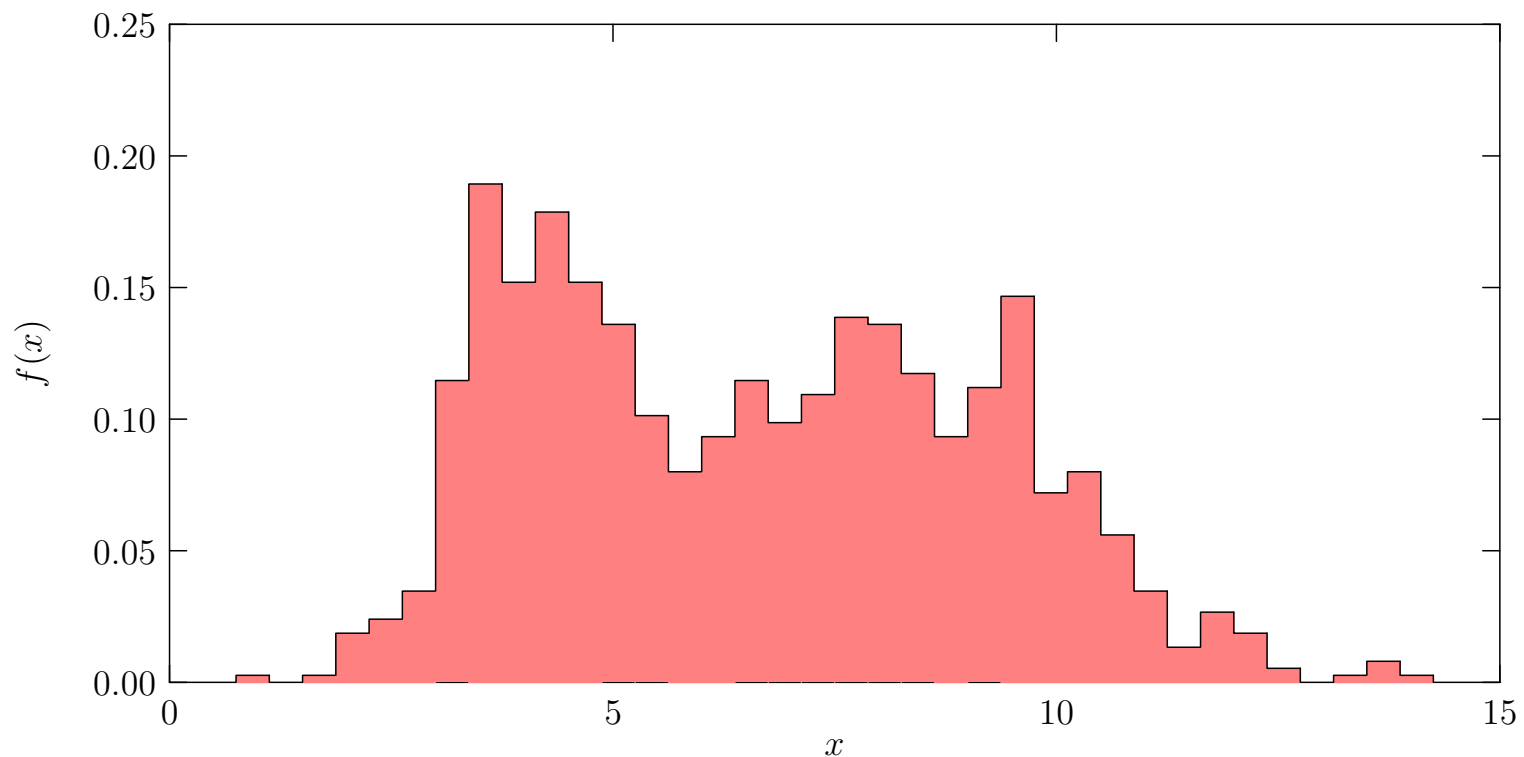
- Suppose we were observing the decays from two types of short-lived particle, A or B
- We observe the half life, X_i , but not the particle type
- We assume X_i is normally distributed with unknown means and variances: $\Theta = \{\mu_A, \sigma_A^2, \mu_B, \sigma_B^2\}$
- Let $Z_i \in \{0,1\}$ be an indicator that particle i is of type A
- The probability of X_i is given by

$$f(X_i|Z_i, \Theta) = Z_i \mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1 - Z_i) \mathcal{N}(X_i|\mu_B, \sigma_B^2)$$

Data

- Note that

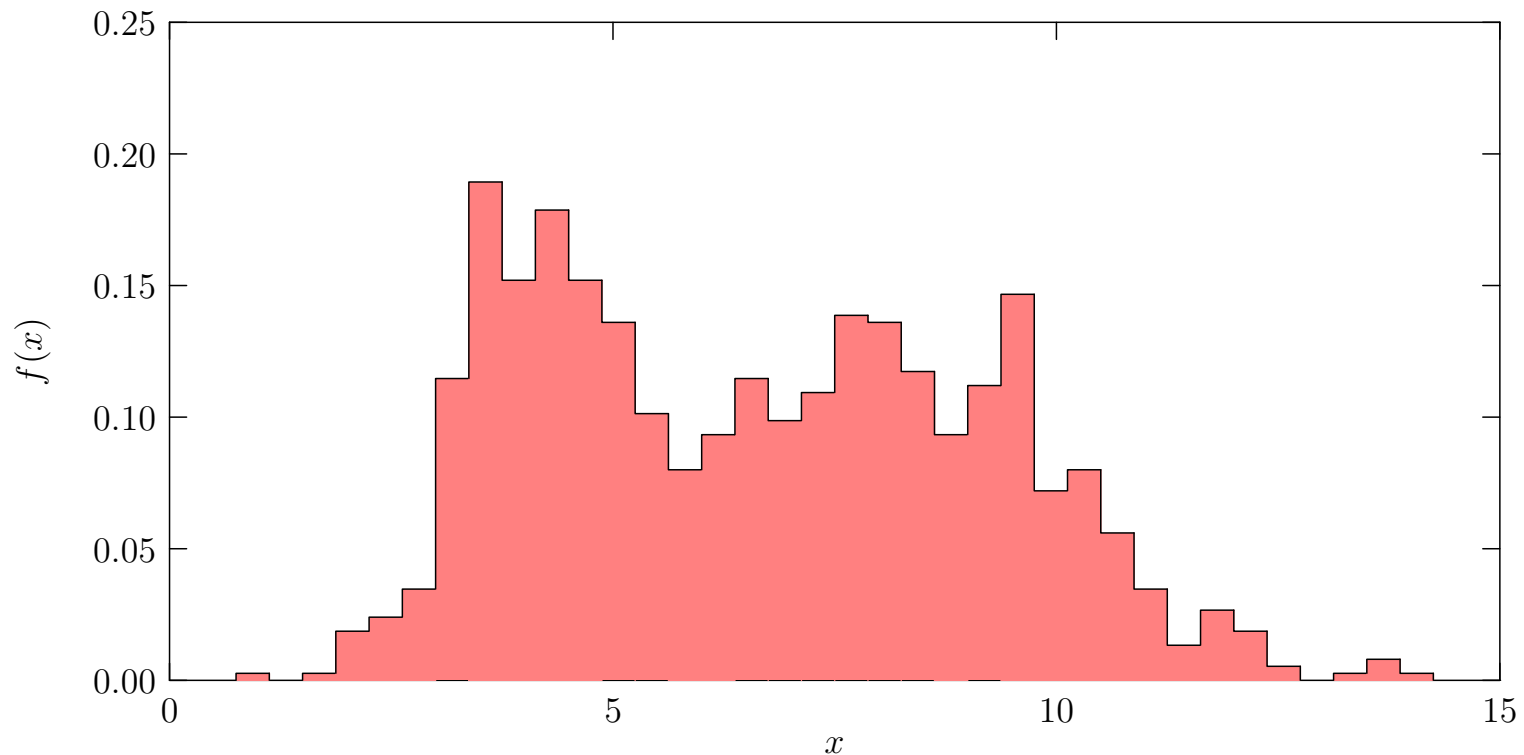
$$\begin{aligned} f(X_i|\Theta) &= \sum_{Z_i \in \{0,1\}} f(X_i, Z_i|\Theta) = \sum_{Z_i \in \{0,1\}} f(X_i|Z_i, \Theta) \mathbb{P}(Z_i) \\ &= \mathbb{E}_{Z_i}[f(X_i|Z_i, \Theta)] = p\mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1-p)\mathcal{N}(X_i|\mu_B, \sigma_B^2) \end{aligned}$$



Data

- Note that

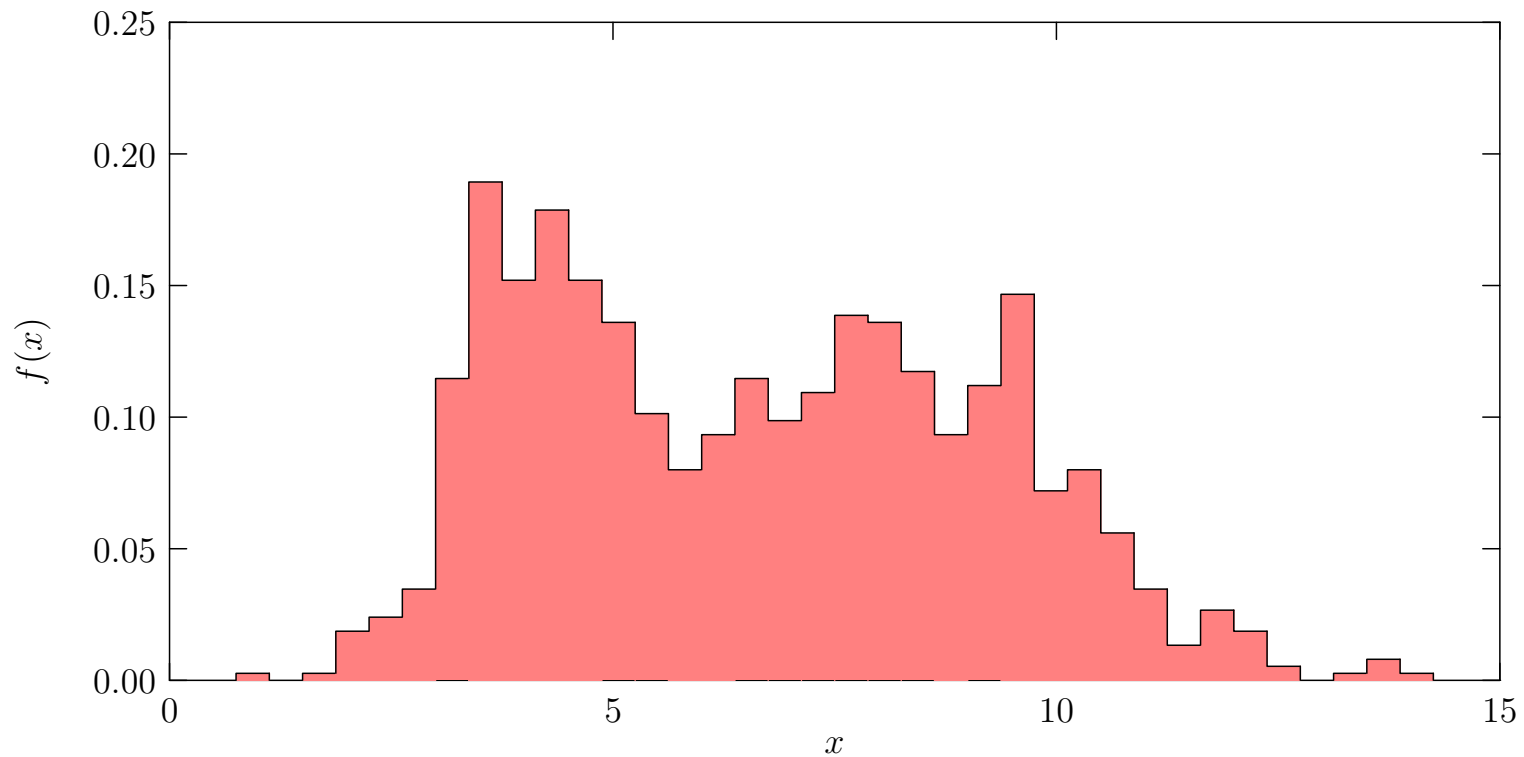
$$\begin{aligned} f(X_i|\Theta) &= \sum_{Z_i \in \{0,1\}} f(X_i, Z_i|\Theta) = \sum_{Z_i \in \{0,1\}} f(X_i|Z_i, \Theta) \mathbb{P}(Z_i) \\ &= \mathbb{E}_{Z_i}[f(X_i|Z_i, \Theta)] = p\mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1-p)\mathcal{N}(X_i|\mu_B, \sigma_B^2) \end{aligned}$$



Data

- Note that

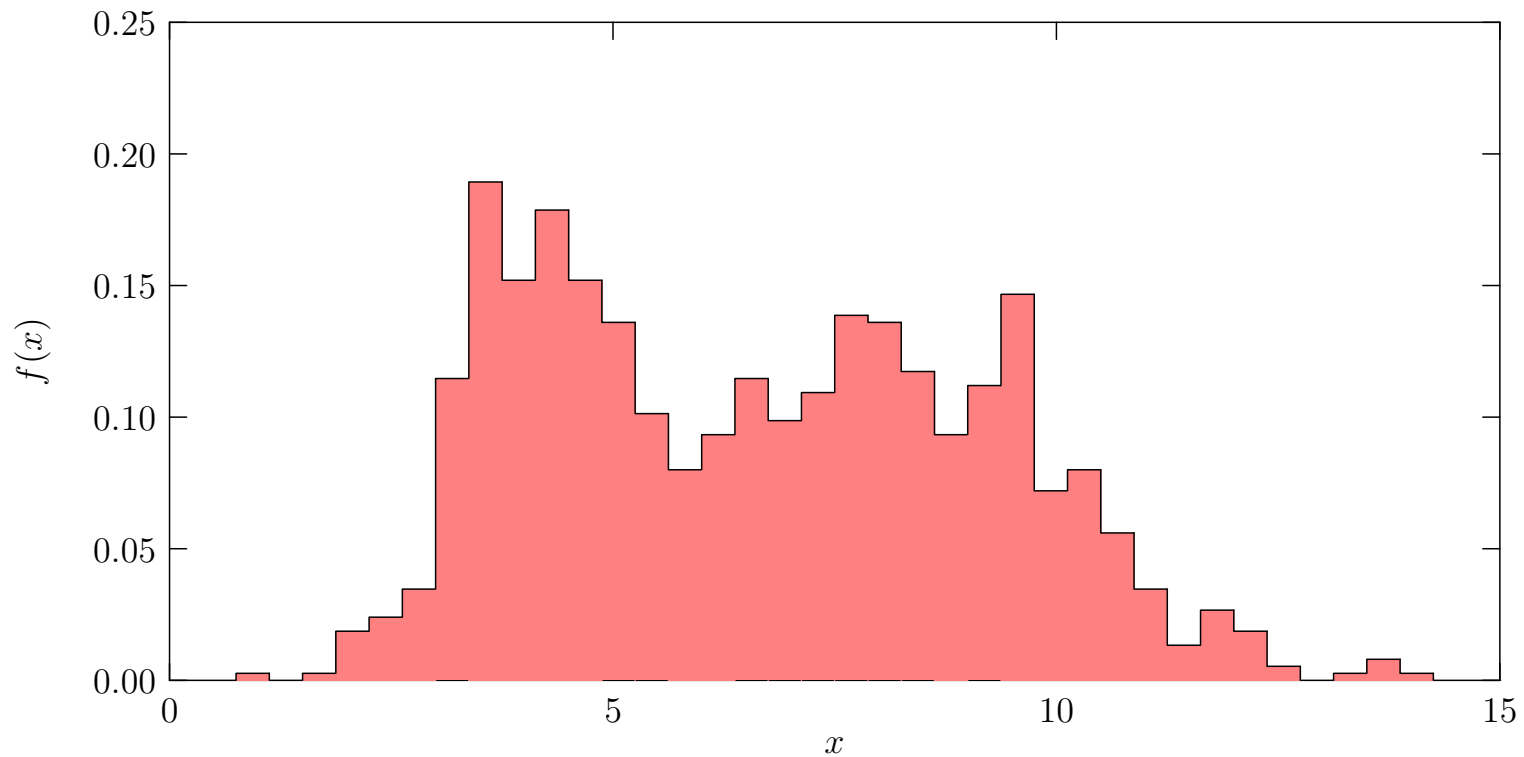
$$\begin{aligned} f(X_i|\Theta) &= \sum_{Z_i \in \{0,1\}} f(X_i, Z_i|\Theta) = \sum_{Z_i \in \{0,1\}} f(X_i|Z_i, \Theta) \mathbb{P}(Z_i) \\ &= \mathbb{E}_{Z_i}[f(X_i|Z_i, \Theta)] = p\mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1-p)\mathcal{N}(X_i|\mu_B, \sigma_B^2) \end{aligned}$$



Data

- Note that

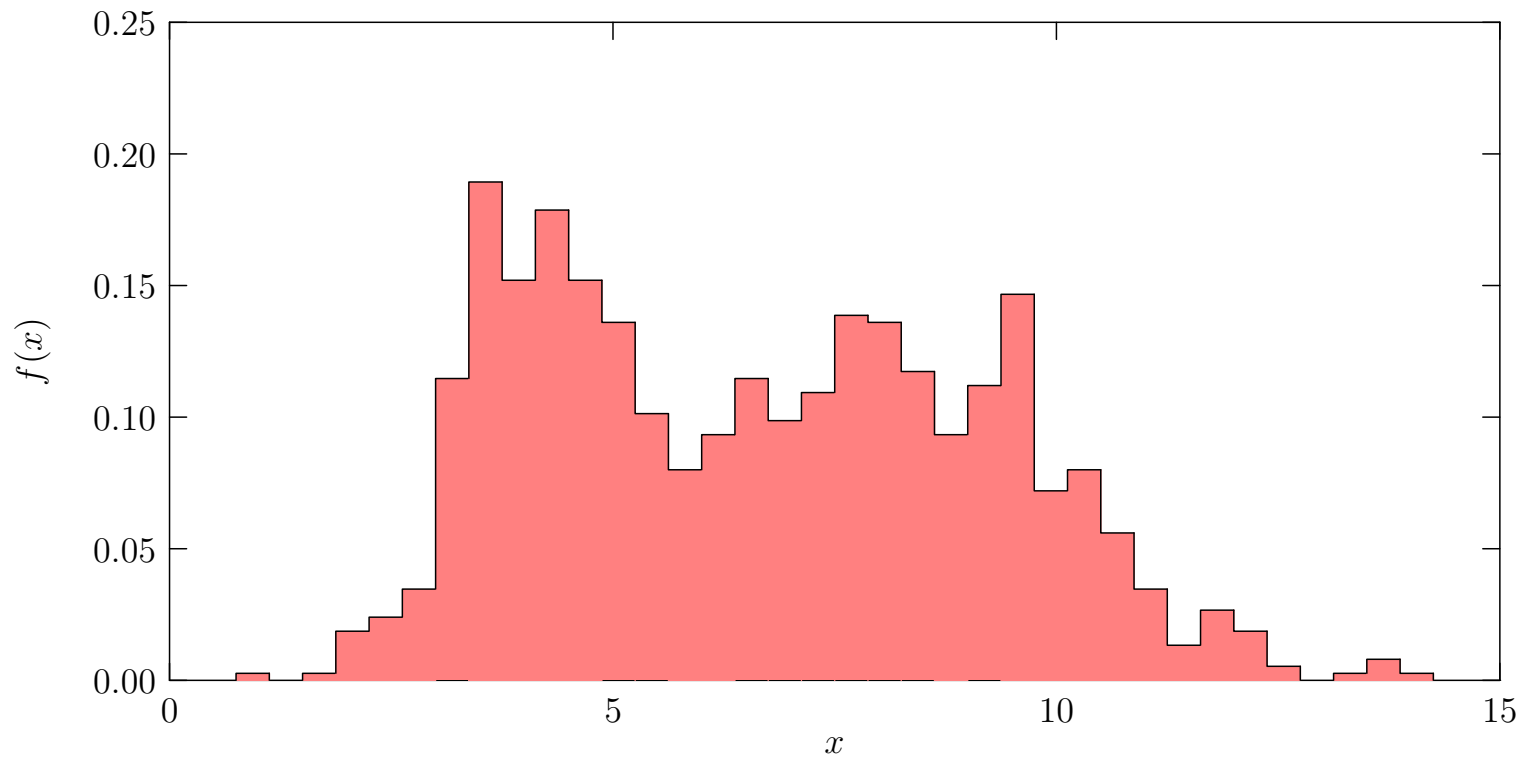
$$\begin{aligned} f(X_i|\Theta) &= \sum_{Z_i \in \{0,1\}} f(X_i, Z_i|\Theta) = \sum_{Z_i \in \{0,1\}} f(X_i|Z_i, \Theta) \mathbb{P}(Z_i) \\ &= \mathbb{E}_{Z_i}[f(X_i|Z_i, \Theta)] = p\mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1-p)\mathcal{N}(X_i|\mu_B, \sigma_B^2) \end{aligned}$$



Data

- Note that

$$\begin{aligned} f(X_i|\Theta) &= \sum_{Z_i \in \{0,1\}} f(X_i, Z_i|\Theta) = \sum_{Z_i \in \{0,1\}} f(X_i|Z_i, \Theta) \mathbb{P}(Z_i) \\ &= \mathbb{E}_{Z_i}[f(X_i|Z_i, \Theta)] = p\mathcal{N}(X_i|\mu_A, \sigma_A^2) + (1-p)\mathcal{N}(X_i|\mu_B, \sigma_B^2) \end{aligned}$$



Maximum Likelihood

- To solve the model as a Bayesian we would have to assign priors to our parameters $\Theta = (\mu_A, \sigma_A, \mu_B, \sigma_B, p)$
- This is doable, but complicated (we would also end up with a distribution for our parameters)
- Often we only want a reasonable estimate for some of our parameters (e.g. the half-lives μ_A and μ_B)
- A reasonable approach is to seek those parameters that maximise the likelihood of our observed data

$$f(\mathcal{D}|\Theta) = \prod_{X \in \mathcal{D}} f(X|\Theta)$$

Maximum Likelihood

- To solve the model as a Bayesian we would have to assign priors to our parameters $\Theta = (\mu_A, \sigma_A, \mu_B, \sigma_B, p)$
- This is doable, but complicated (we would also end up with a distribution for our parameters)
- Often we only want a reasonable estimate for some of our parameters (e.g. the half-lives μ_A and μ_B)
- A reasonable approach is to seek those parameters that maximise the likelihood of our observed data

$$f(\mathcal{D}|\Theta) = \prod_{X \in \mathcal{D}} f(X|\Theta)$$

Maximum Likelihood

- To solve the model as a Bayesian we would have to assign priors to our parameters $\Theta = (\mu_A, \sigma_A, \mu_B, \sigma_B, p)$
- This is doable, but complicated (we would also end up with a distribution for our parameters)
- Often we only want a reasonable estimate for some of our parameters (e.g. the half-lives μ_A and μ_B)
- A reasonable approach is to seek those parameters that maximise the likelihood of our observed data

$$f(\mathcal{D}|\Theta) = \prod_{X \in \mathcal{D}} f(X|\Theta)$$

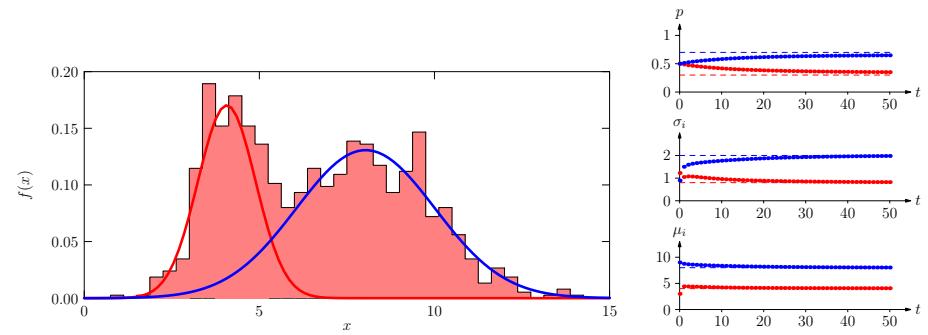
Maximum Likelihood

- To solve the model as a Bayesian we would have to assign priors to our parameters $\Theta = (\mu_A, \sigma_A, \mu_B, \sigma_B, p)$
- This is doable, but complicated (we would also end up with a distribution for our parameters)
- Often we only want a reasonable estimate for some of our parameters (e.g. the half-lives μ_A and μ_B)
- A reasonable approach is to seek those parameters that maximise the likelihood of our observed data

$$f(\mathcal{D}|\Theta) = \prod_{X \in \mathcal{D}} f(X|\Theta)$$

Outline

1. Building Probabilistic Models
2. Mixture of Gaussians
3. **Expectation Maximisation**



Maximum Likelihood with Latent Variables

- The maximum likelihood is a non-linear function of the parameters so cannot be immediately maximised
- If we knew which type of particle a data-point belongs to (Z_i) then it would be straightforward to maximise the likelihood
- As we don't we need to estimate $\mathbb{P}(Z_i = 1)$, but this depends on $\mu_A, \sigma_A^2, \mu_B, \sigma_B^2$ and p
- We could use a standard optimiser, but this is slightly inelegant

Maximum Likelihood with Latent Variables

- The maximum likelihood is a non-linear function of the parameters so cannot be immediately maximised
- If we knew which type of particle a data-point belongs to (Z_i) then it would be straightforward to maximise the likelihood
- As we don't we need to estimate $\mathbb{P}(Z_i = 1)$, but this depends on $\mu_A, \sigma_A^2, \mu_B, \sigma_B^2$ and p
- We could use a standard optimiser, but this is slightly inelegant

Maximum Likelihood with Latent Variables

- The maximum likelihood is a non-linear function of the parameters so cannot be immediately maximised
- If we knew which type of particle a data-point belongs to (Z_i) then it would be straightforward to maximise the likelihood
- As we don't we need to estimate $\mathbb{P}(Z_i = 1)$, but this depends on $\mu_A, \sigma_A^2, \mu_B, \sigma_B^2$ and p
- We could use a standard optimiser, but this is slightly inelegant

Maximum Likelihood with Latent Variables

- The maximum likelihood is a non-linear function of the parameters so cannot be immediately maximised
- If we knew which type of particle a data-point belongs to (Z_i) then it would be straightforward to maximise the likelihood
- As we don't we need to estimate $\mathbb{P}(Z_i = 1)$, but this depends on $\mu_A, \sigma_A^2, \mu_B, \sigma_B^2$ and p
- We could use a standard optimiser, but this is slightly inelegant

EM Algorithm

- Instead we can use an **expectation-maximisation algorithm** usually known as an **EM algorithm**
- We proceed iteratively by maximising the expected log-likelihood with respect to the current set of parameters

$$\Theta^{(t+1)} = \operatorname{argmax}_{\Theta} \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z} | \mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D} | \mathbf{Z}, \Theta))$$

- It isn't obvious why this works

EM Algorithm

- Instead we can use an **expectation-maximisation algorithm** usually known as an **EM algorithm**
- We proceed iteratively by maximising the expected log-likelihood with respect to the current set of parameters

$$\Theta^{(t+1)} = \operatorname{argmax}_{\Theta} \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z} | \mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D} | \mathbf{Z}, \Theta))$$

- It isn't obvious why this works

EM Algorithm

- Instead we can use an **expectation-maximisation algorithm** usually known as an **EM algorithm**
- We proceed iteratively by maximising the expected log-likelihood with respect to the current set of parameters

$$\Theta^{(t+1)} = \operatorname{argmax}_{\Theta} \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z} | \mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D} | \mathbf{Z}, \Theta))$$

- It isn't obvious why this works

Why EM Algorithm Works

- The argument around why this works is quite involved
- Note that at each step we maximise

$$Q(\Theta|\Theta^{(t)}) = \sum_{\mathbf{Z} \in \{0,1\}^m} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta))$$

- We can show that the maximum, $\Theta^{(t+1)}$, is such that

$$\log(f(\mathcal{D}|\Theta^{(t+1)})) - \log(f(\mathcal{D}|\Theta^{(t)})) \geq Q(\Theta^{(t+1)}|\Theta^{(t)}) - Q(\Theta^{(t)}|\Theta^{(t)}) \geq 0$$

- The details are given in the supplemental notes

Why EM Algorithm Works

- The argument around why this works is quite involved
- Note that at each step we maximise

$$Q(\Theta|\Theta^{(t)}) = \sum_{\mathbf{Z} \in \{0,1\}^m} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta))$$

- We can show that the maximum, $\Theta^{(t+1)}$, is such that

$$\log(f(\mathcal{D}|\Theta^{(t+1)})) - \log(f(\mathcal{D}|\Theta^{(t)})) \geq Q(\Theta^{(t+1)}|\Theta^{(t)}) - Q(\Theta^{(t)}|\Theta^{(t)}) \geq 0$$

- The details are given in the supplemental notes

Why EM Algorithm Works

- The argument around why this works is quite involved
- Note that at each step we maximise

$$Q(\Theta|\Theta^{(t)}) = \sum_{\mathbf{Z} \in \{0,1\}^m} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta))$$

- We can show that the maximum, $\Theta^{(t+1)}$, is such that

$$\log(f(\mathcal{D}|\Theta^{(t+1)})) - \log(f(\mathcal{D}|\Theta^{(t)})) \geq Q(\Theta^{(t+1)}|\Theta^{(t)}) - Q(\Theta^{(t)}|\Theta^{(t)}) \geq 0$$

- The details are given in the supplemental notes

Why EM Algorithm Works

- The argument around why this works is quite involved
- Note that at each step we maximise

$$Q(\Theta|\Theta^{(t)}) = \sum_{\mathbf{Z} \in \{0,1\}^m} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta))$$

- We can show that the maximum, $\Theta^{(t+1)}$, is such that

$$\log(f(\mathcal{D}|\Theta^{(t+1)})) - \log(f(\mathcal{D}|\Theta^{(t)})) \geq Q(\Theta^{(t+1)}|\Theta^{(t)}) - Q(\Theta^{(t)}|\Theta^{(t)}) \geq 0$$

- The details are given in the supplemental notes

Why EM Algorithm Works

- The argument around why this works is quite involved
- Note that at each step we maximise

$$Q(\Theta|\Theta^{(t)}) = \sum_{\mathbf{Z} \in \{0,1\}^m} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta))$$

- We can show that the maximum, $\Theta^{(t+1)}$, is such that

$$\log(f(\mathcal{D}|\Theta^{(t+1)})) - \log(f(\mathcal{D}|\Theta^{(t)})) \geq Q(\Theta^{(t+1)}|\Theta^{(t)}) - Q(\Theta^{(t)}|\Theta^{(t)}) \geq 0$$

- The details are given in the supplemental notes

Conditional Latent Variables

- We need to compute the distribution of latent variables conditioned on the data and current estimated parameters
- For our problem

$$\mathbb{P}\left(\mathbf{Z}|\mathcal{D},\Theta^{(t)}\right) = \prod_{i=1}^m \mathbb{P}\left(Z_i|X_i,\Theta^{(t)}\right)$$

where

$$\mathbb{P}\left(Z_i = 1|X_i,\Theta^{(t)}\right) = \frac{p^{(t)}\mathcal{N}\left(X_i|\mu_A^{(t)},\sigma_A^{2(t)}\right)}{p^{(t)}\mathcal{N}\left(X_i|\mu_A^{(t)},\sigma_A^{2(t)}\right) + (1 - p^{(t)})\mathcal{N}\left(X_i|\mu_B^{(t)},\sigma_B^{2(t)}\right)}$$

$$\mathbb{P}\left(Z_i = 0|X_i,\Theta^{(t)}\right) = 1 - \mathbb{P}\left(Z_i = 1|X_i,\Theta^{(t)}\right)$$

Conditional Latent Variables

- We need to compute the distribution of latent variables conditioned on the data and current estimated parameters
- For our problem

$$\mathbb{P}\left(\mathbf{Z}|\mathcal{D},\Theta^{(t)}\right) = \prod_{i=1}^m \mathbb{P}\left(Z_i|X_i,\Theta^{(t)}\right)$$

where

$$\mathbb{P}\left(Z_i = 1|X_i,\Theta^{(t)}\right) = \frac{p^{(t)} \mathcal{N}\left(X_i|\mu_A^{(t)},\sigma_A^{2(t)}\right)}{p^{(t)} \mathcal{N}\left(X_i|\mu_A^{(t)},\sigma_A^{2(t)}\right) + (1 - p^{(t)}) \mathcal{N}\left(X_i|\mu_B^{(t)},\sigma_B^{2(t)}\right)}$$

$$\mathbb{P}\left(Z_i = 0|X_i,\Theta^{(t)}\right) = 1 - \mathbb{P}\left(Z_i = 1|X_i,\Theta^{(t)}\right)$$

Conditional Latent Variables

- We need to compute the distribution of latent variables conditioned on the data and current estimated parameters
- For our problem

$$\mathbb{P}\left(\mathbf{Z}|\mathcal{D},\Theta^{(t)}\right) = \prod_{i=1}^m \mathbb{P}\left(Z_i|X_i,\Theta^{(t)}\right)$$

where

$$\mathbb{P}\left(Z_i = 1|X_i,\Theta^{(t)}\right) = \frac{p^{(t)} \mathcal{N}\left(X_i|\mu_A^{(t)},\sigma_A^{2(t)}\right)}{p^{(t)} \mathcal{N}\left(X_i|\mu_A^{(t)},\sigma_A^{2(t)}\right) + (1 - p^{(t)}) \mathcal{N}\left(X_i|\mu_B^{(t)},\sigma_B^{2(t)}\right)}$$

$$\mathbb{P}\left(Z_i = 0|X_i,\Theta^{(t)}\right) = 1 - \mathbb{P}\left(Z_i = 1|X_i,\Theta^{(t)}\right)$$

EM for Mixture of Gaussians

- Maximise with respect to parameters θ

$$\begin{aligned} Q(\theta|\theta^{(t)}) &= \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta)) = \sum_{i=1}^m \sum_{Z_i} \mathbb{P}(Z_i|\mathcal{D}, \Theta^{(t)}) \log(f(X_i|Z_i, \Theta)) \\ &= \sum_{i=1}^m \sum_{Z_i \in \{0,1\}} \mathbb{P}(Z_i|X_i, \theta_i^{(t)}) (Z_i \log(p) + (1 - Z_i) \log(1 - p) \\ &\quad - \frac{(X_i - \mu_{Z_i})^2}{2\sigma_{Z_i}^2} - \log(\sqrt{2\pi}\sigma_{Z_i})) \end{aligned}$$

- Compute update equations

$$\frac{\partial Q(\theta|\theta^{(t)})}{\partial \mu_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial \sigma_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial p} = 0$$

EM for Mixture of Gaussians

- Maximise with respect to parameters θ

$$\begin{aligned} Q(\theta|\theta^{(t)}) &= \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta)) = \sum_{i=1}^m \sum_{Z_i} \mathbb{P}(Z_i|\mathcal{D}, \Theta^{(t)}) \log(f(X_i|Z_i, \Theta)) \\ &= \sum_{i=1}^m \sum_{Z_i \in \{0,1\}} \mathbb{P}(Z_i|X_i, \theta_i^{(t)}) (Z_i \log(p) + (1 - Z_i) \log(1 - p) \\ &\quad - \frac{(X_i - \mu_{Z_i})^2}{2\sigma_{Z_i}^2} - \log(\sqrt{2\pi}\sigma_{Z_i})) \end{aligned}$$

- Compute update equations

$$\frac{\partial Q(\theta|\theta^{(t)})}{\partial \mu_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial \sigma_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial p} = 0$$

EM for Mixture of Gaussians

- Maximise with respect to parameters θ

$$\begin{aligned} Q(\theta|\theta^{(t)}) &= \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta)) = \sum_{i=1}^m \sum_{Z_i} \mathbb{P}(Z_i|\mathcal{D}, \Theta^{(t)}) \log(f(X_i|Z_i, \Theta)) \\ &= \sum_{i=1}^m \sum_{Z_i \in \{0,1\}} \mathbb{P}(Z_i|X_i, \theta_i^{(t)}) (Z_i \log(p) + (1 - Z_i) \log(1 - p) \\ &\quad - \frac{(X_i - \mu_{Z_i})^2}{2\sigma_{Z_i}^2} - \log(\sqrt{2\pi}\sigma_{Z_i})) \end{aligned}$$

- Compute update equations

$$\frac{\partial Q(\theta|\theta^{(t)})}{\partial \mu_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial \sigma_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial p} = 0$$

EM for Mixture of Gaussians

- Maximise with respect to parameters θ

$$\begin{aligned} Q(\theta|\theta^{(t)}) &= \sum_{\mathbf{Z}} \mathbb{P}(\mathbf{Z}|\mathcal{D}, \Theta^{(t)}) \log(f(\mathcal{D}|\mathbf{Z}, \Theta)) = \sum_{i=1}^m \sum_{Z_i} \mathbb{P}(Z_i|\mathcal{D}, \Theta^{(t)}) \log(f(X_i|Z_i, \Theta)) \\ &= \sum_{i=1}^m \sum_{Z_i \in \{0,1\}} \mathbb{P}(Z_i|X_i, \theta_i^{(t)}) (Z_i \log(p) + (1 - Z_i) \log(1 - p) \\ &\quad - \frac{(X_i - \mu_{Z_i})^2}{2\sigma_{Z_i}^2} - \log(\sqrt{2\pi}\sigma_{Z_i})) \end{aligned}$$

- Compute update equations

$$\frac{\partial Q(\theta|\theta^{(t)})}{\partial \mu_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial \sigma_k} = 0, \quad \frac{\partial Q(\theta|\theta^{(t)})}{\partial p} = 0$$

Update Equations

- Means

$$\mu_{Z_i}^{(t+1)} = \frac{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)}) X_i}{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)})},$$

- Variances

$$(\sigma_{Z_i}^{(t+1)})^2 = \frac{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)}) (X_i - \mu_{Z_i}^{(t+1)})^2}{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)})}$$

- Probability of being type 1

$$p^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(Z_i = 1 | X_i, \boldsymbol{\theta}_i^{(t)})$$

Update Equations

- Means

$$\mu_{Z_i}^{(t+1)} = \frac{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)}) X_i}{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)})},$$

- Variances

$$(\sigma_{Z_i}^{(t+1)})^2 = \frac{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)}) (X_i - \mu_{Z_i}^{(t+1)})^2}{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)})}$$

- Probability of being type 1

$$p^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(Z_i = 1 | X_i, \boldsymbol{\theta}_i^{(t)})$$

Update Equations

- Means

$$\mu_{Z_i}^{(t+1)} = \frac{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)}) X_i}{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)})},$$

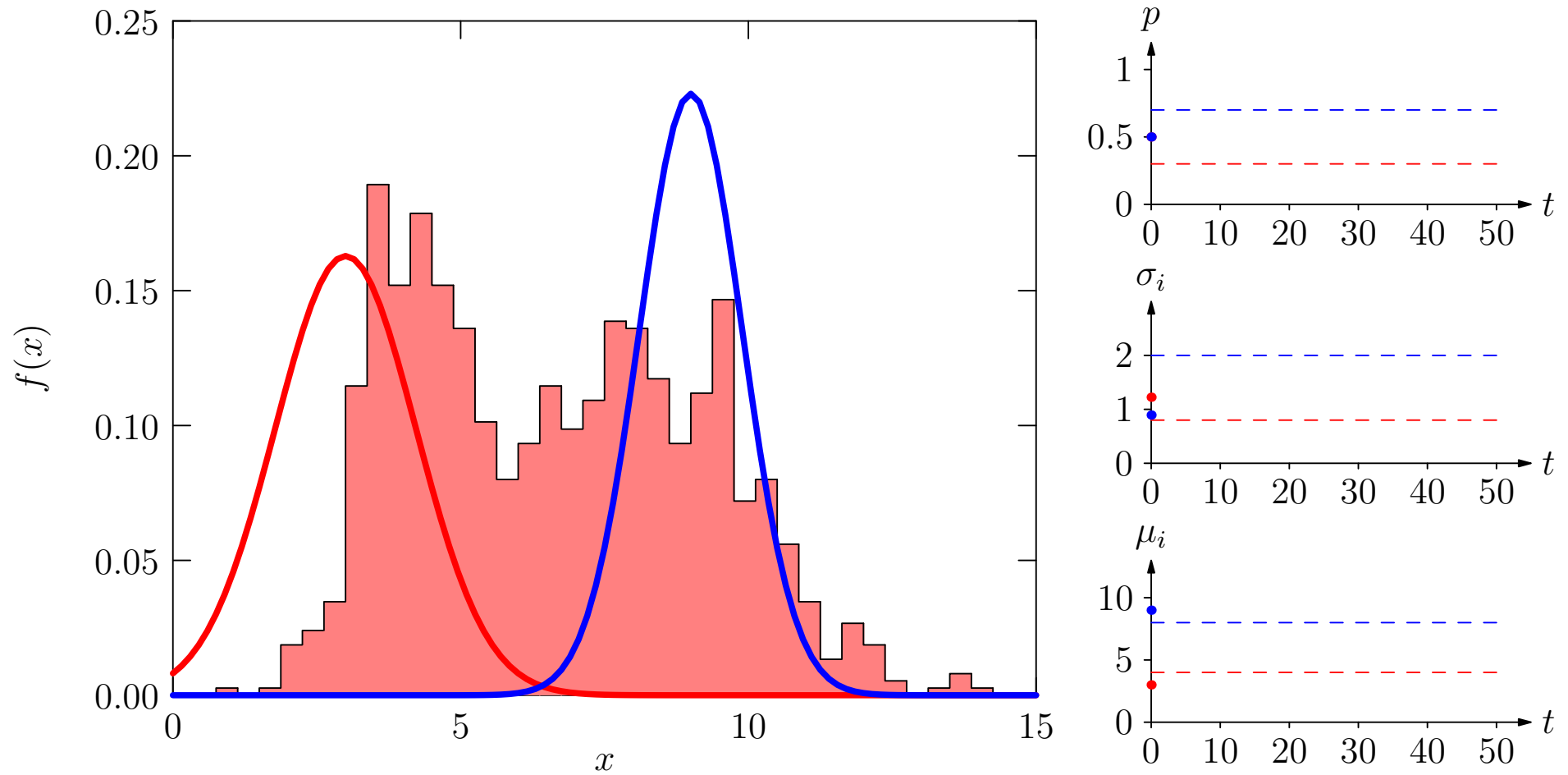
- Variances

$$(\sigma_{Z_i}^{(t+1)})^2 = \frac{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)}) (X_i - \mu_{Z_i}^{(t+1)})^2}{\sum_{i=1}^n \mathbb{P}(Z_i | X_i, \boldsymbol{\theta}^{(t)})}$$

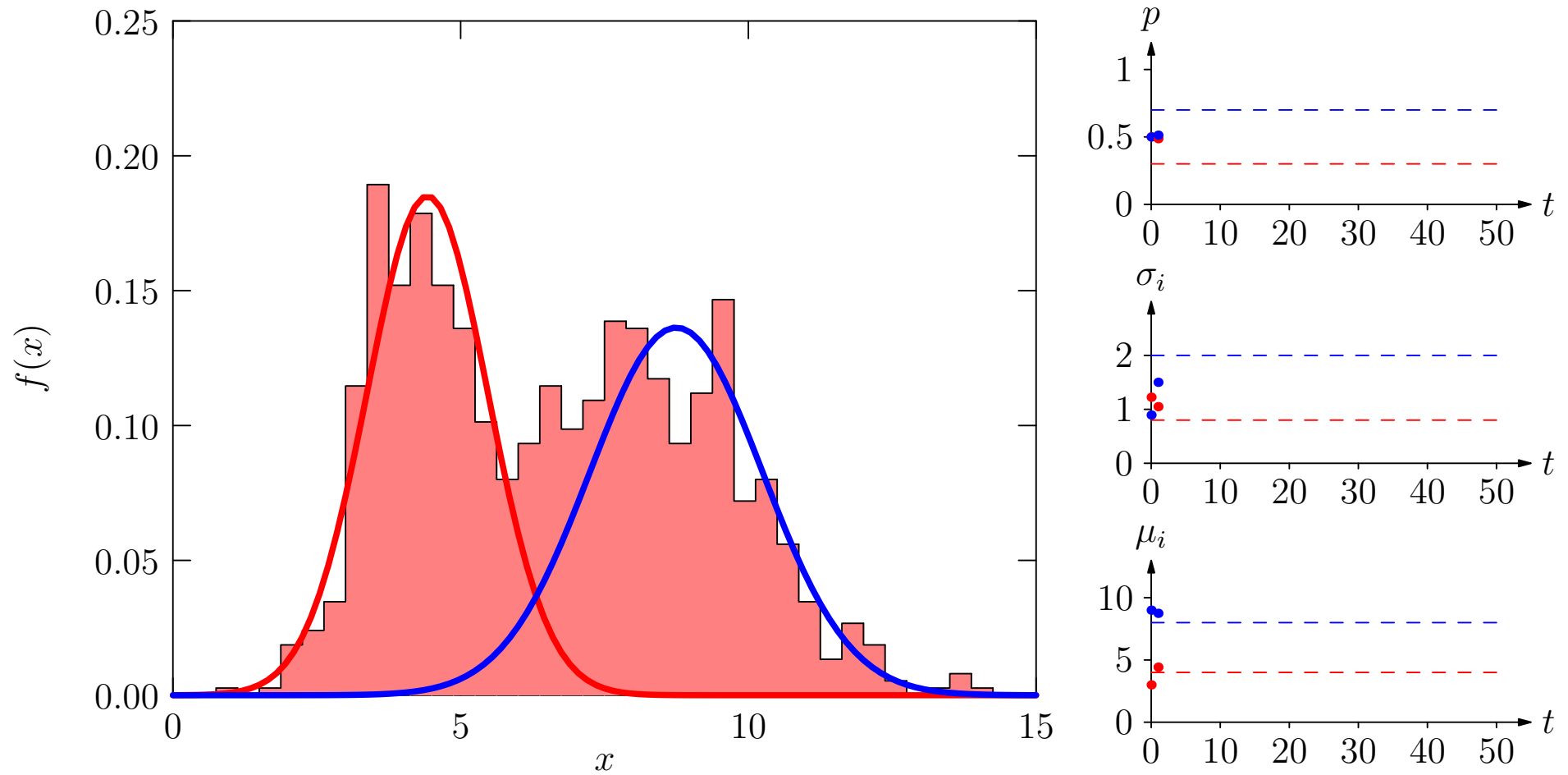
- Probability of being type 1

$$p^{(t+1)} = \frac{1}{n} \sum_{i=1}^n \mathbb{P}(Z_i = 1 | X_i, \boldsymbol{\theta}_i^{(t)})$$

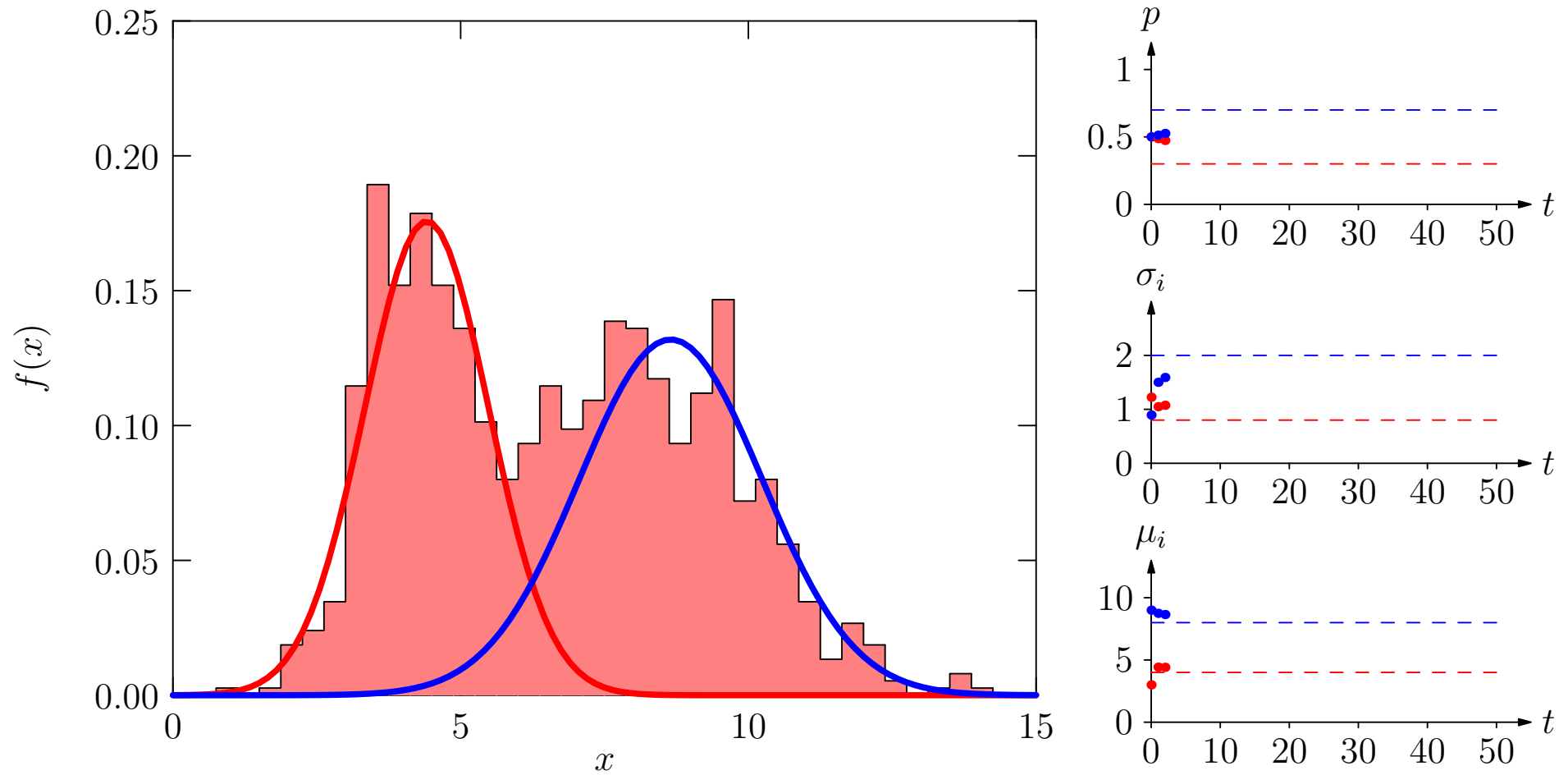
Example



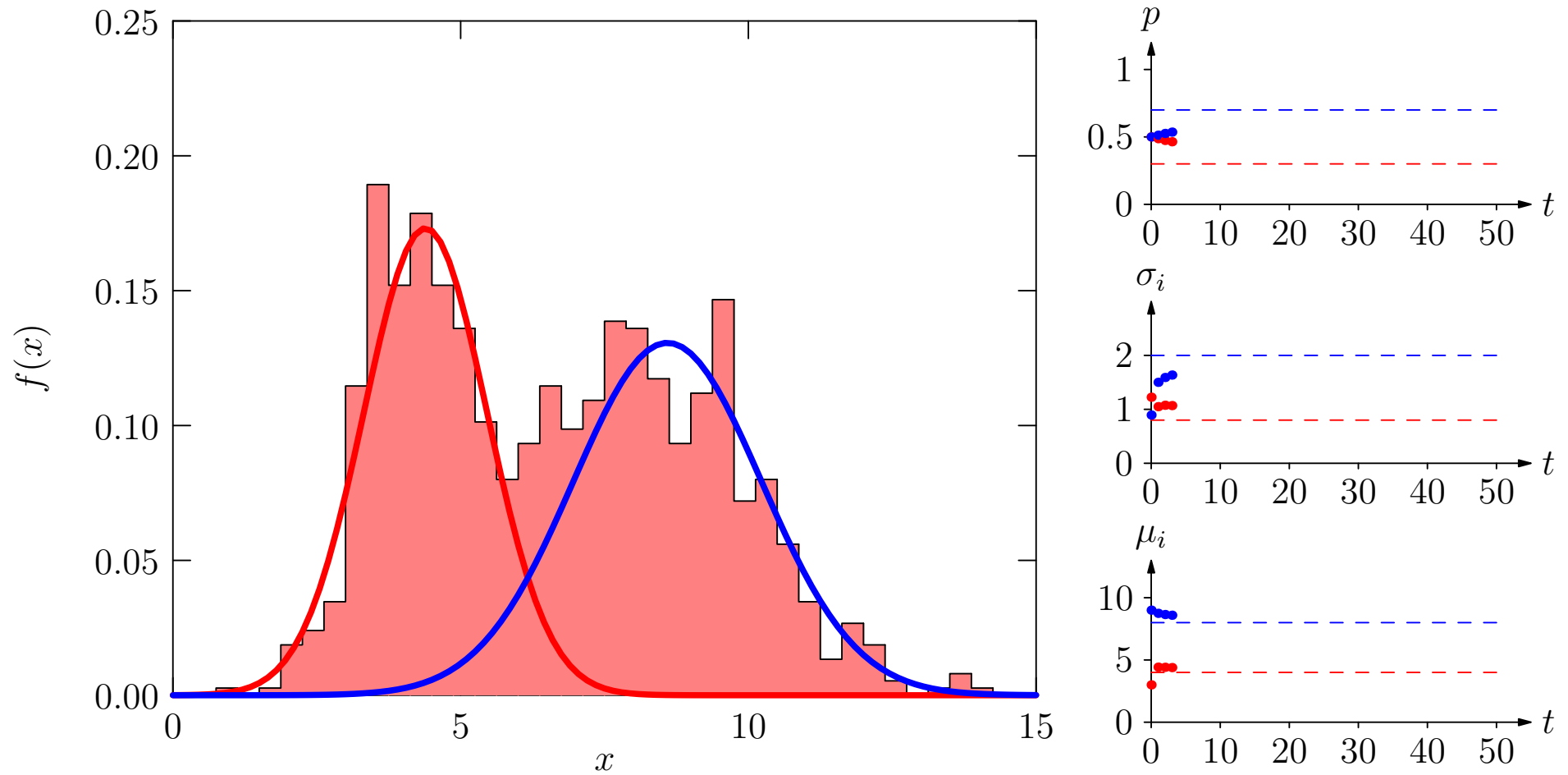
Example



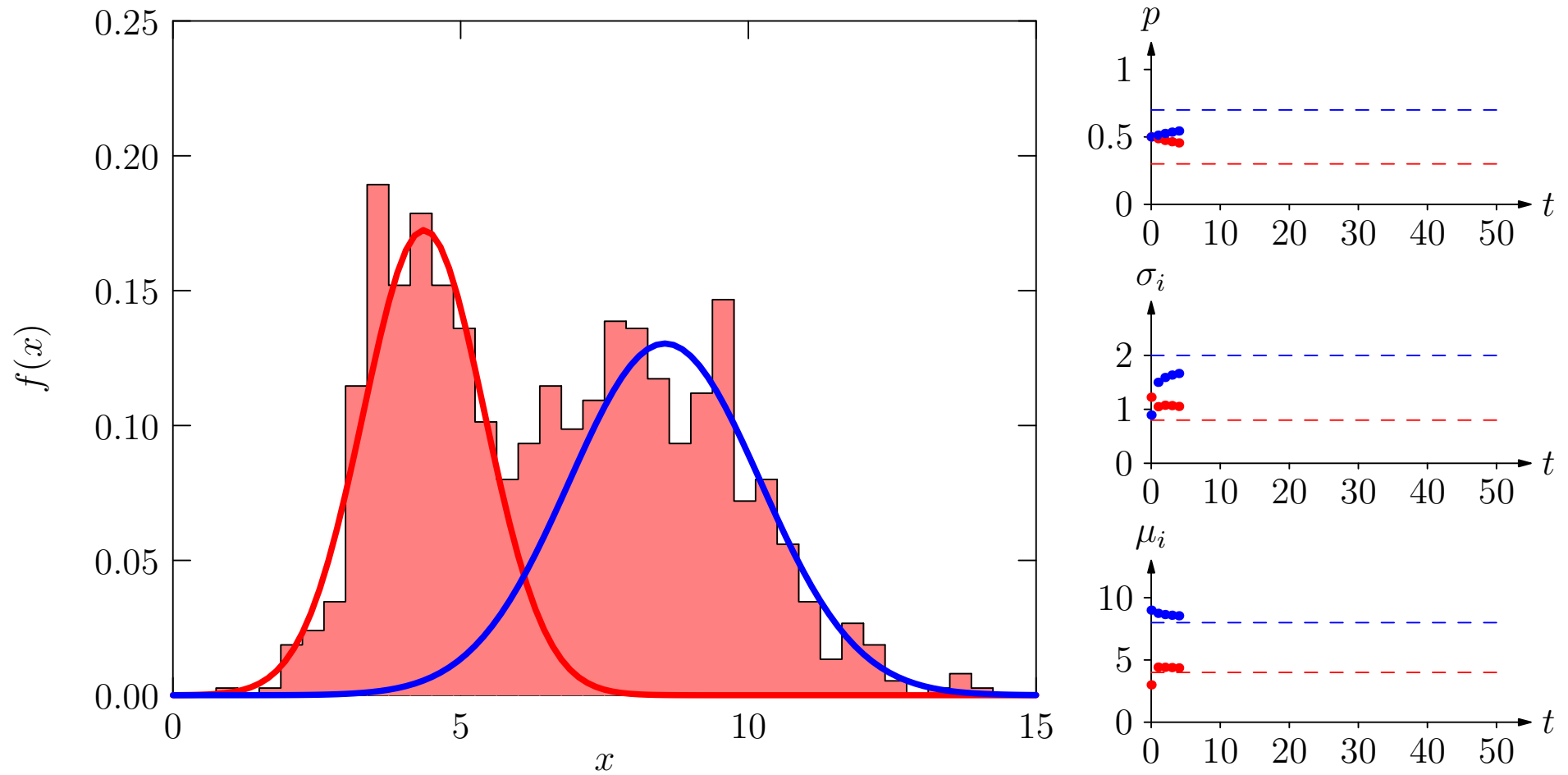
Example



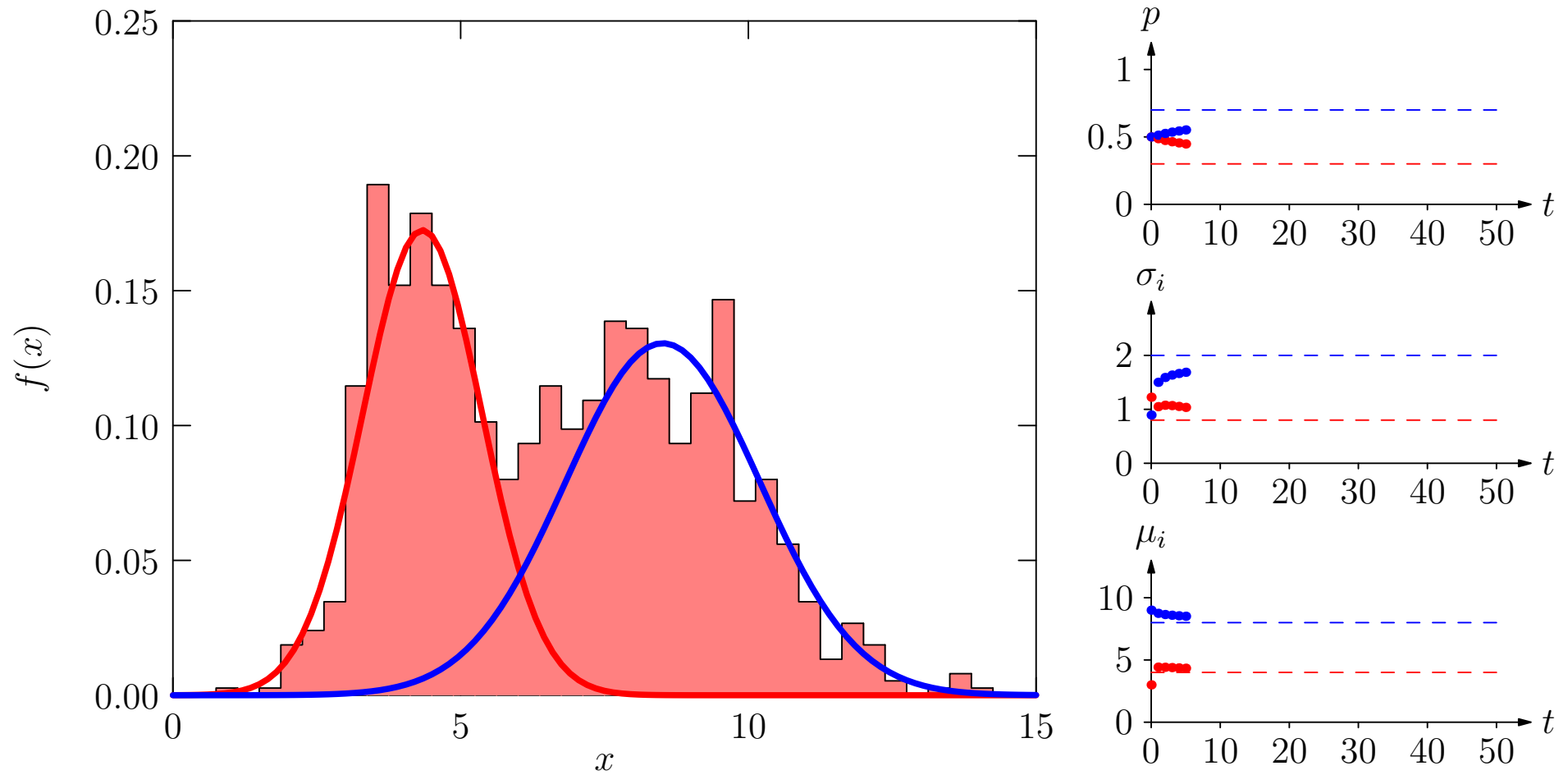
Example



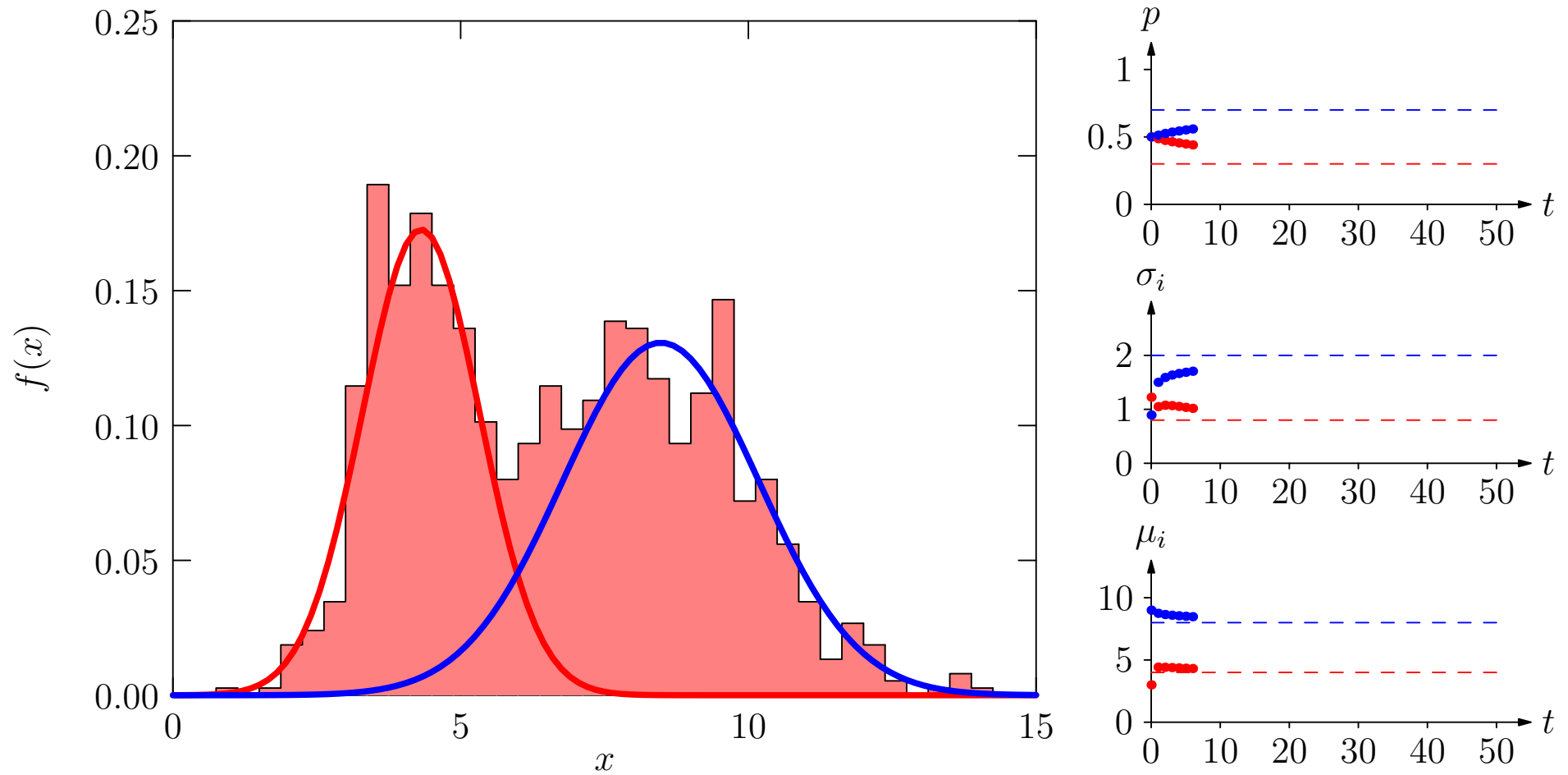
Example



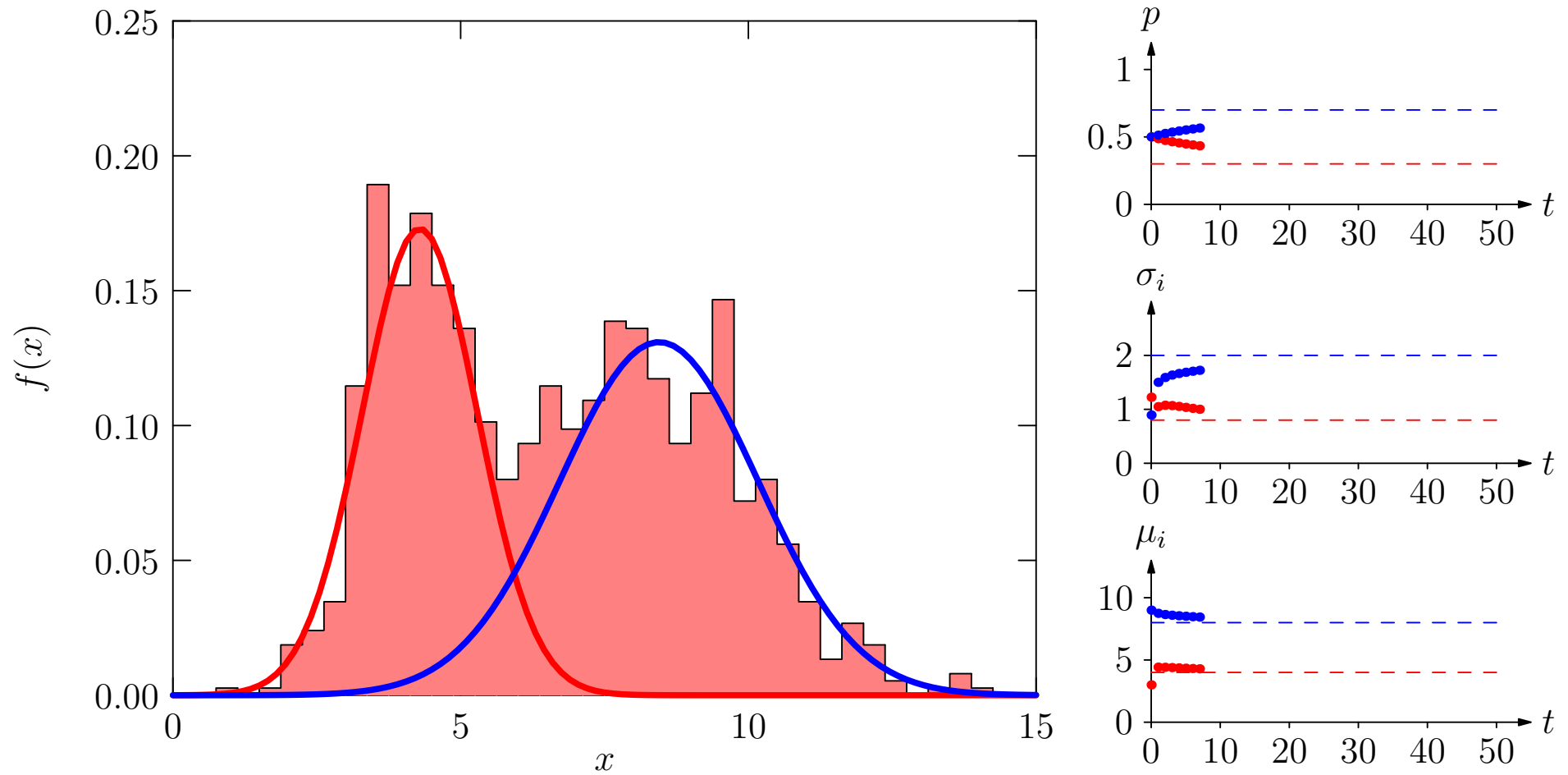
Example



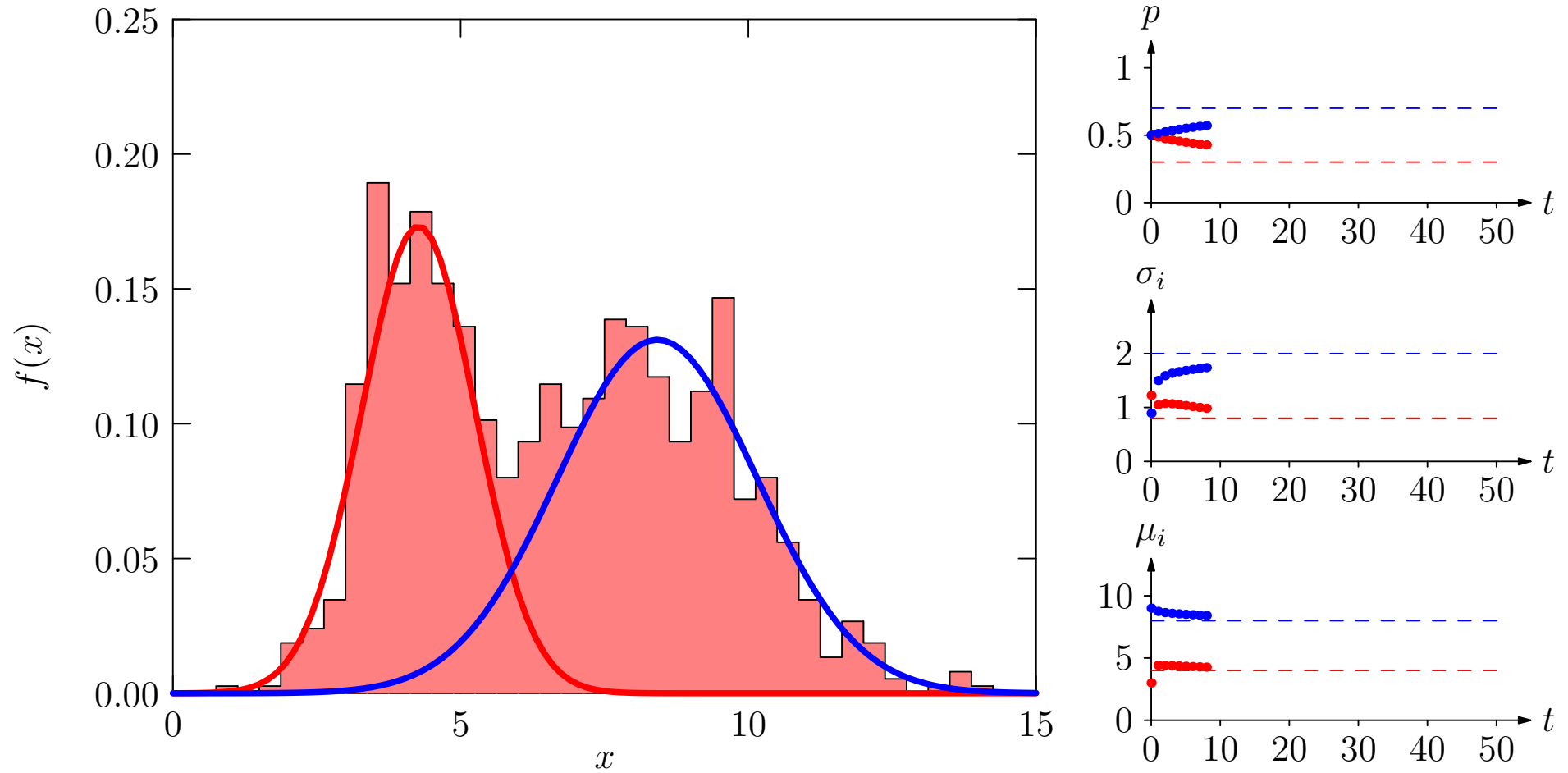
Example



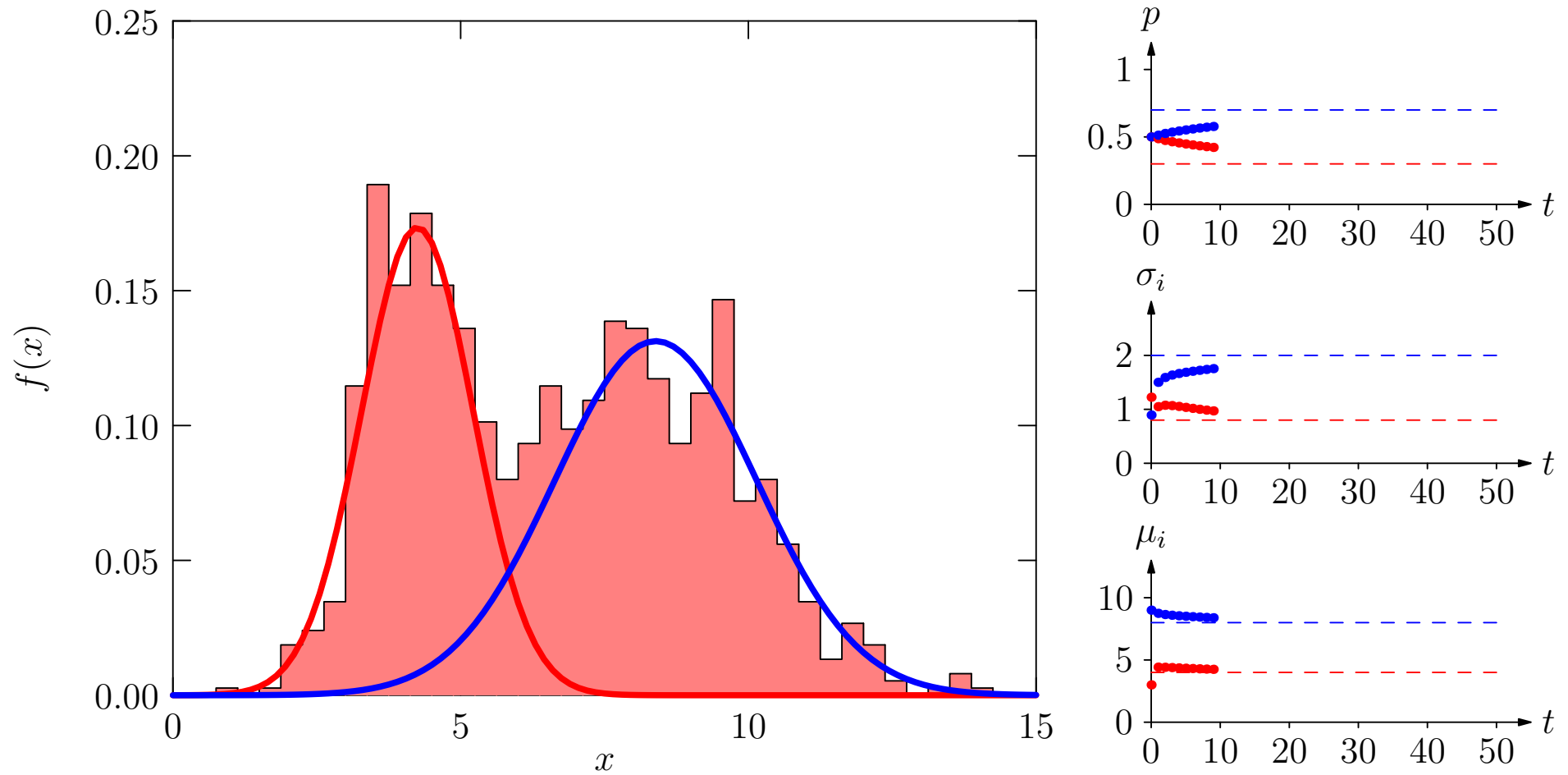
Example



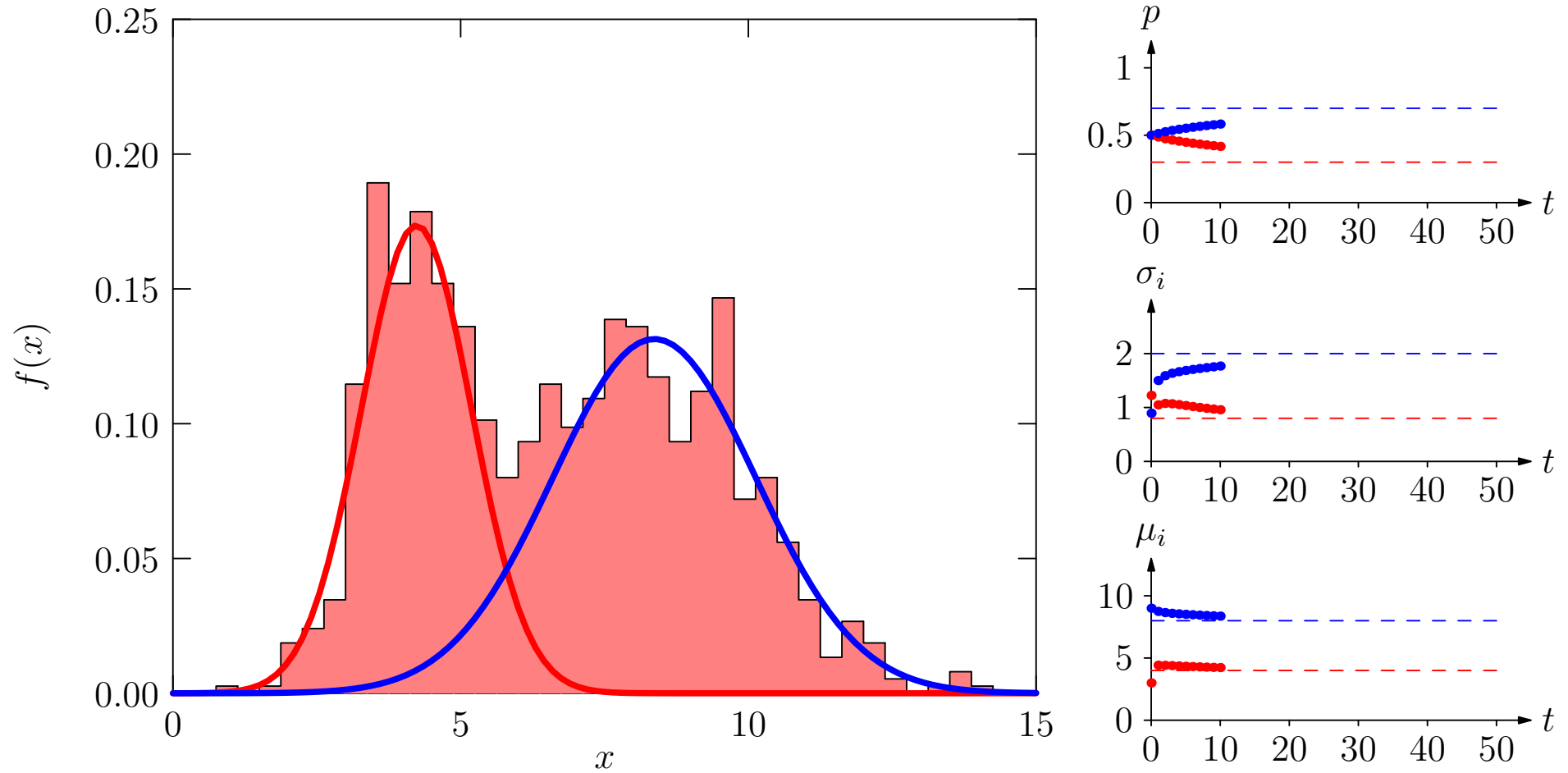
Example



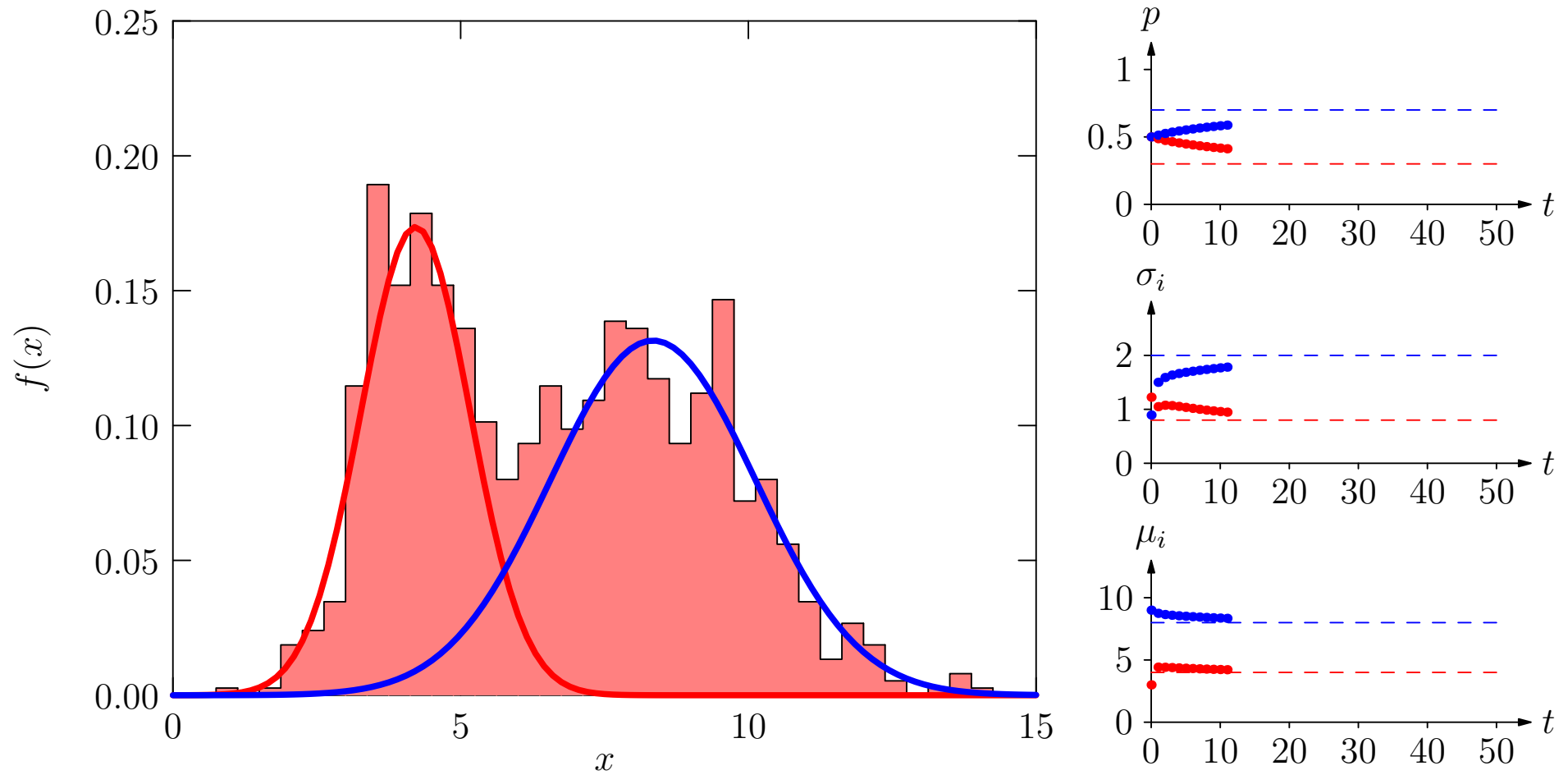
Example



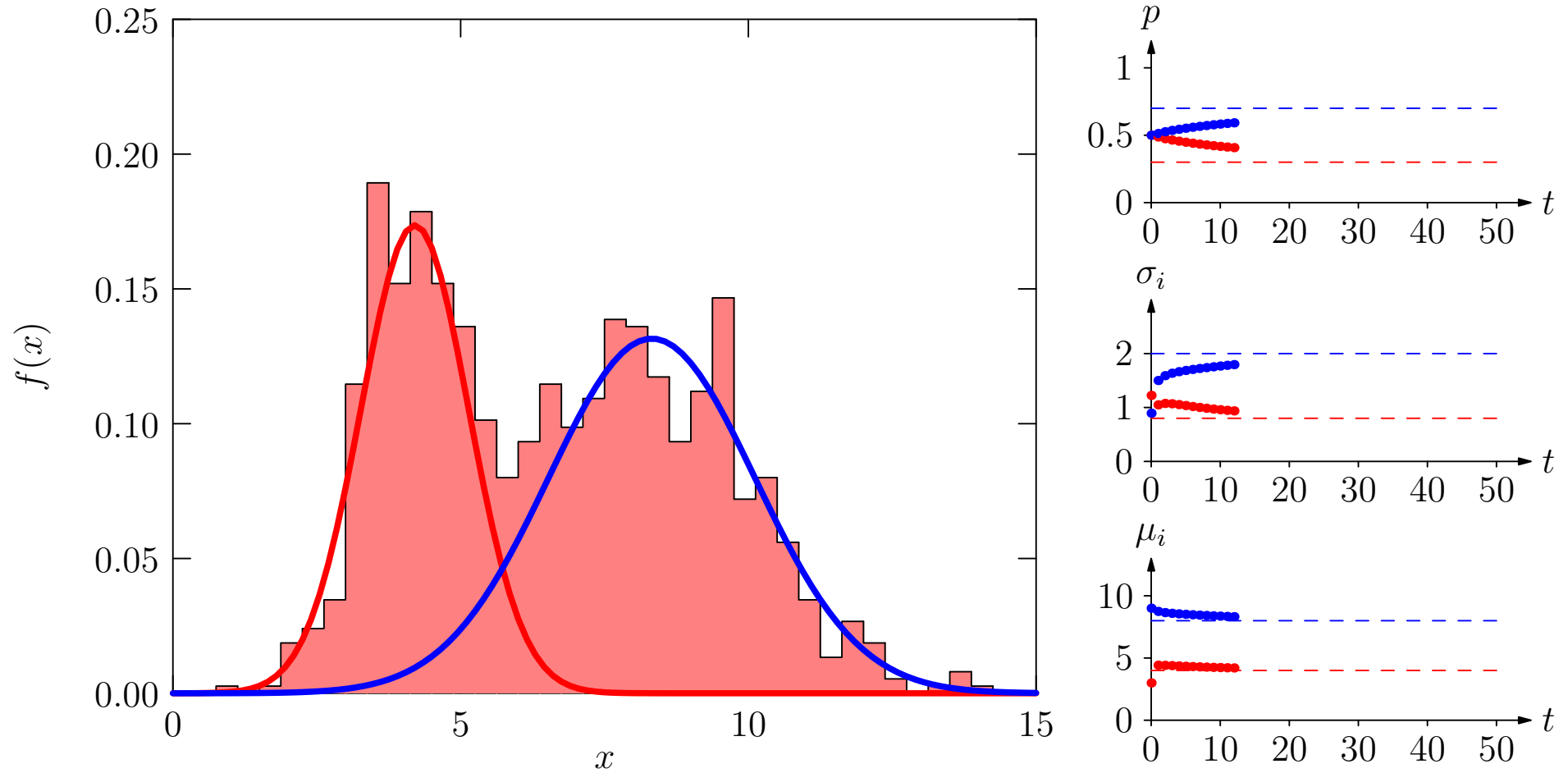
Example



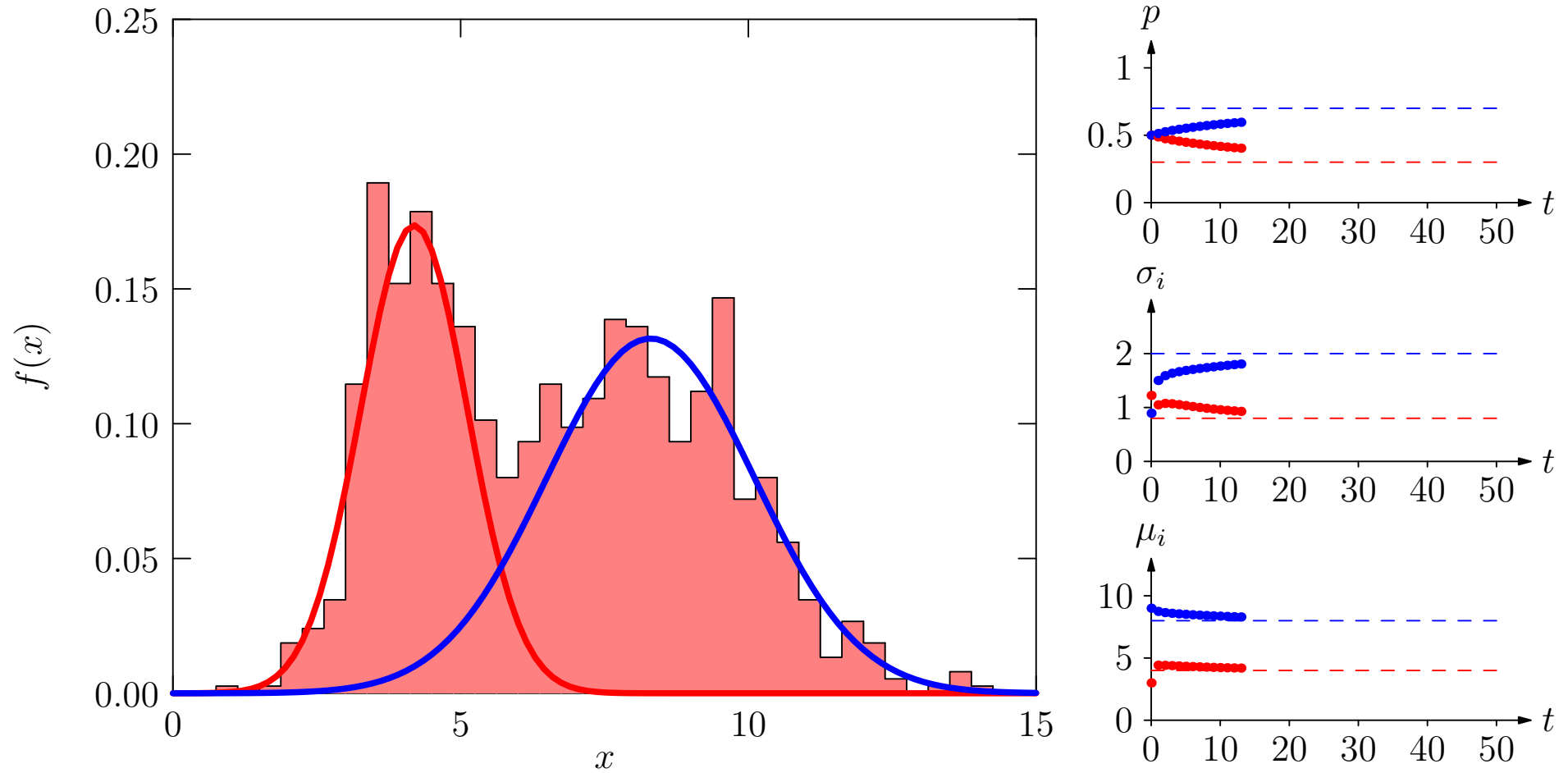
Example



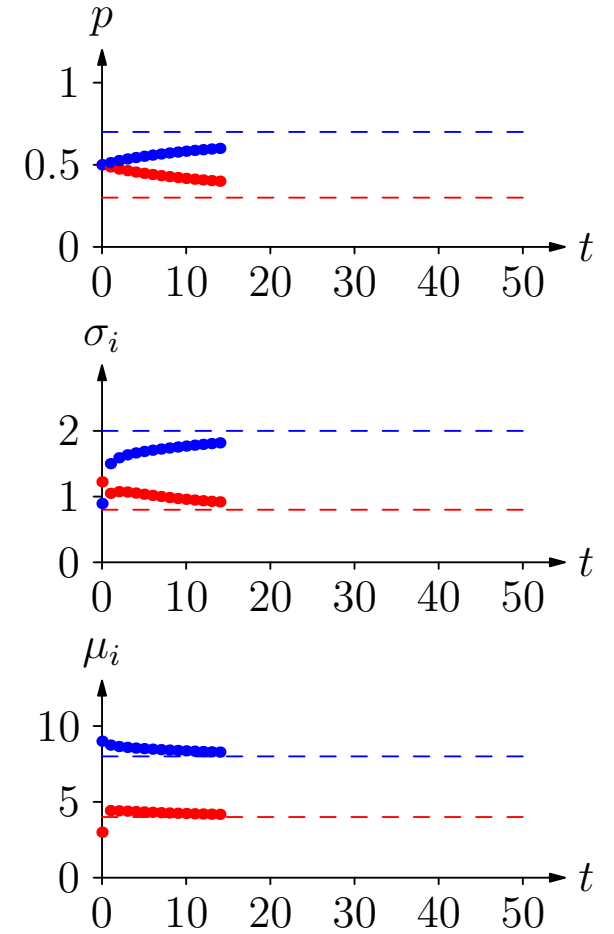
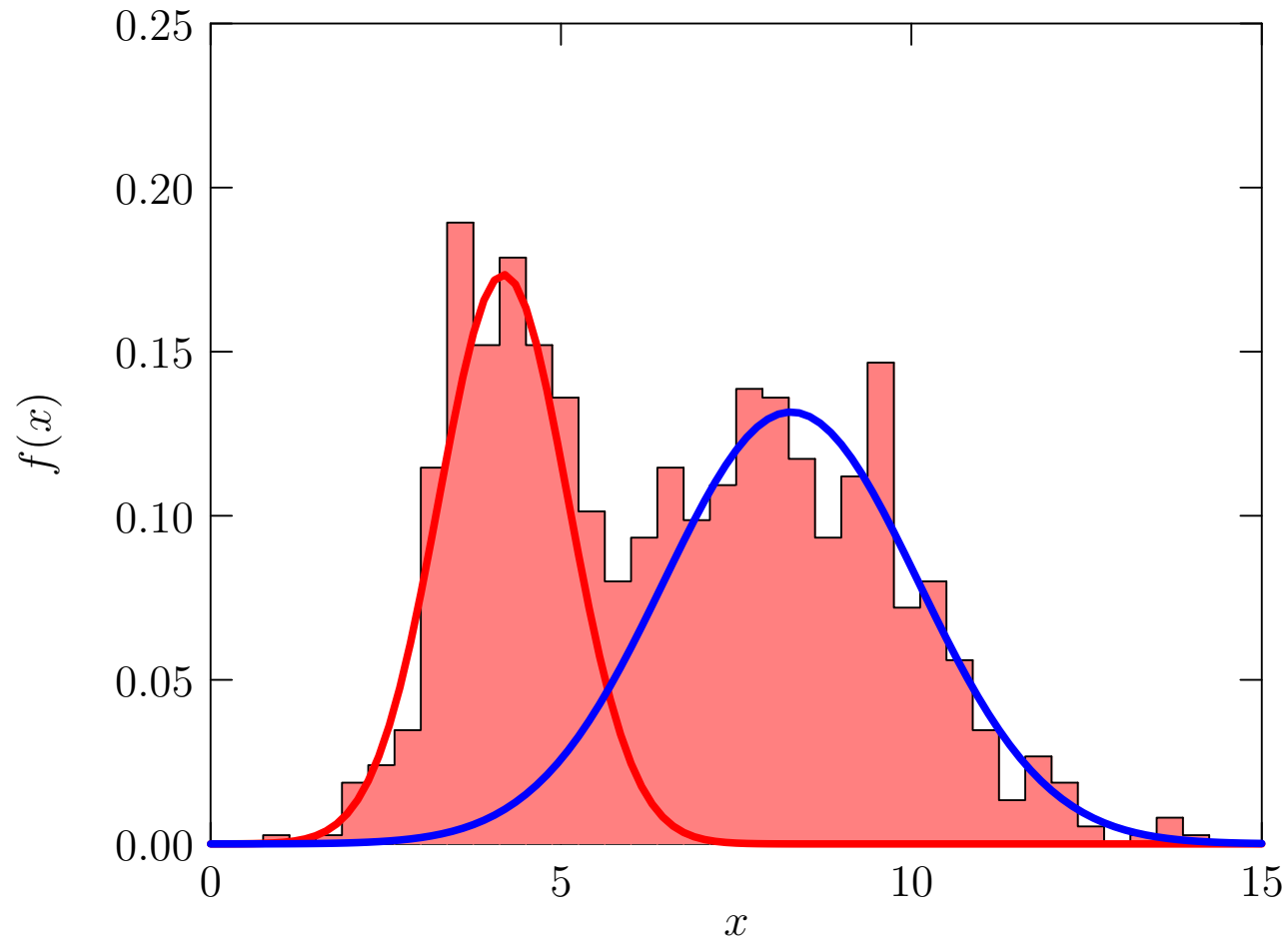
Example



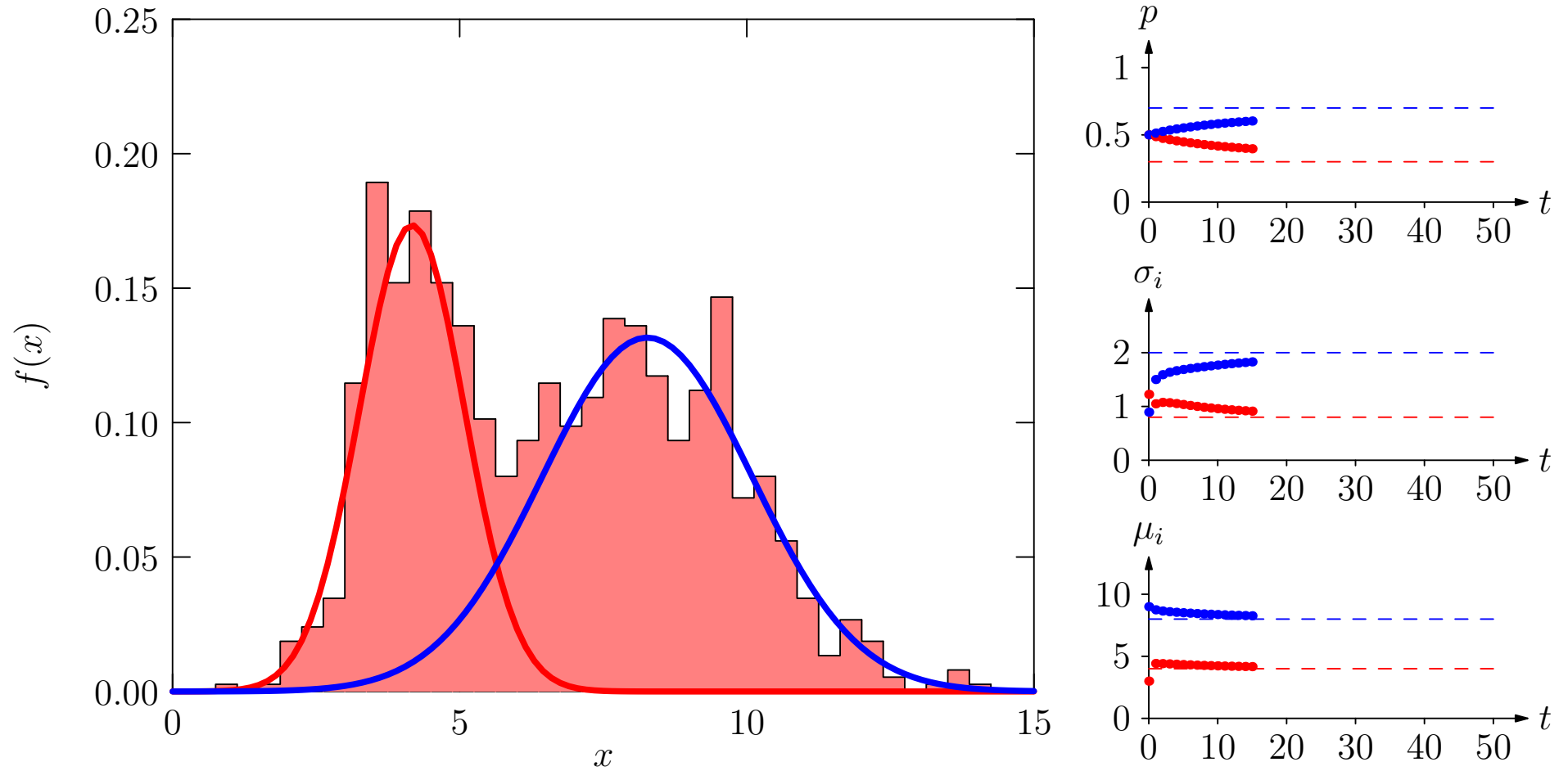
Example



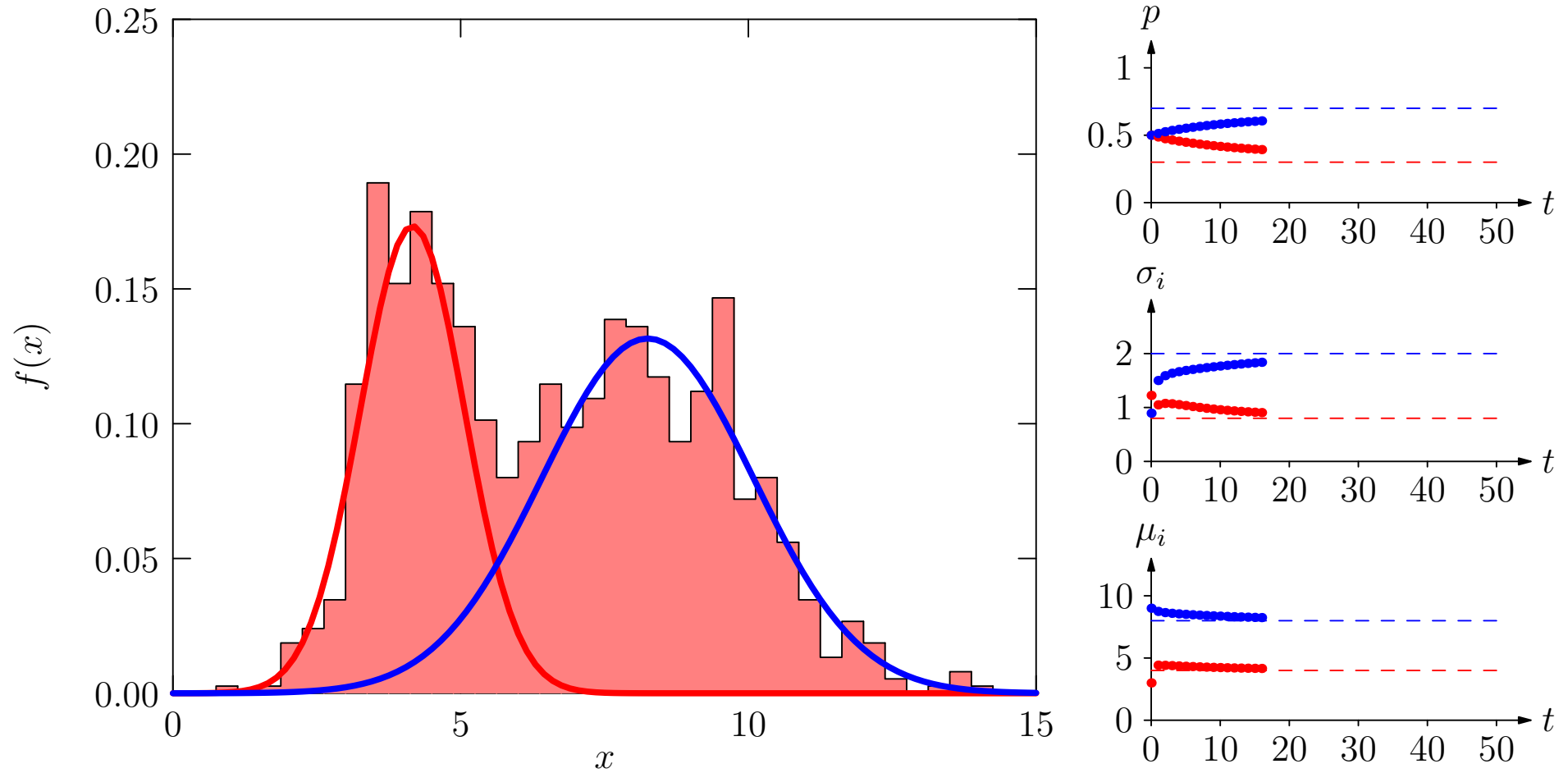
Example



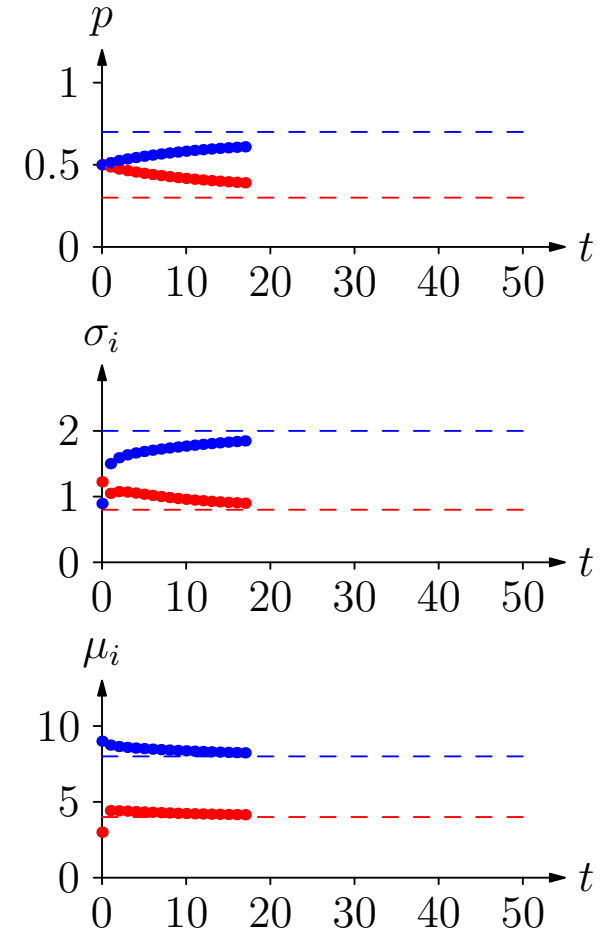
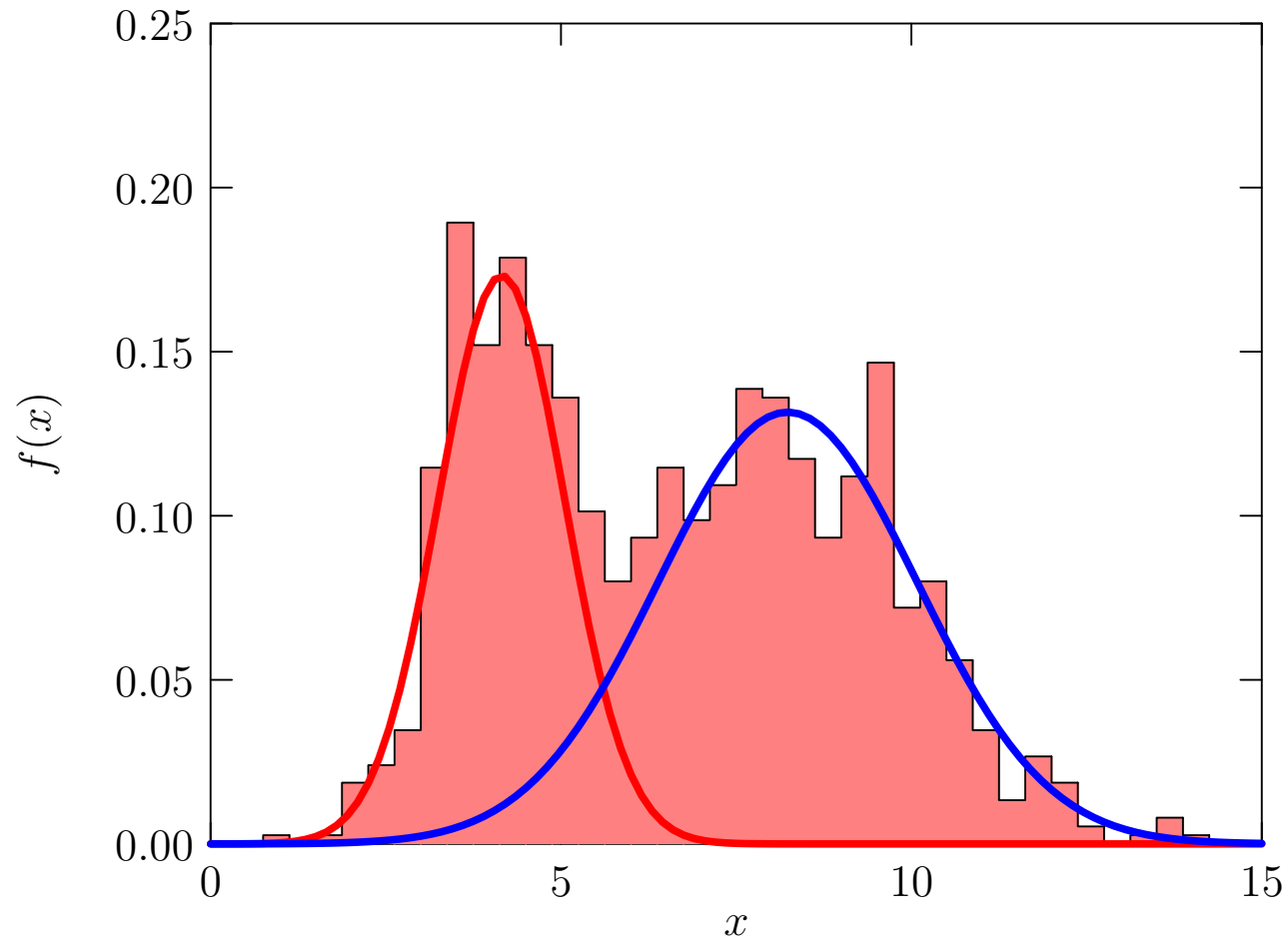
Example



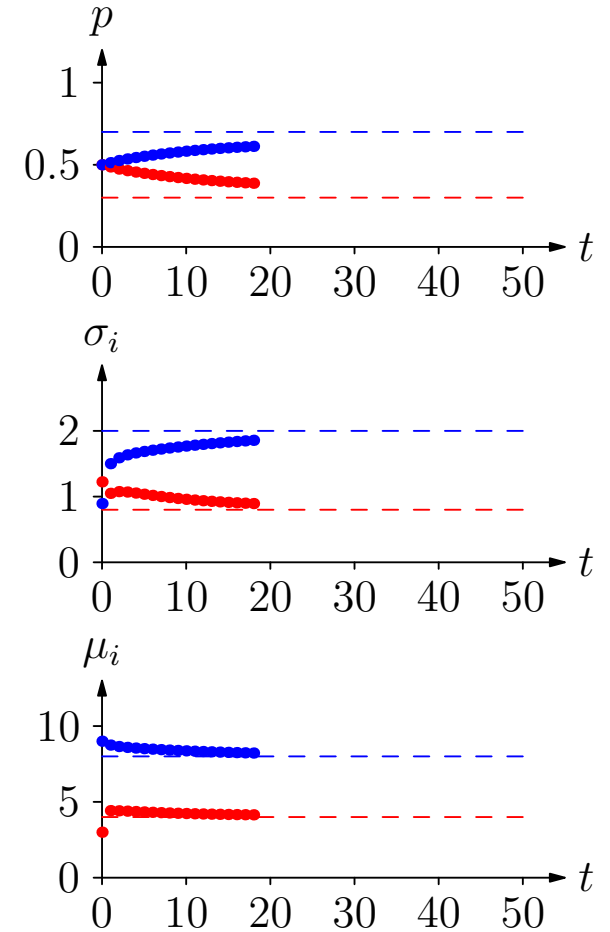
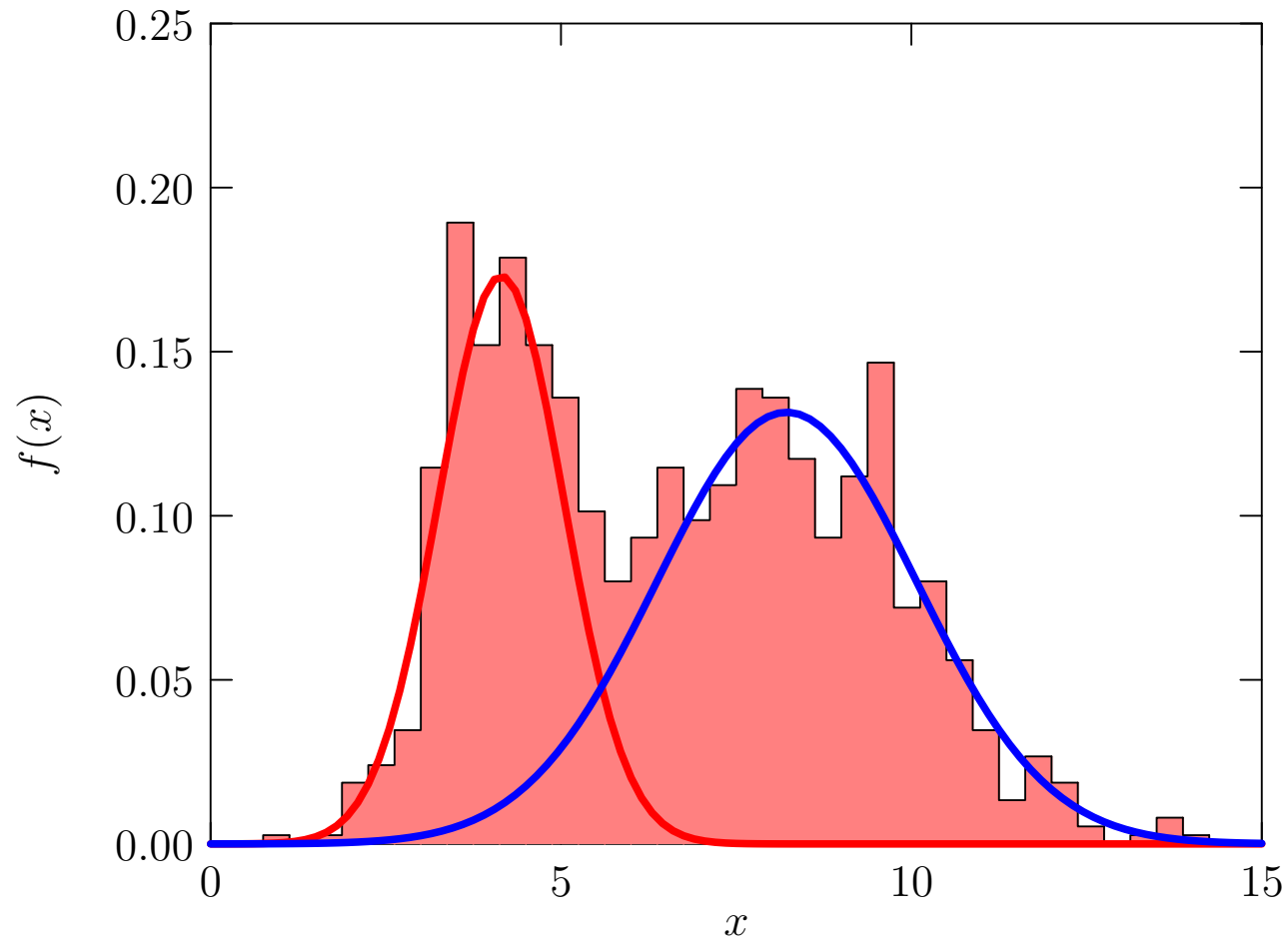
Example



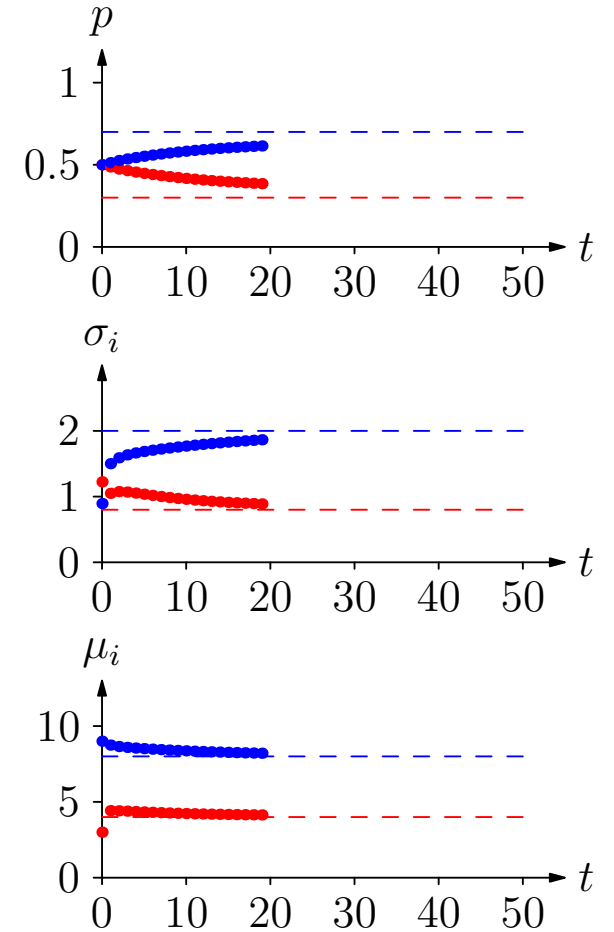
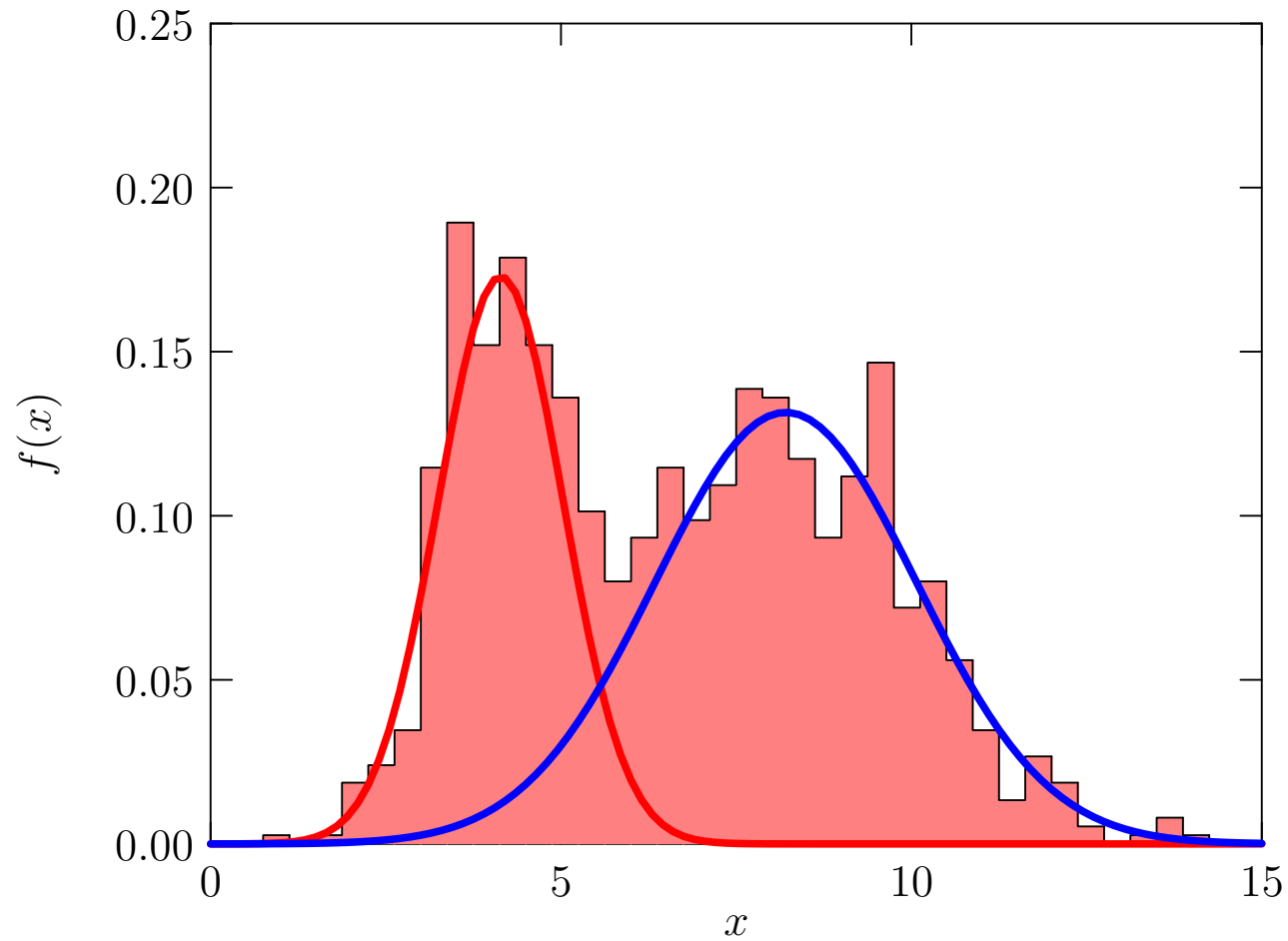
Example



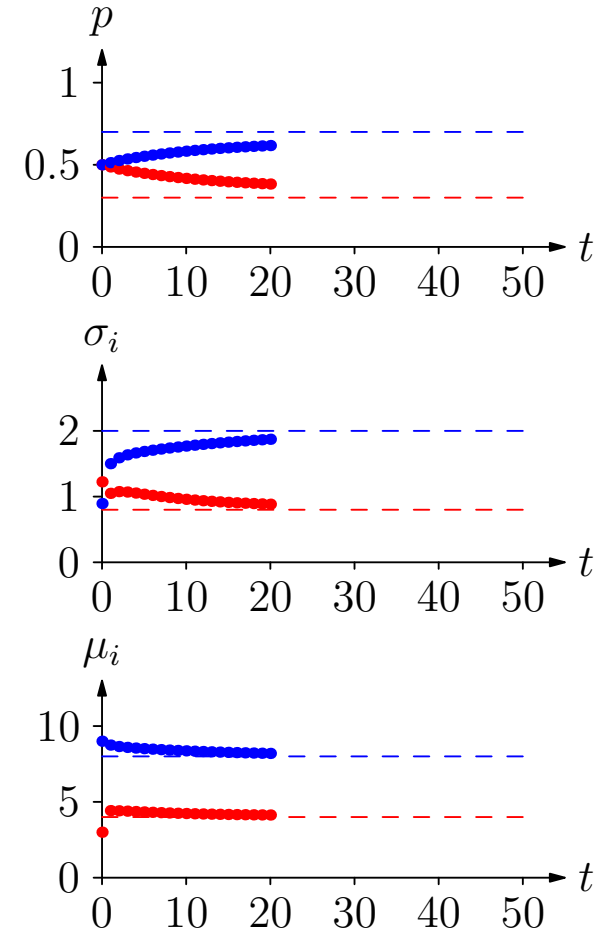
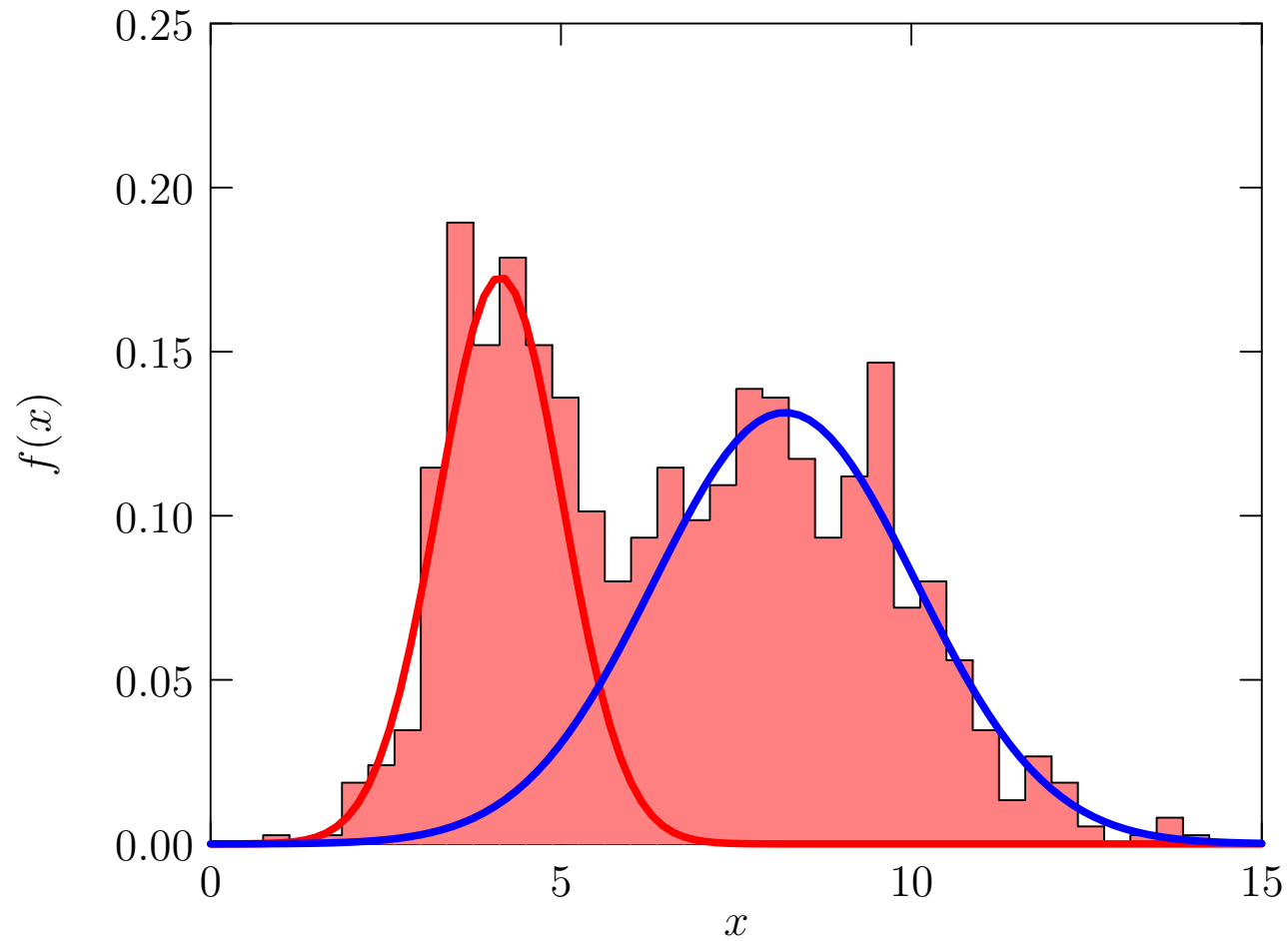
Example



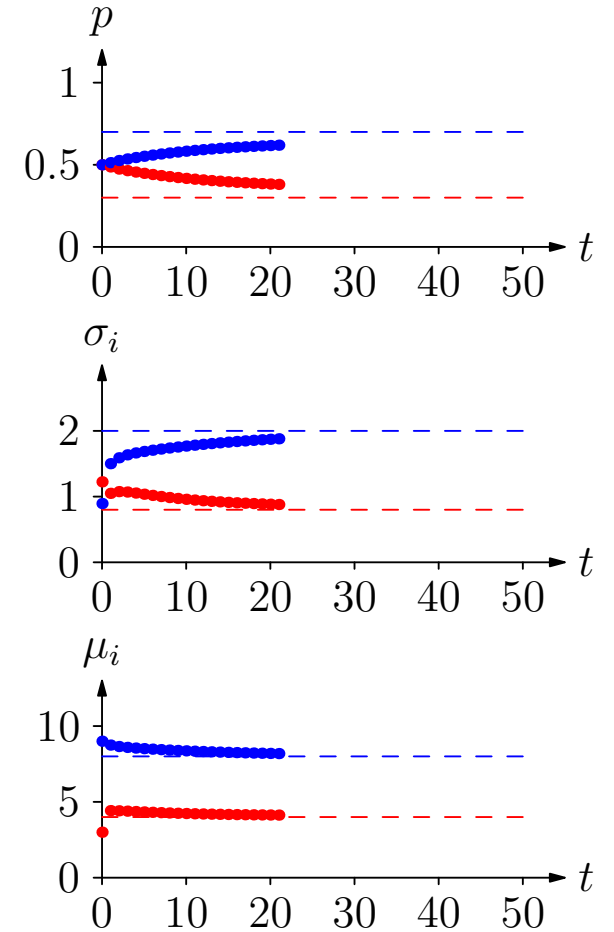
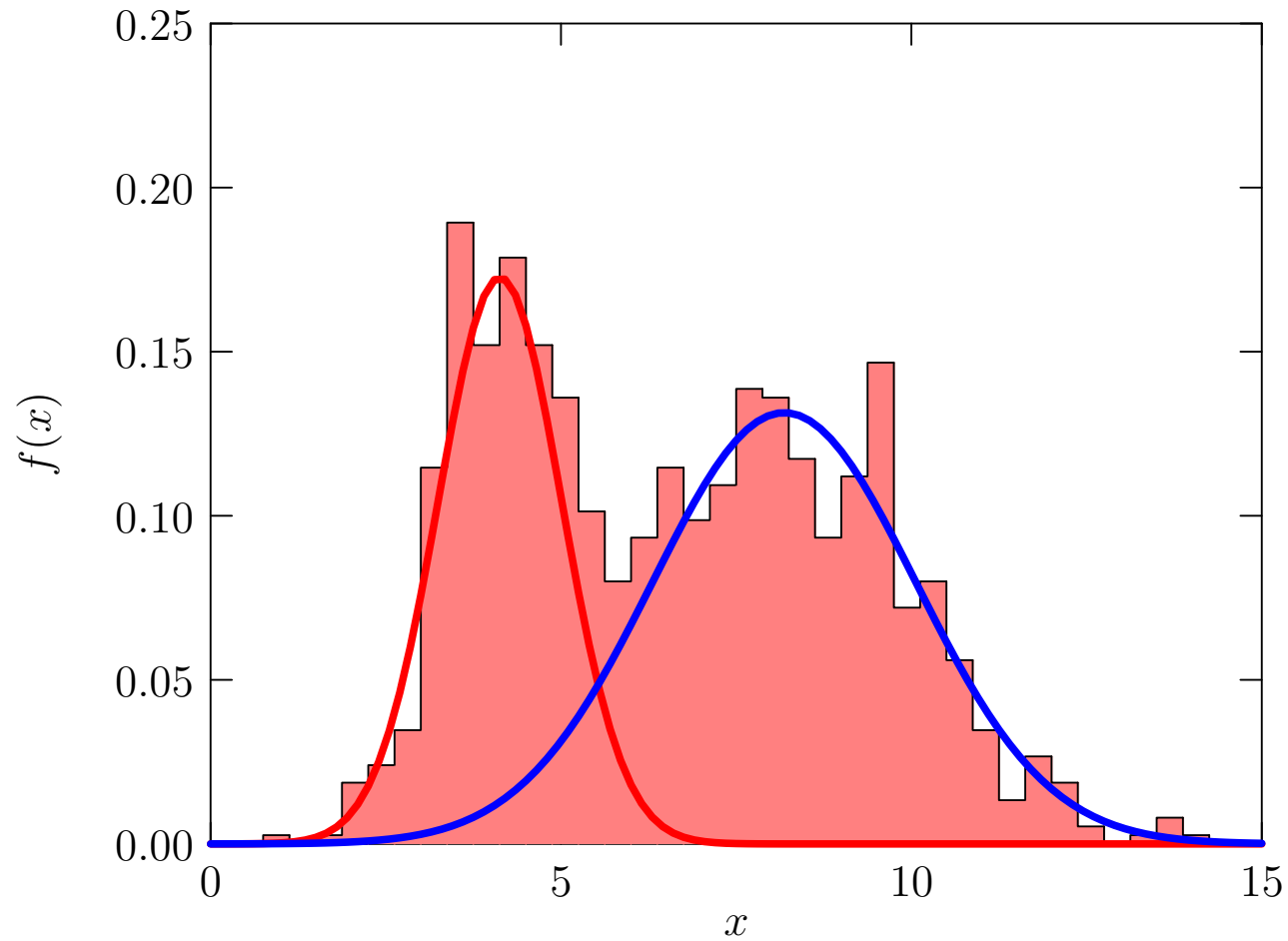
Example



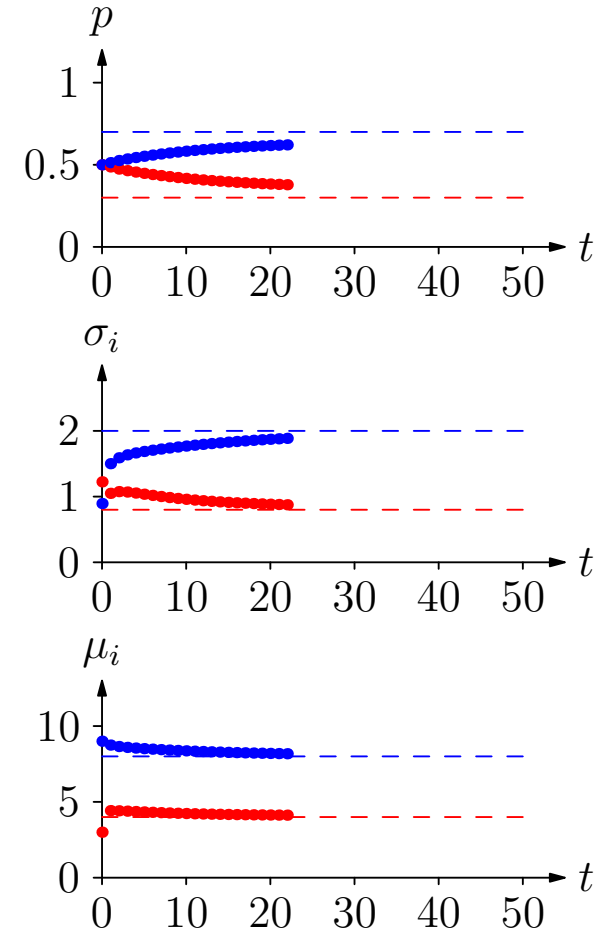
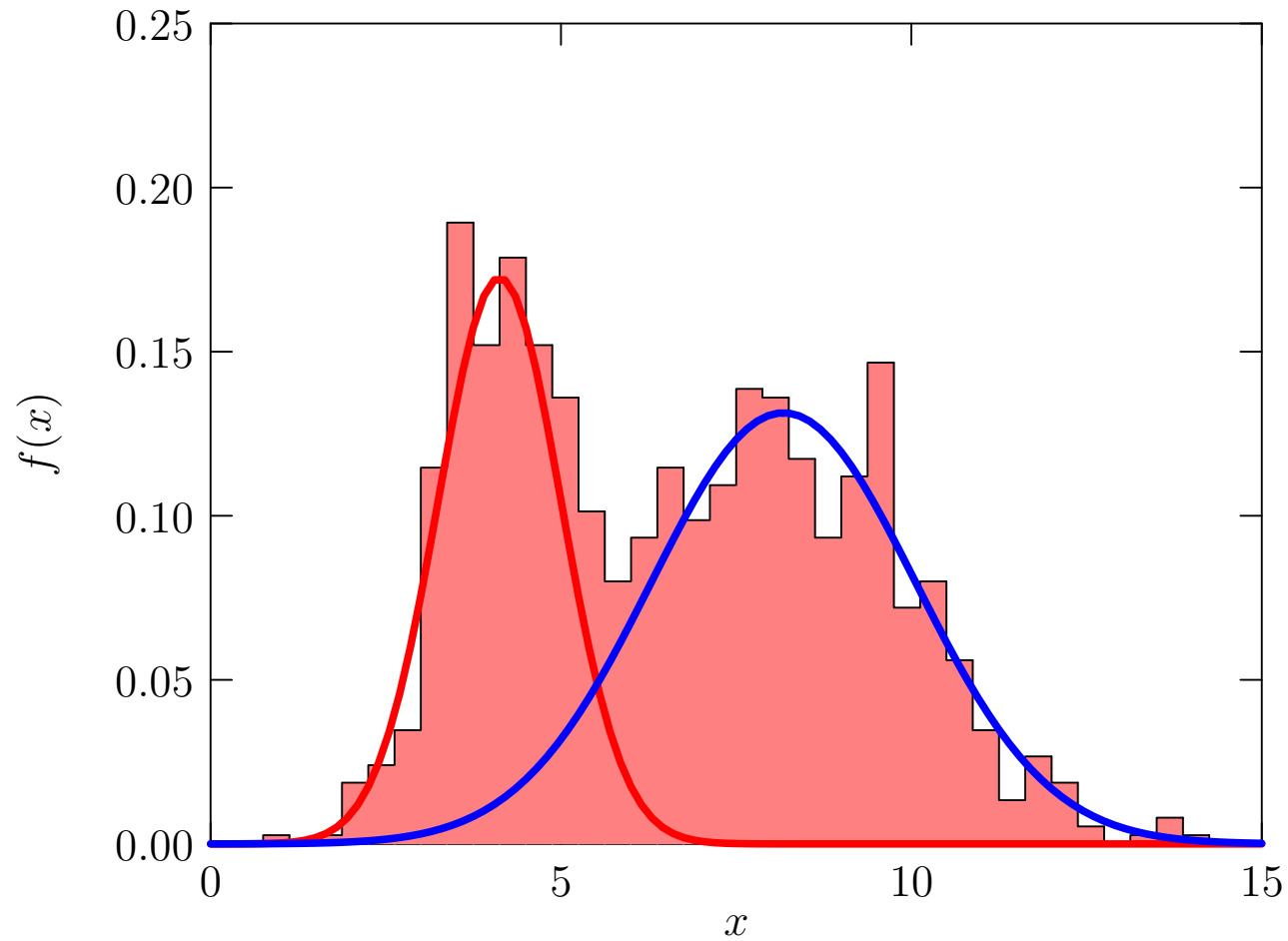
Example



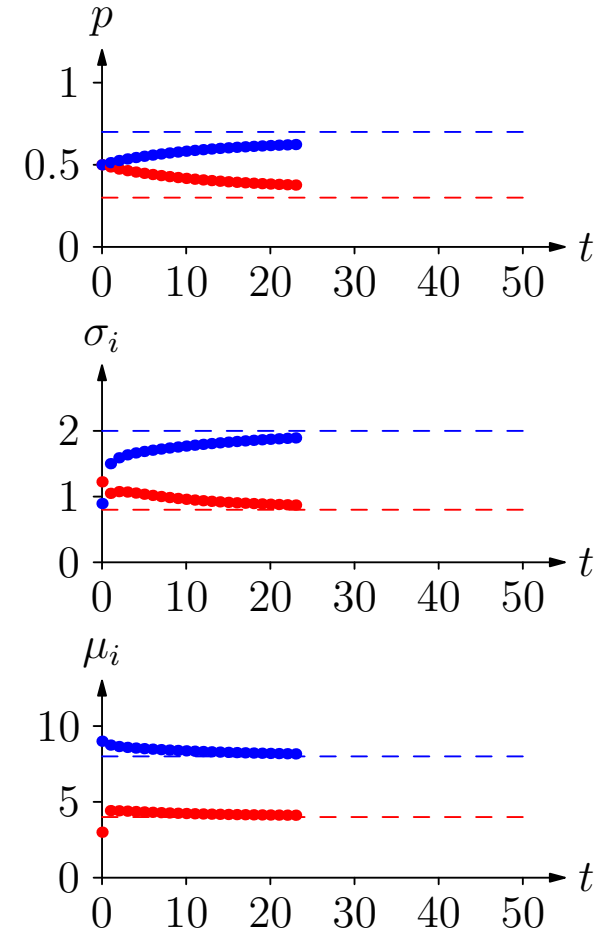
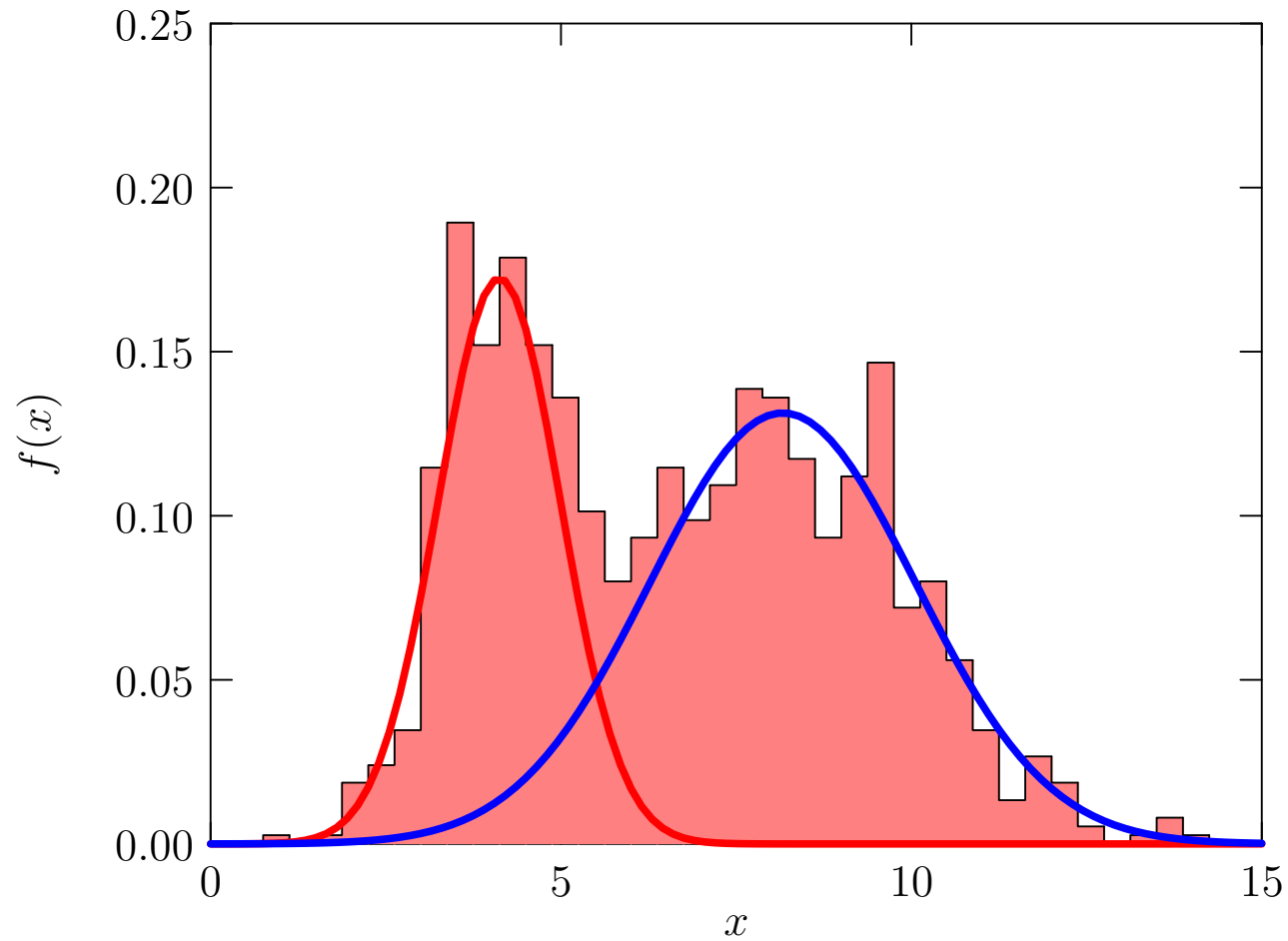
Example



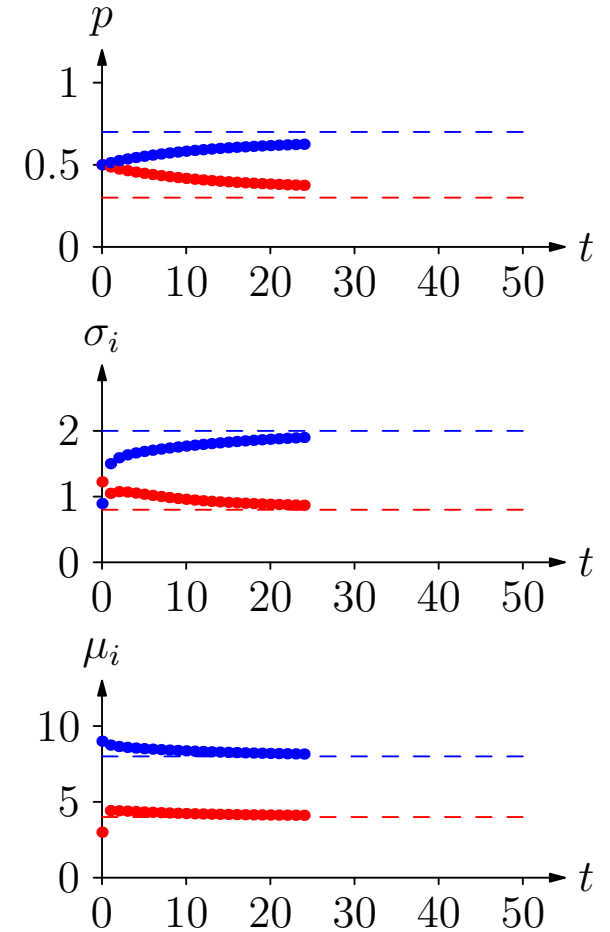
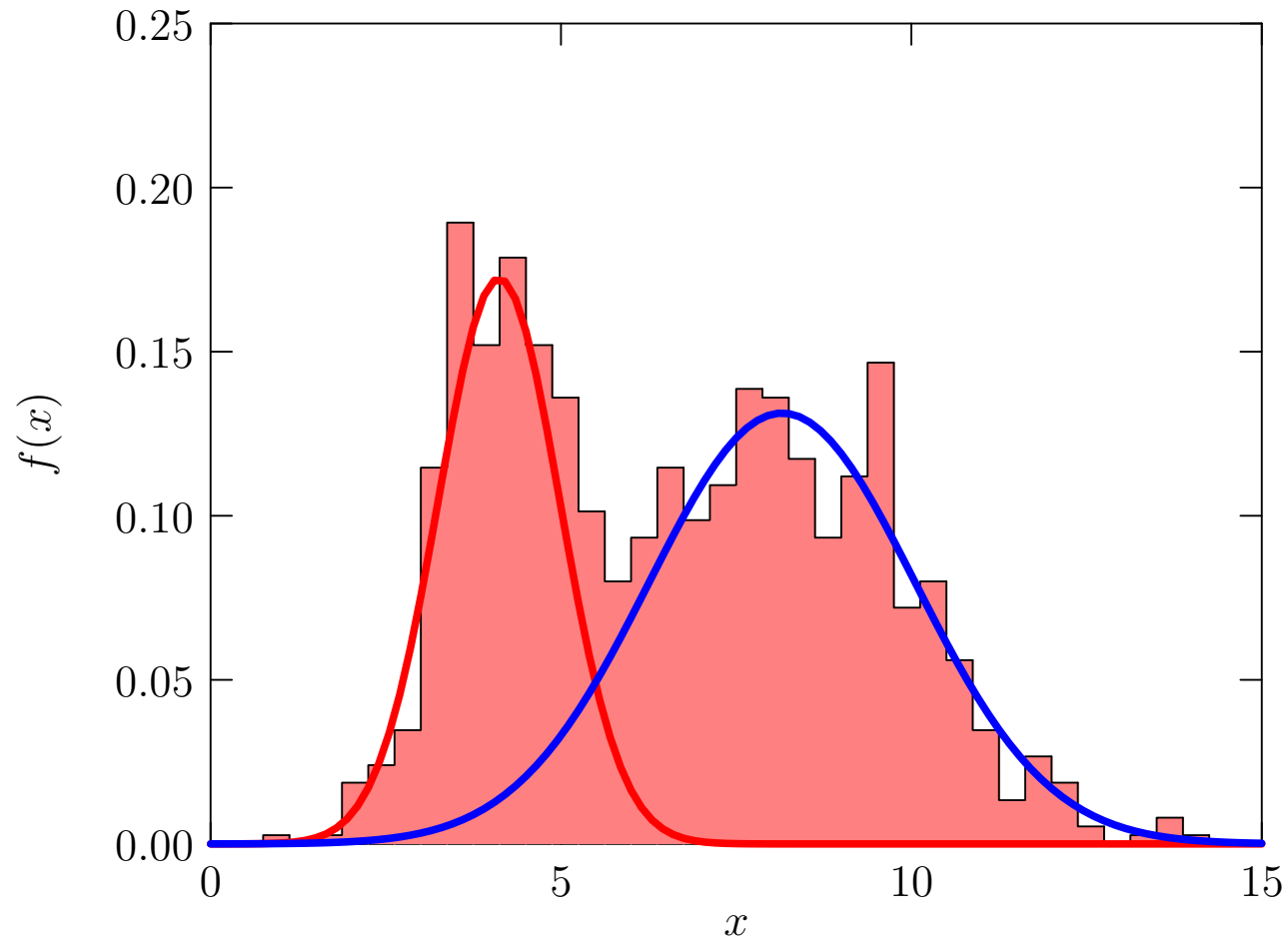
Example



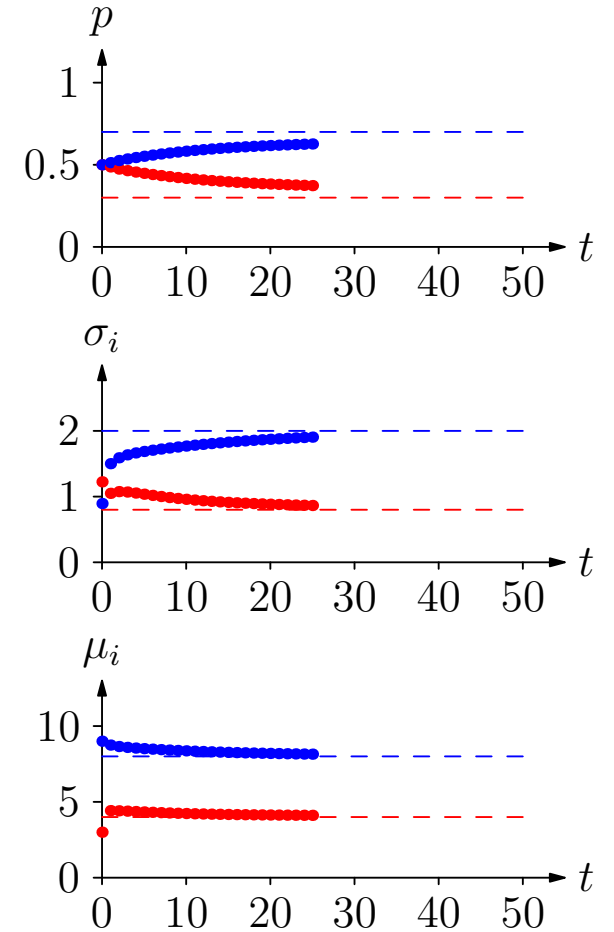
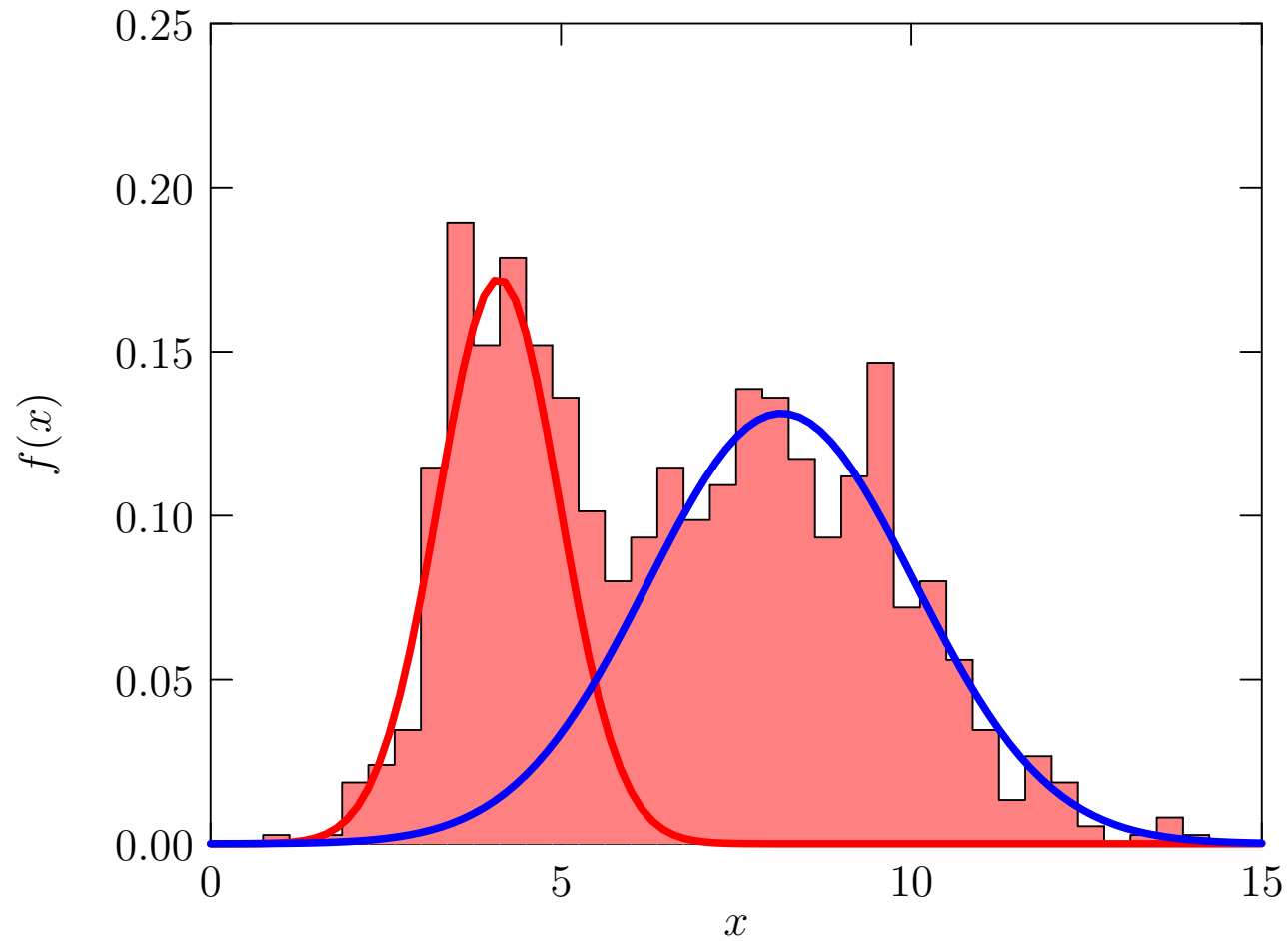
Example



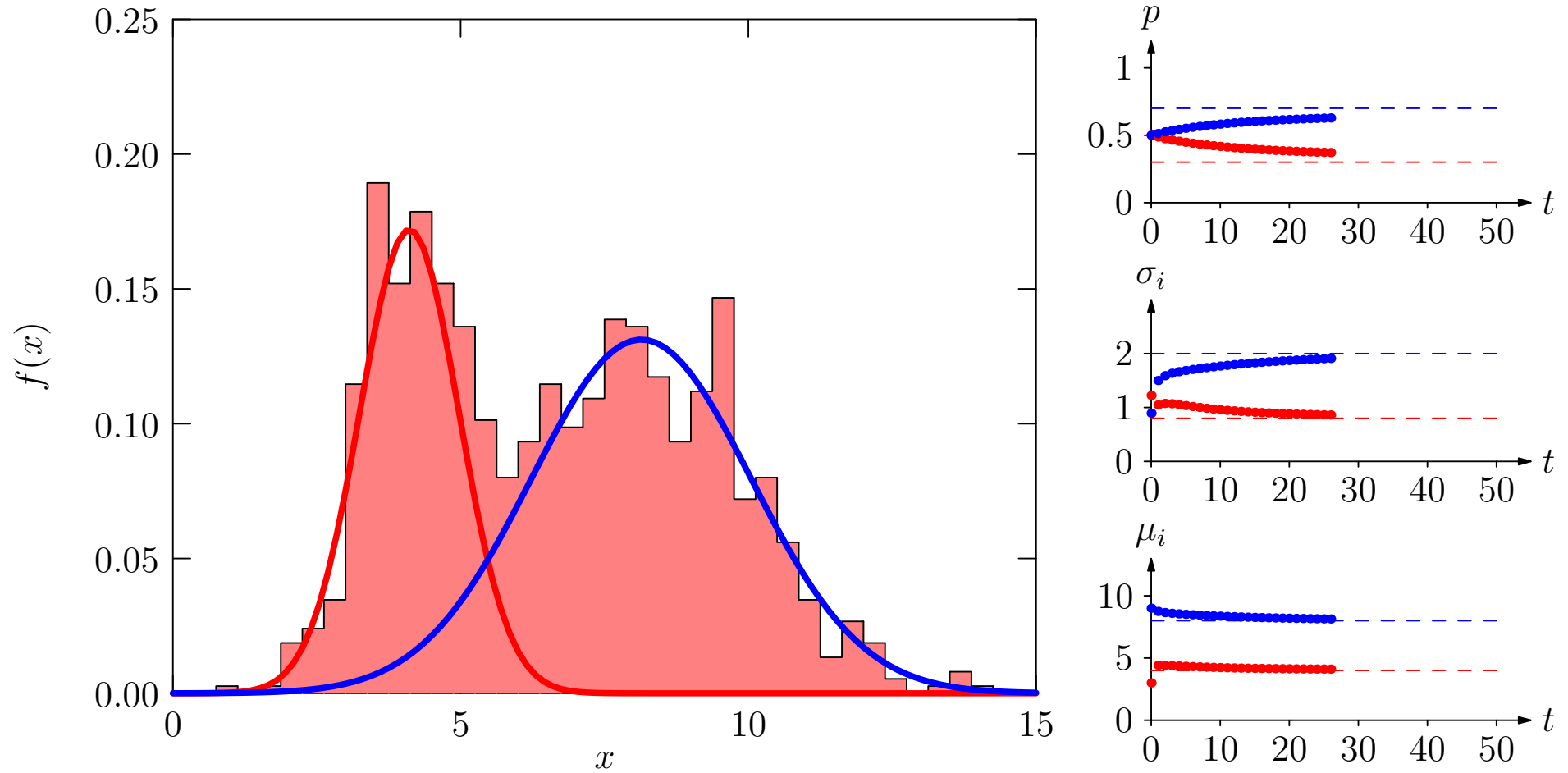
Example



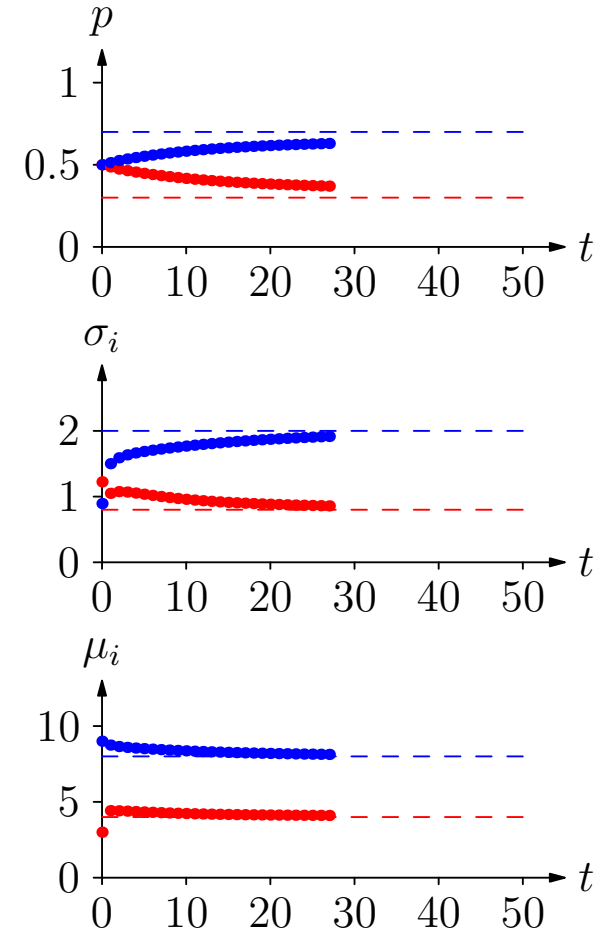
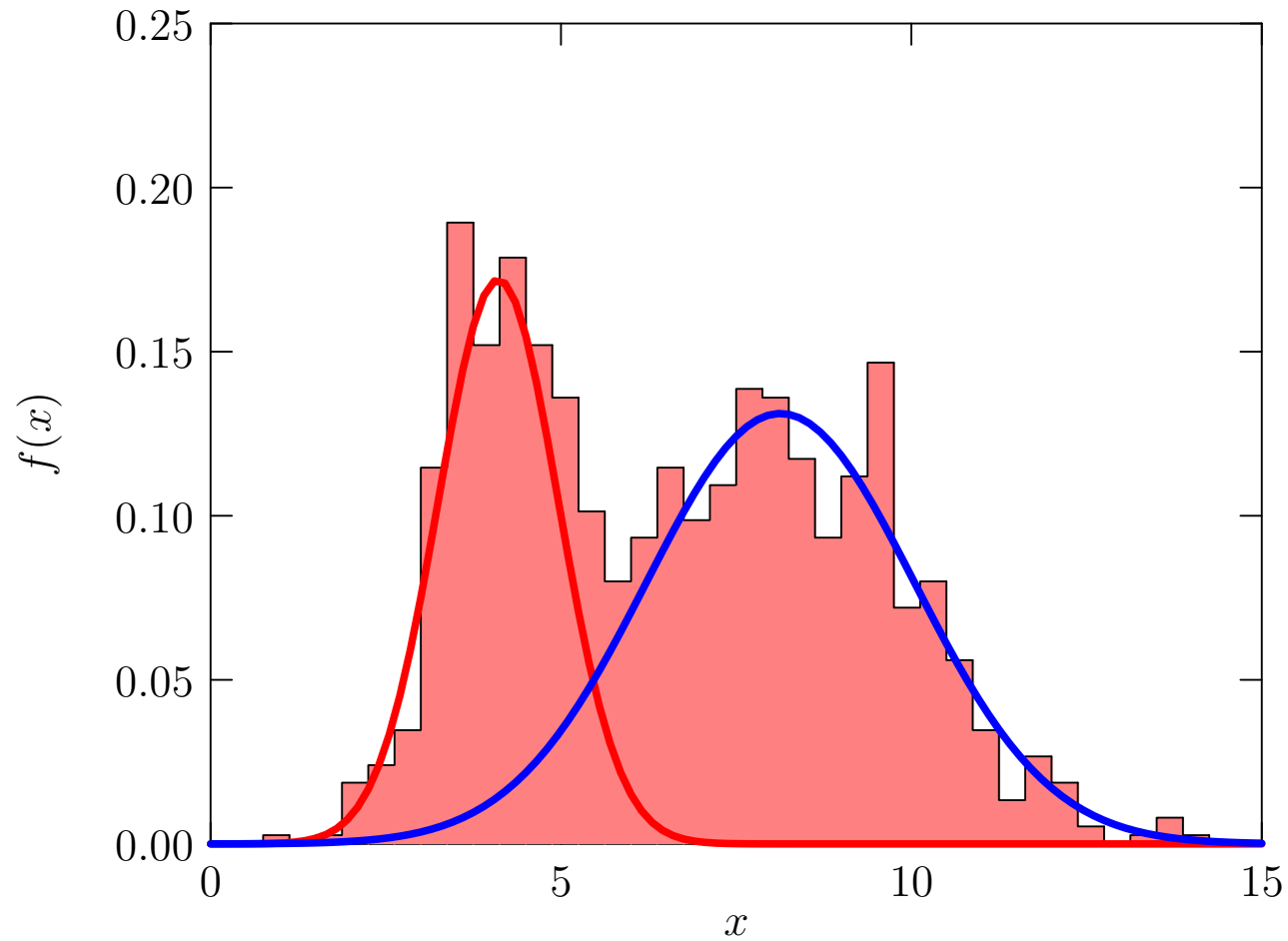
Example



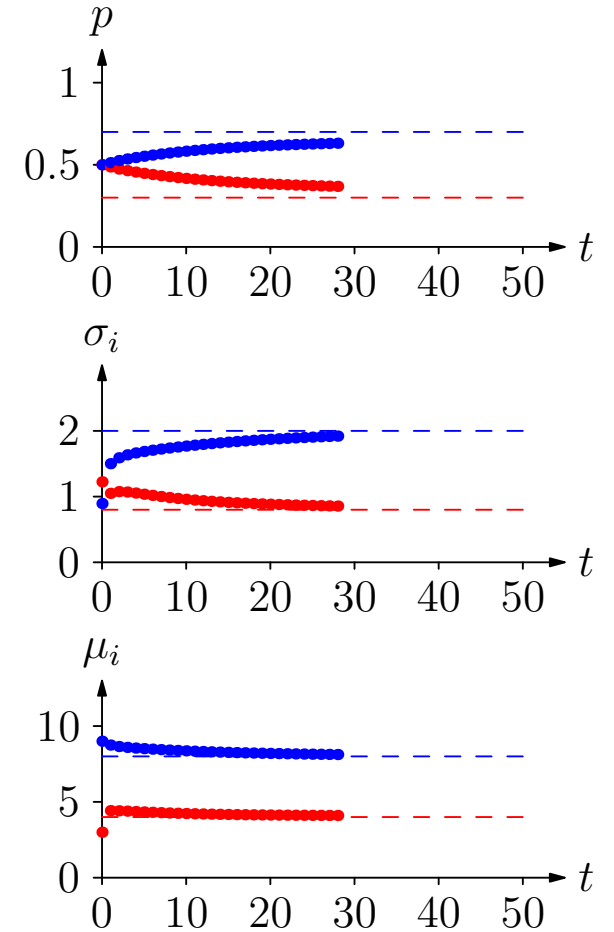
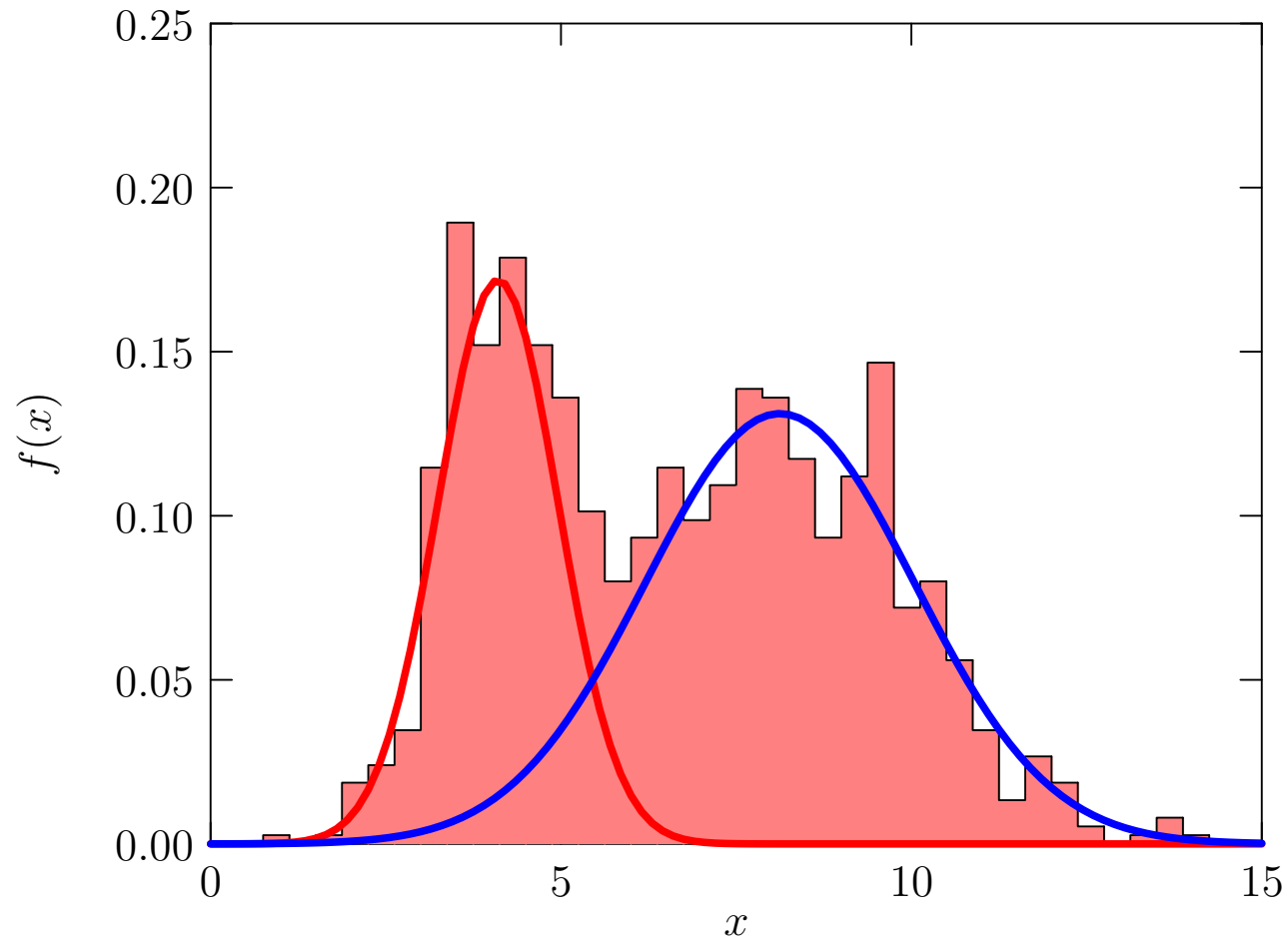
Example



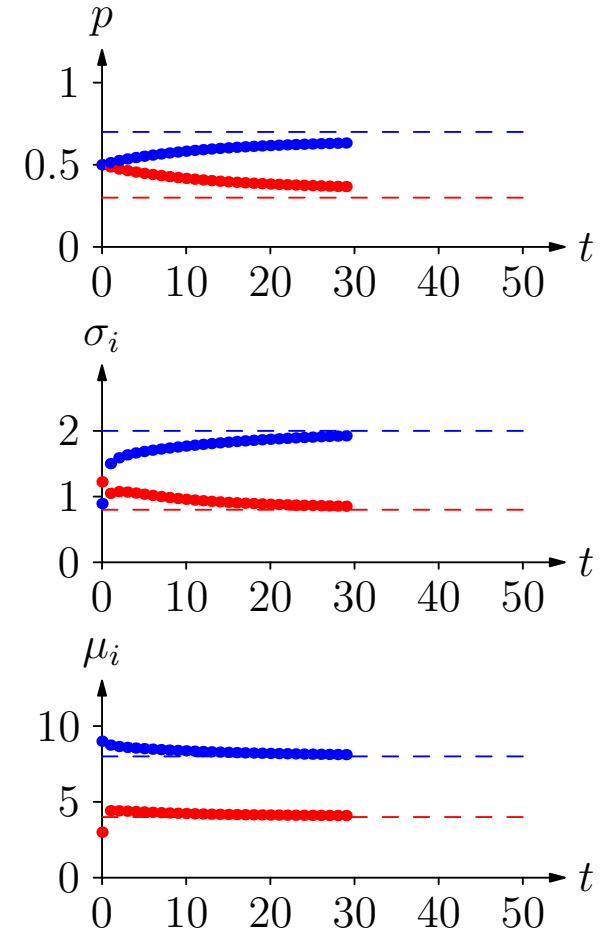
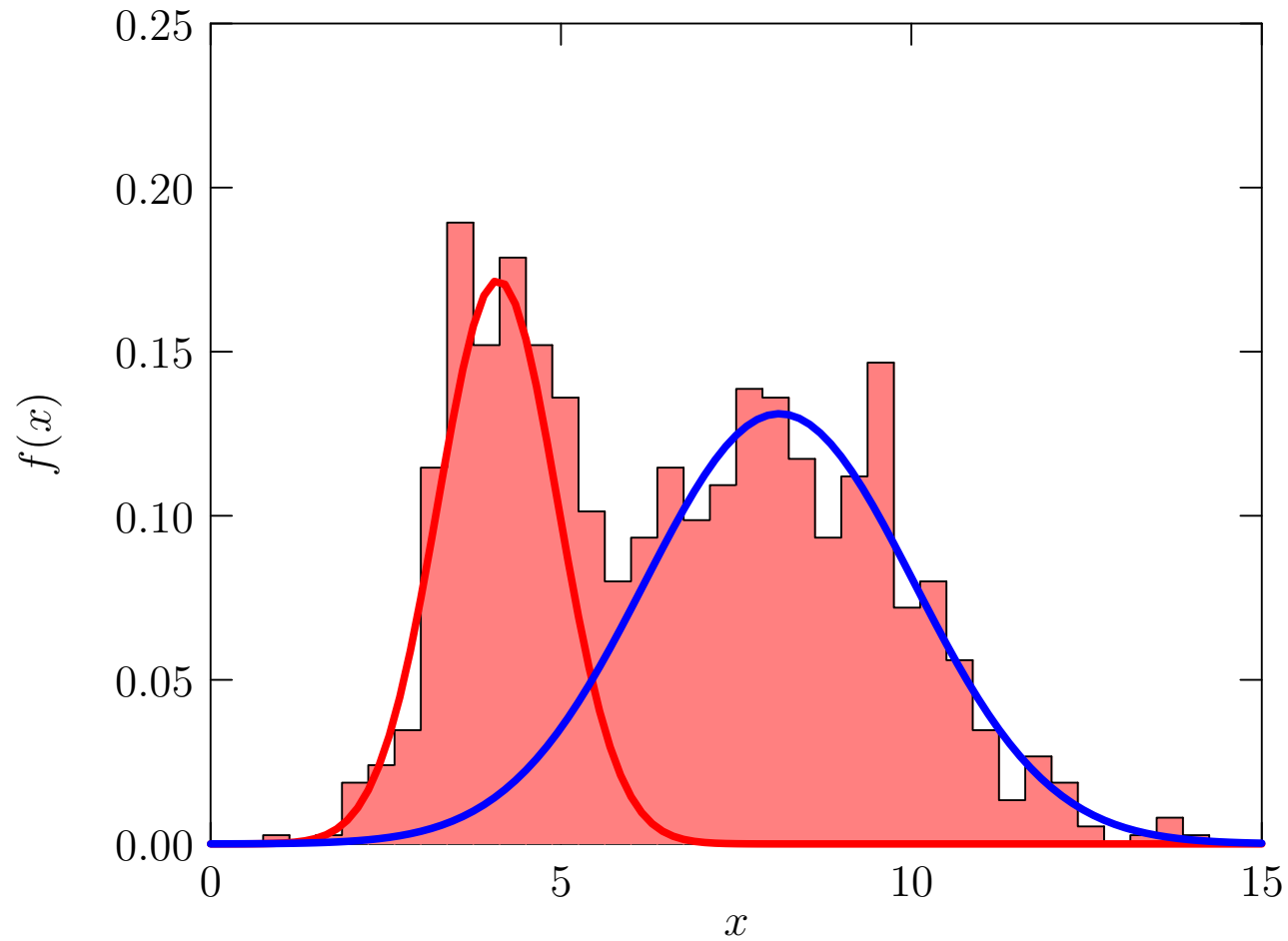
Example



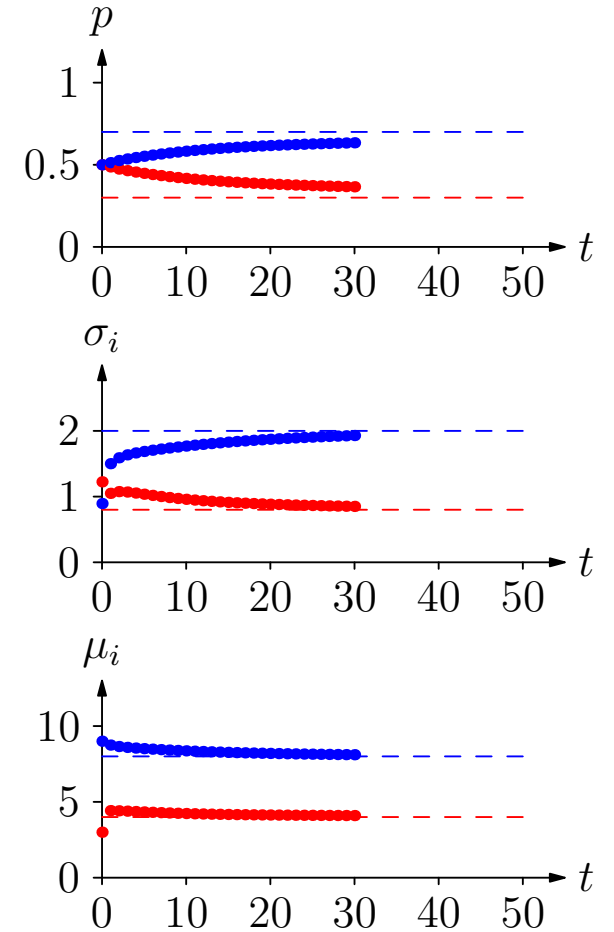
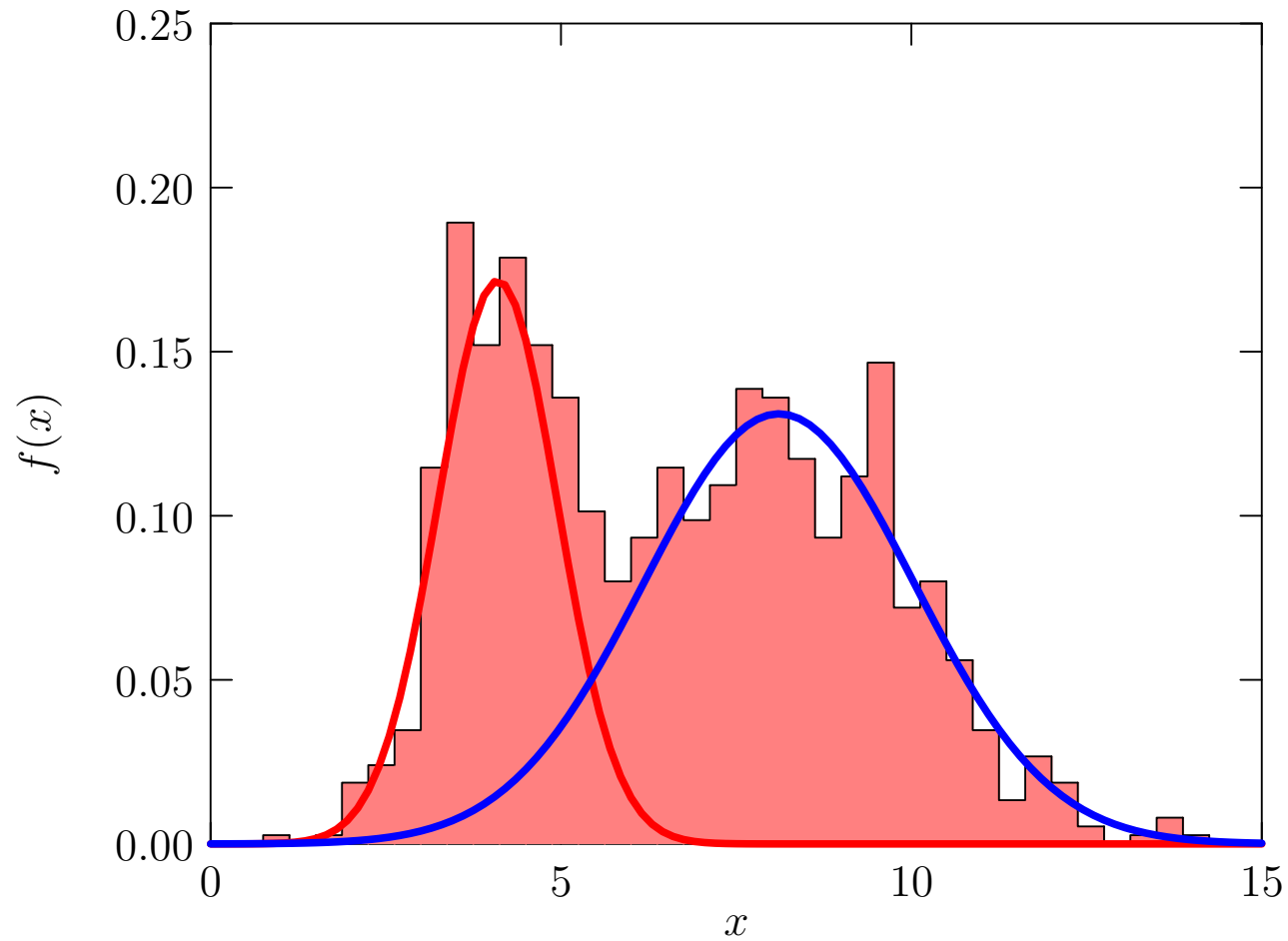
Example



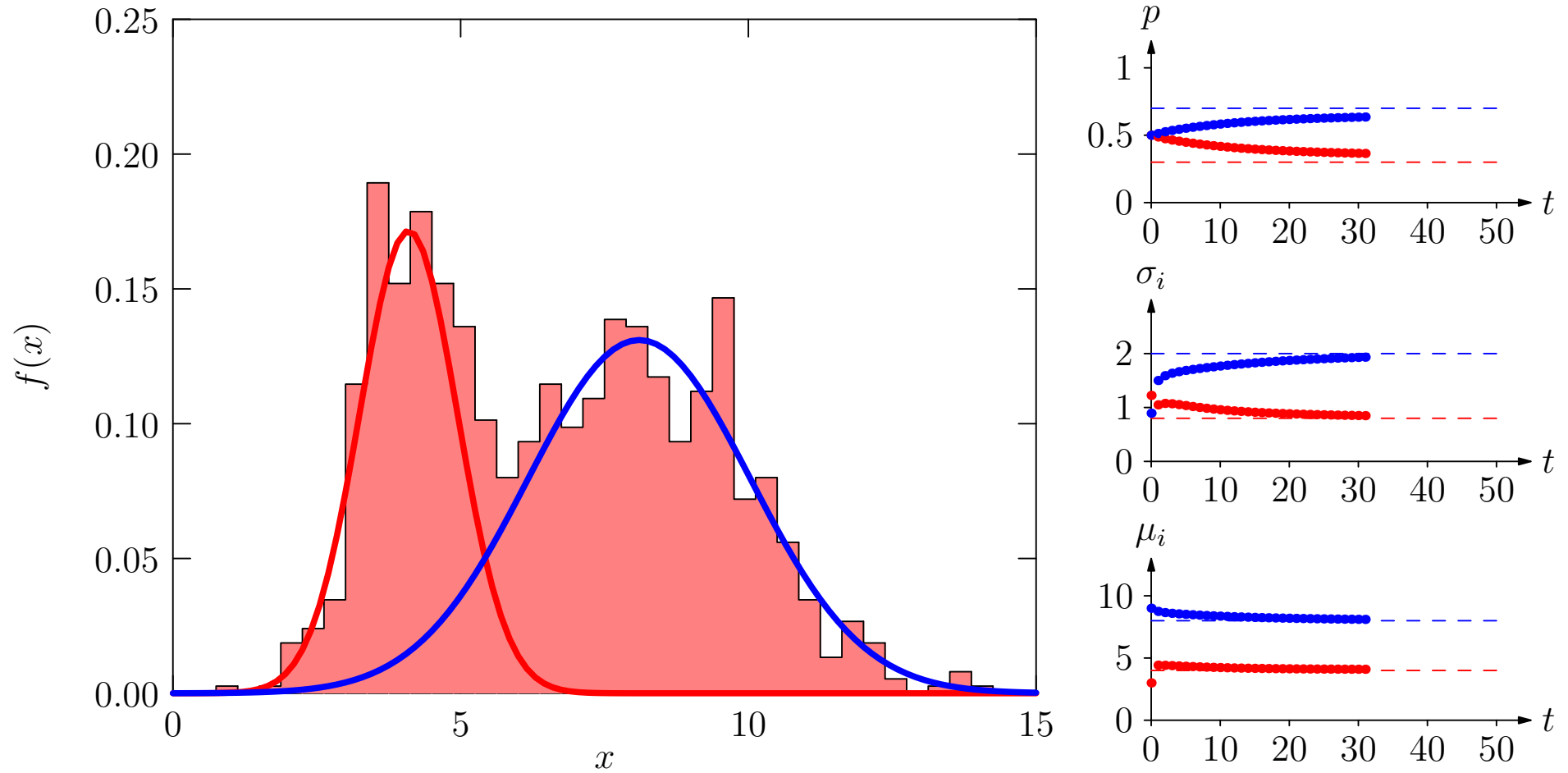
Example



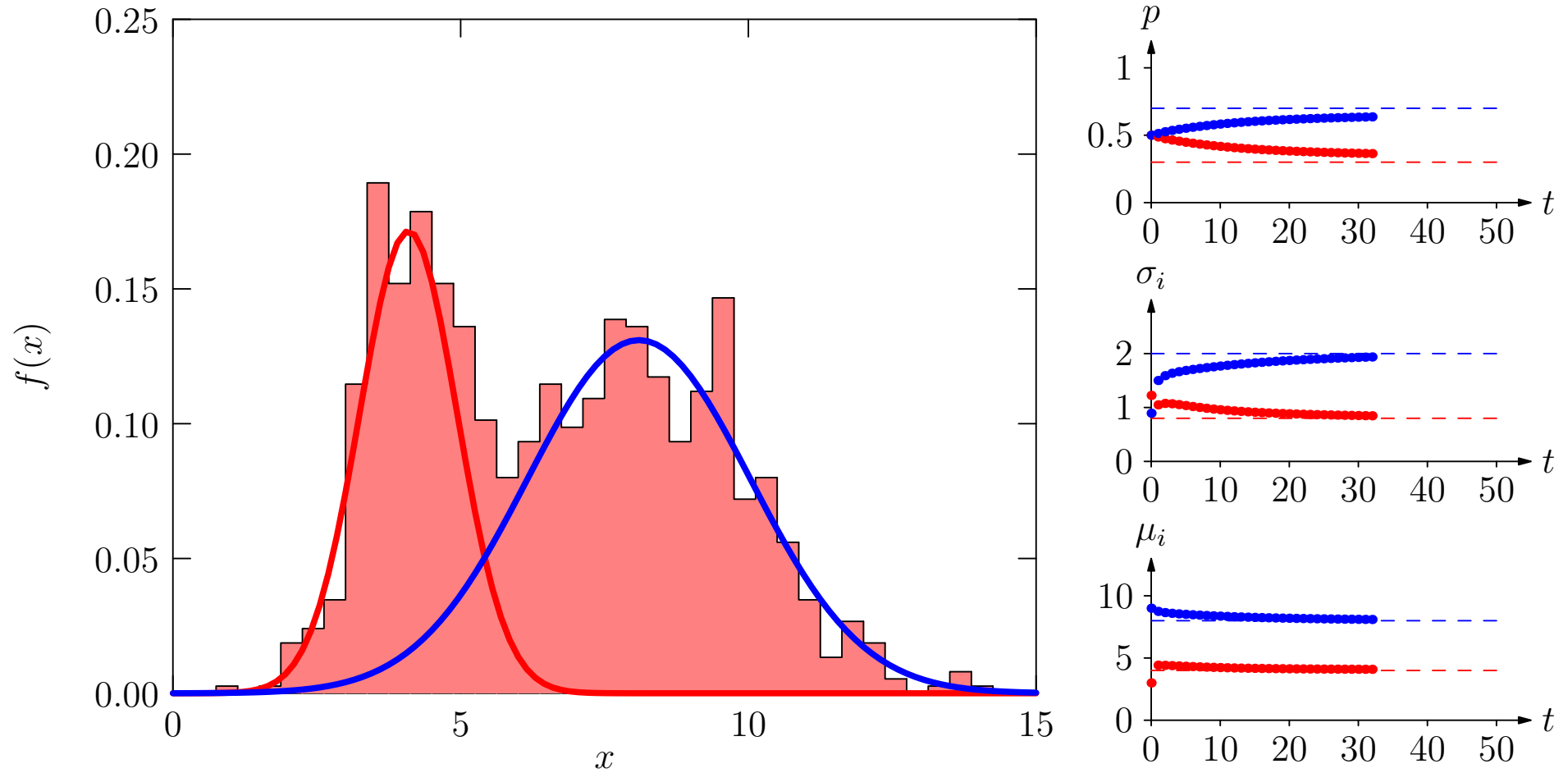
Example



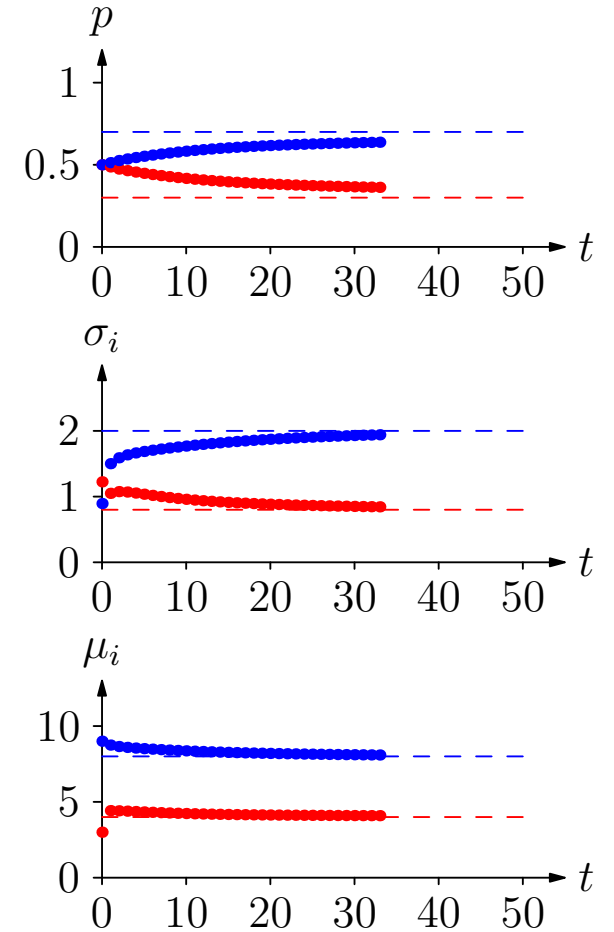
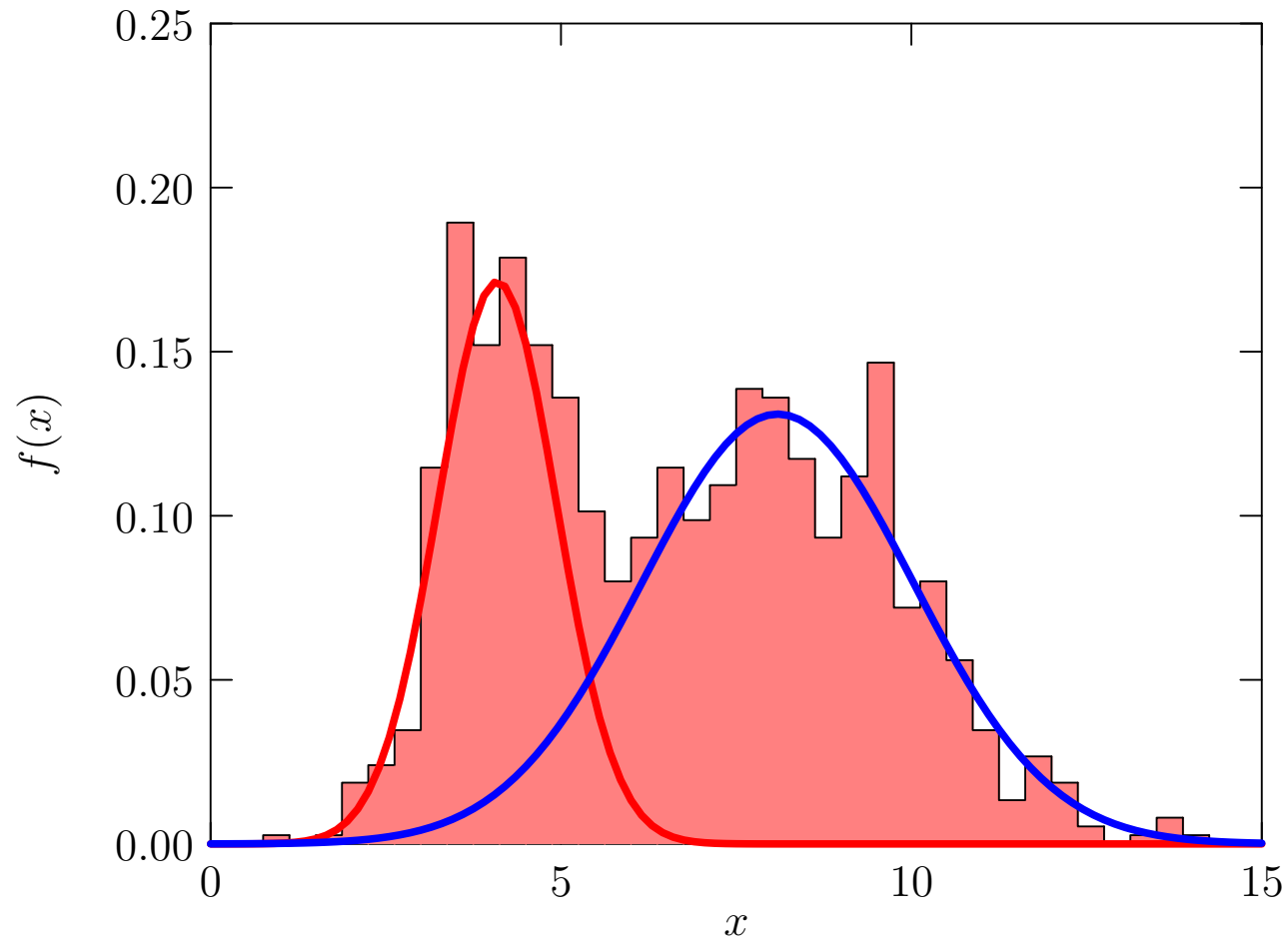
Example



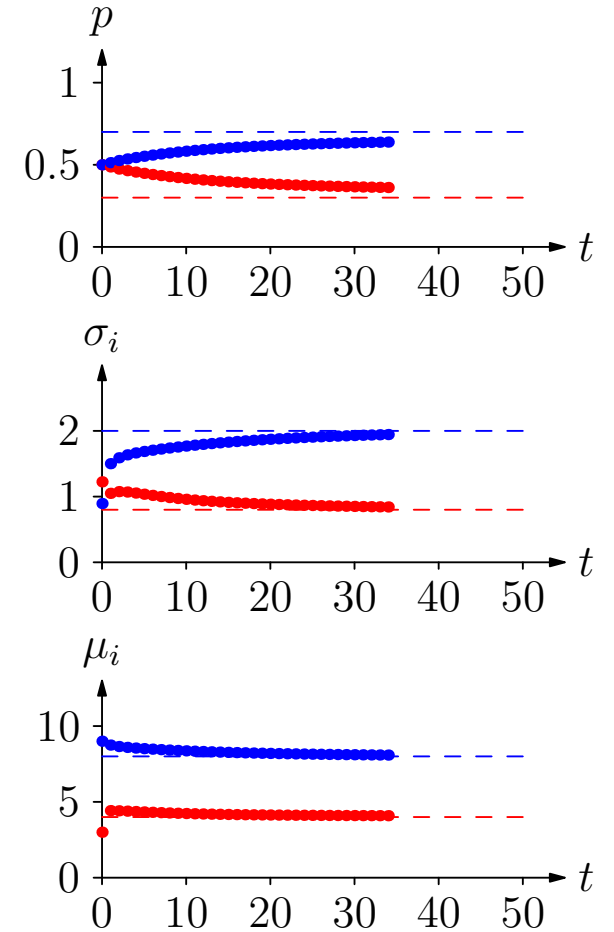
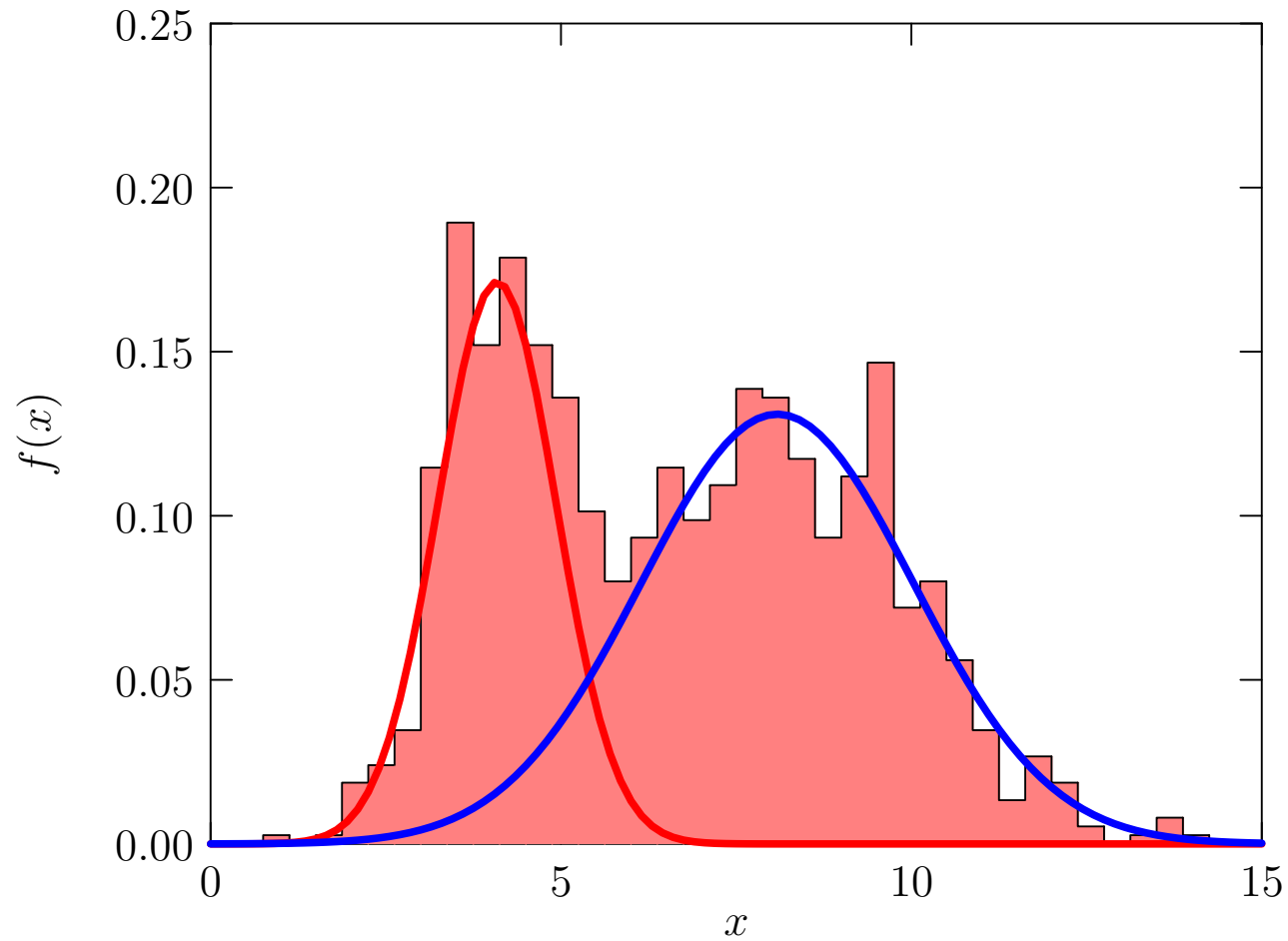
Example



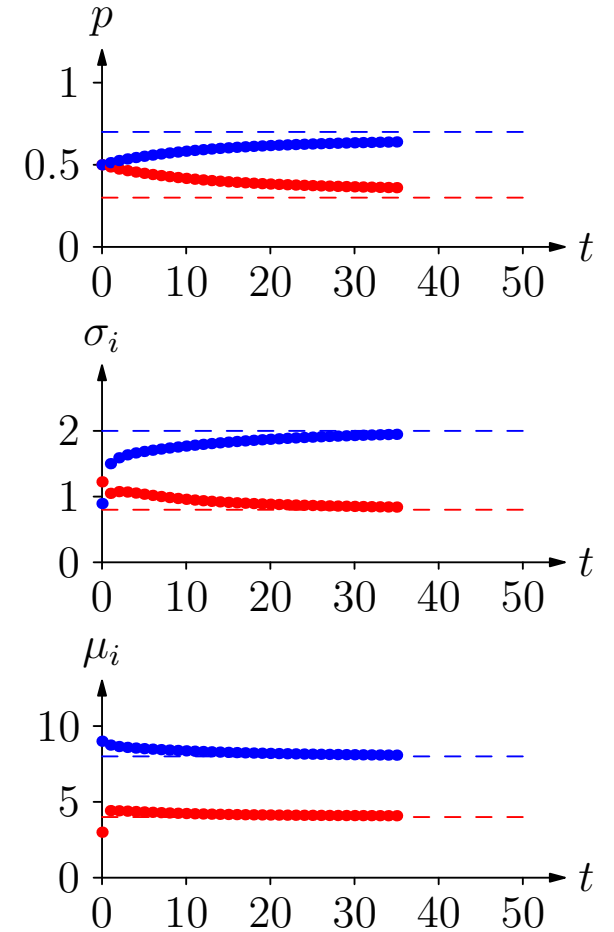
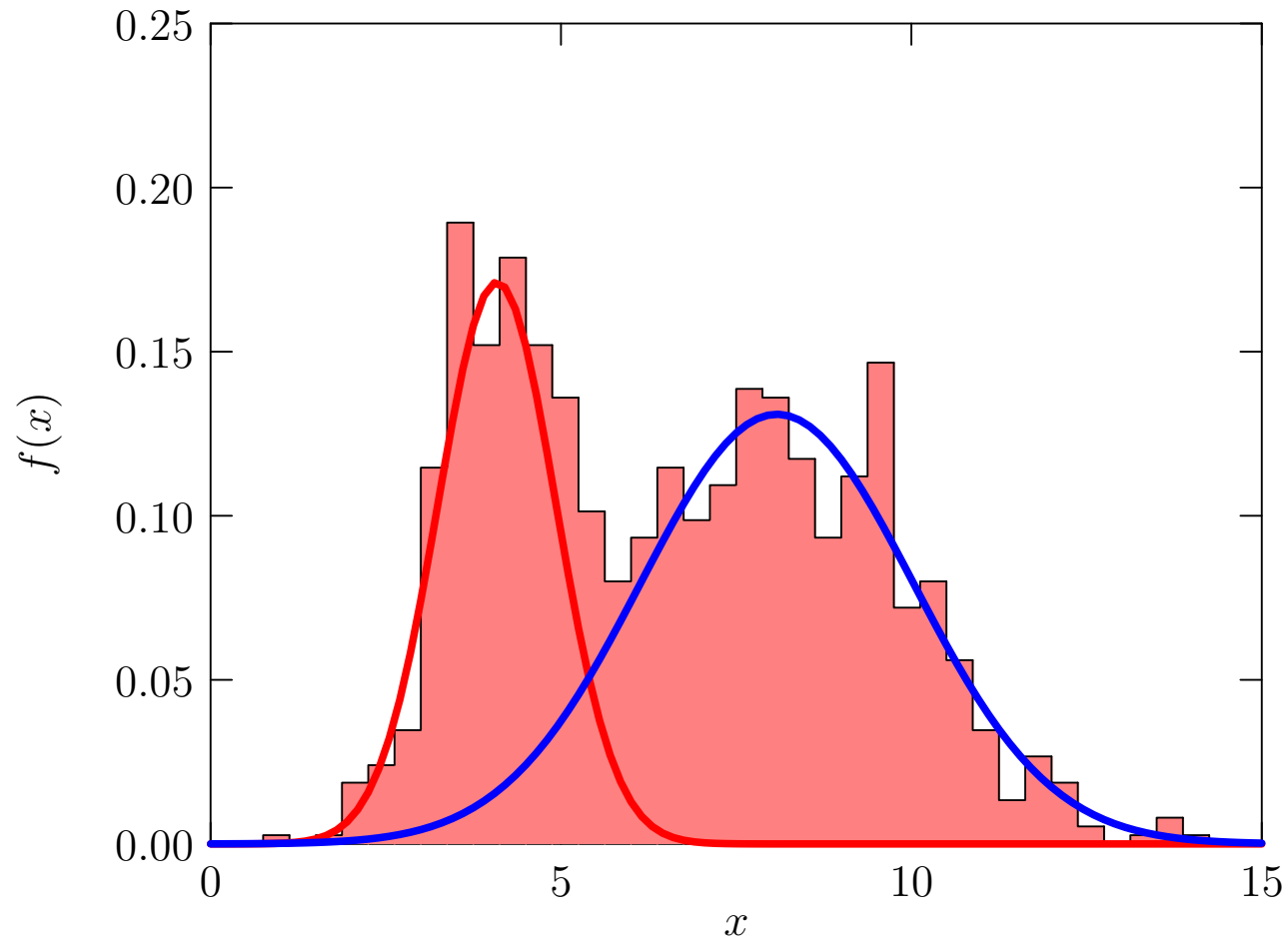
Example



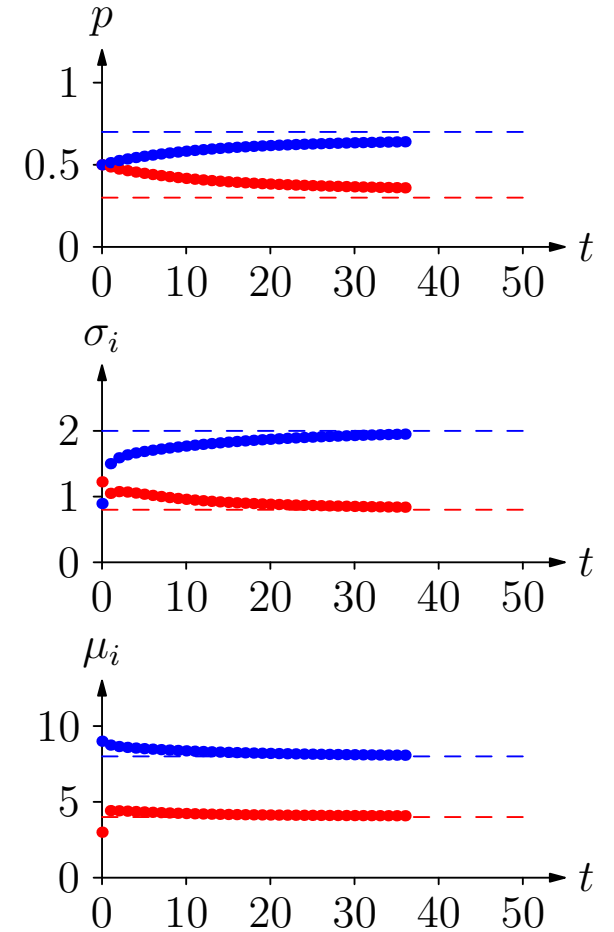
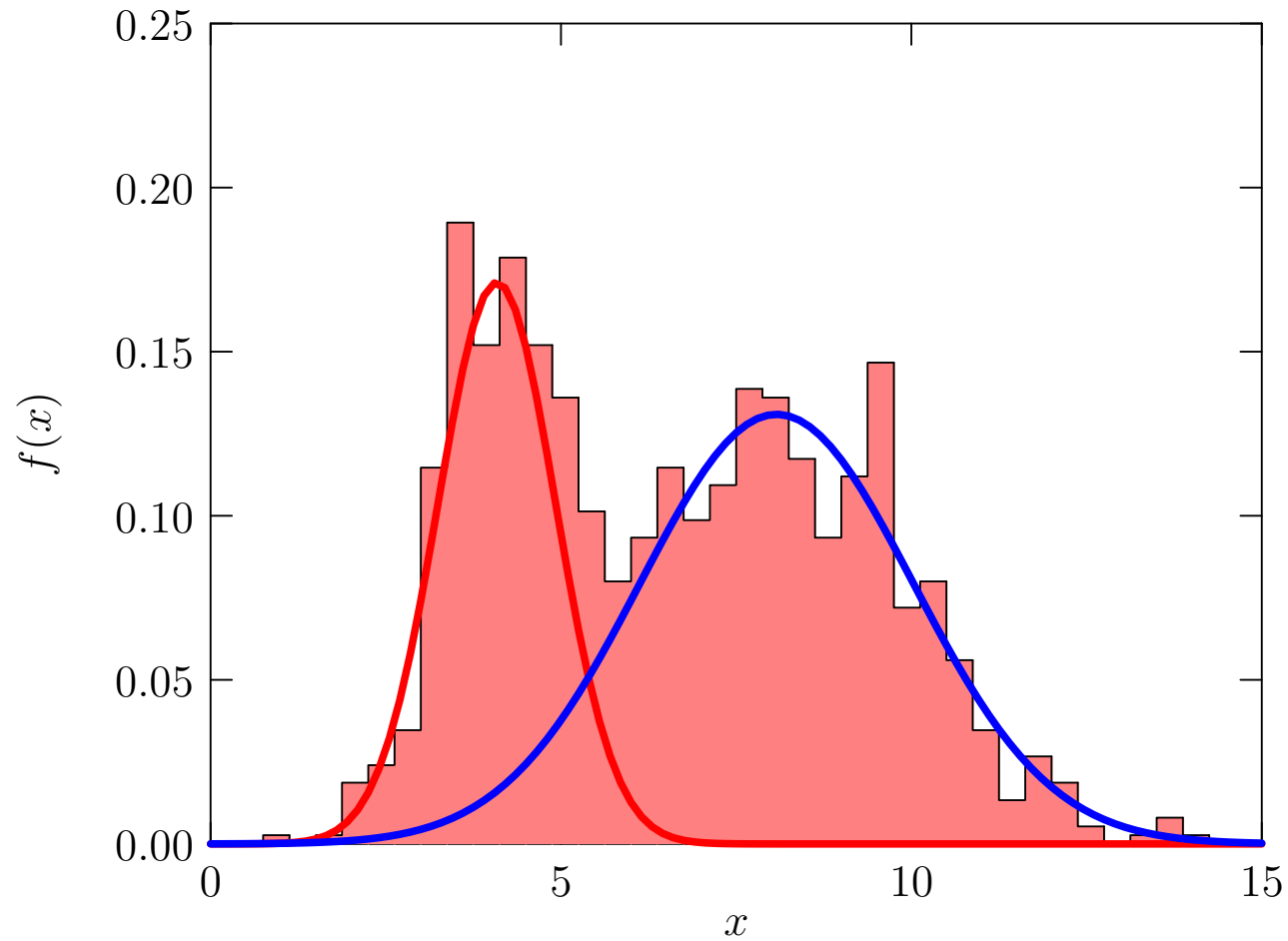
Example



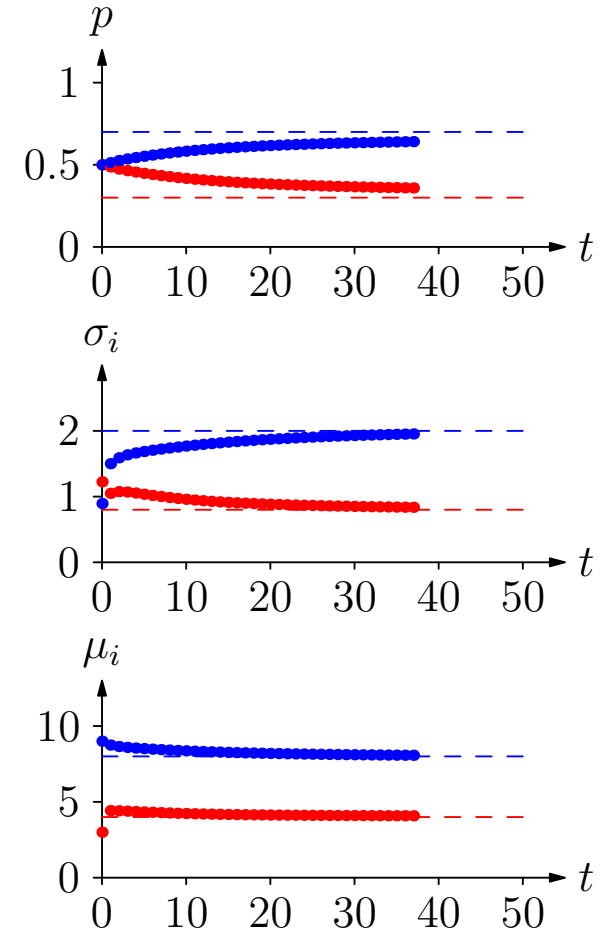
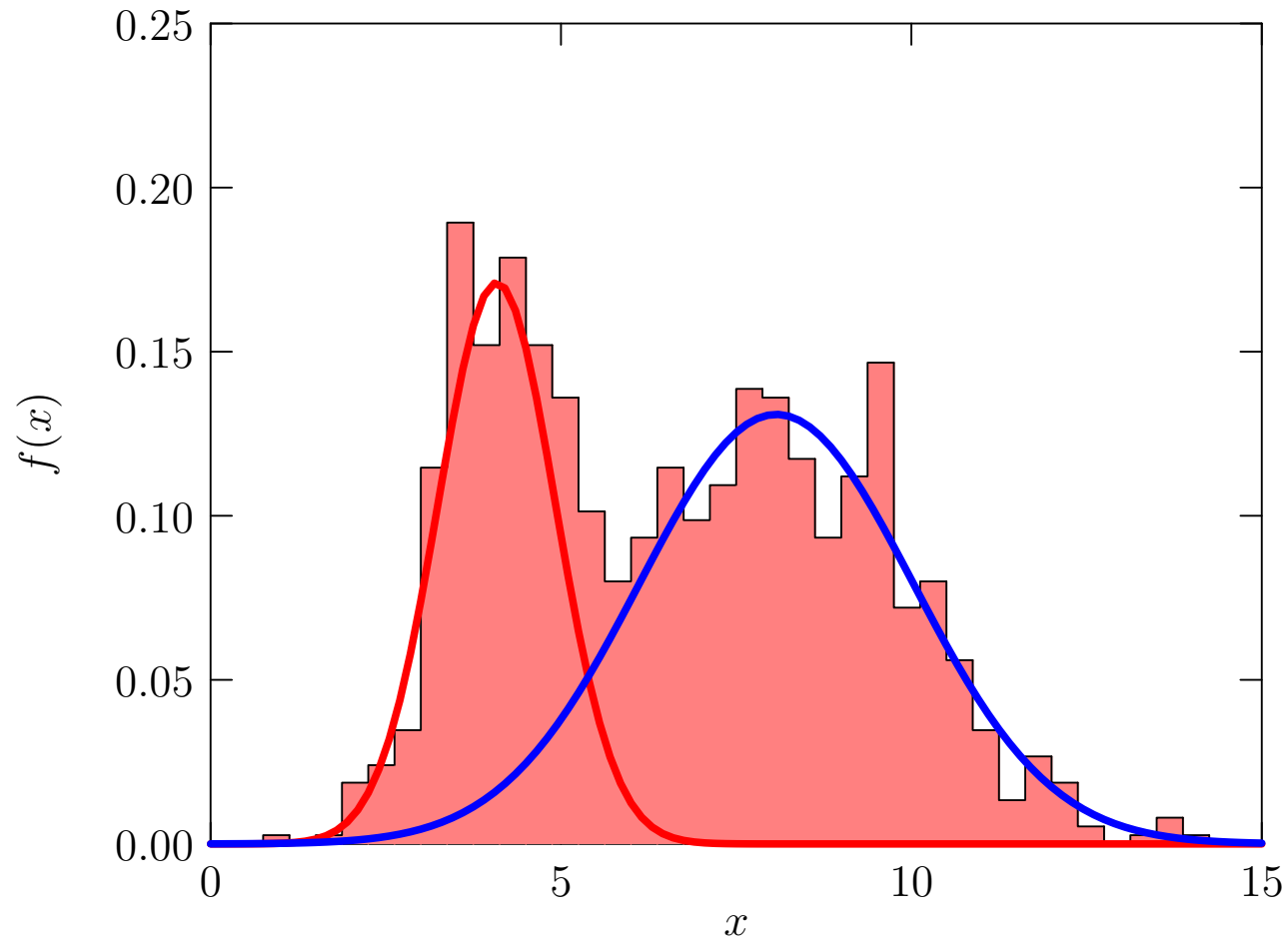
Example



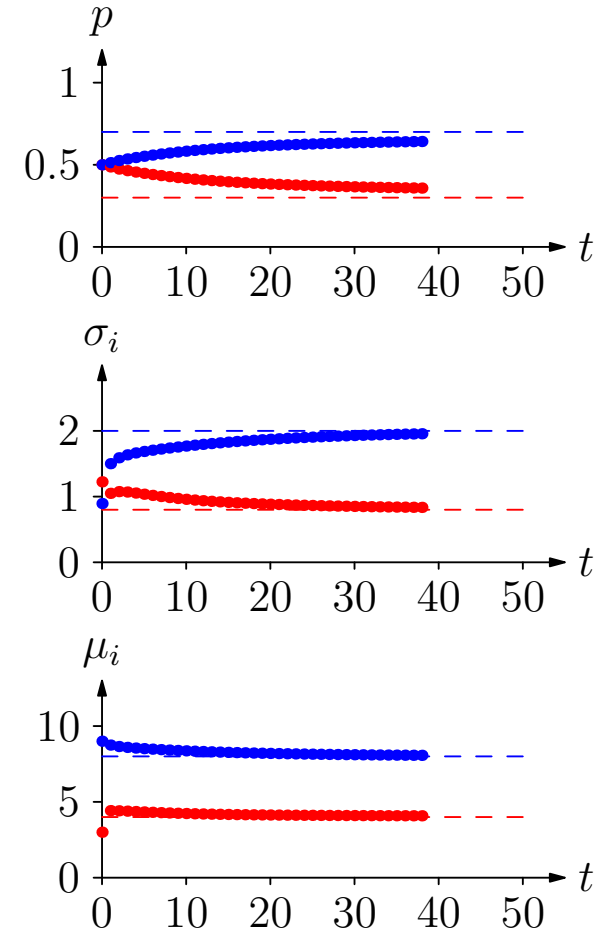
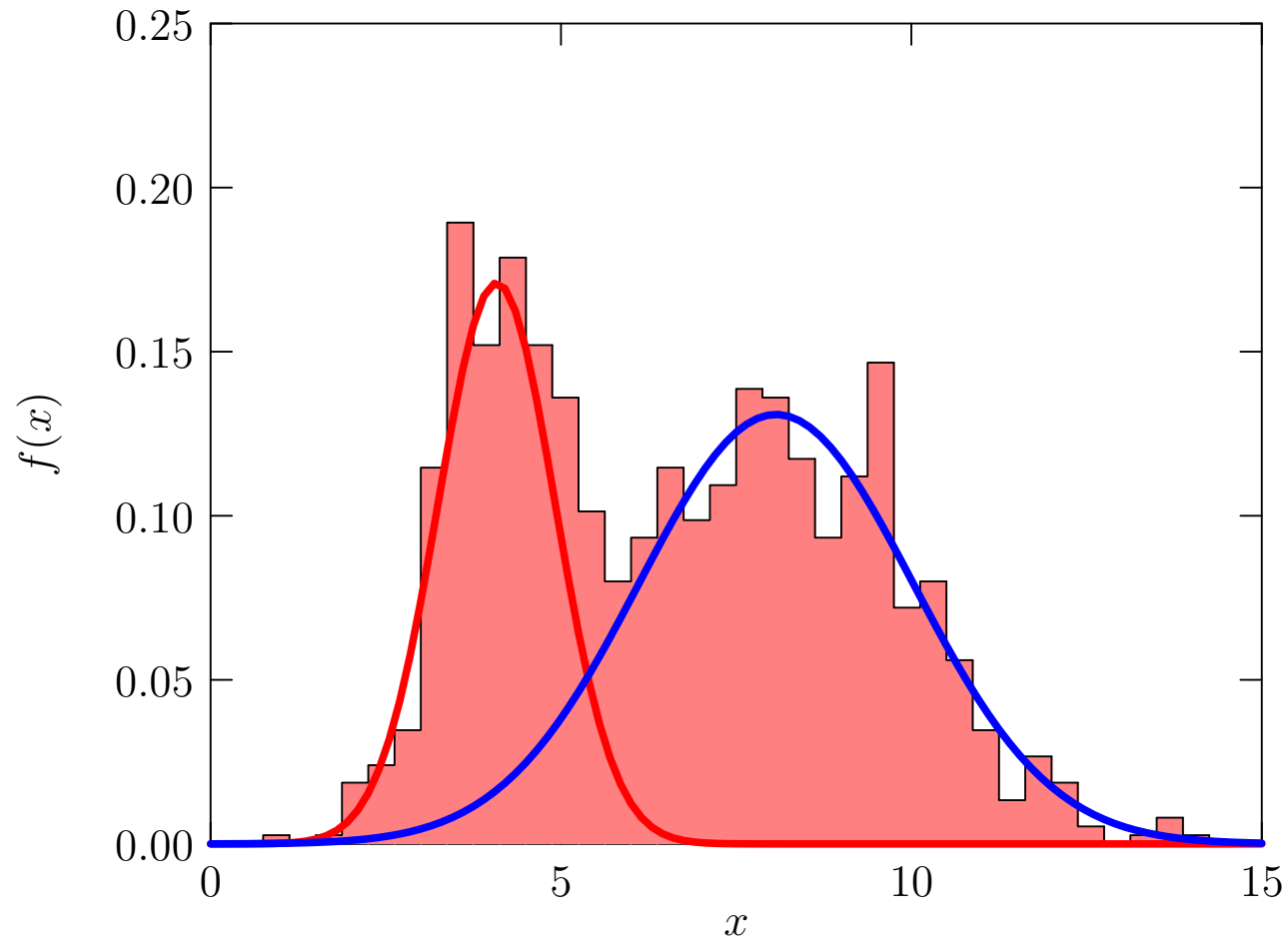
Example



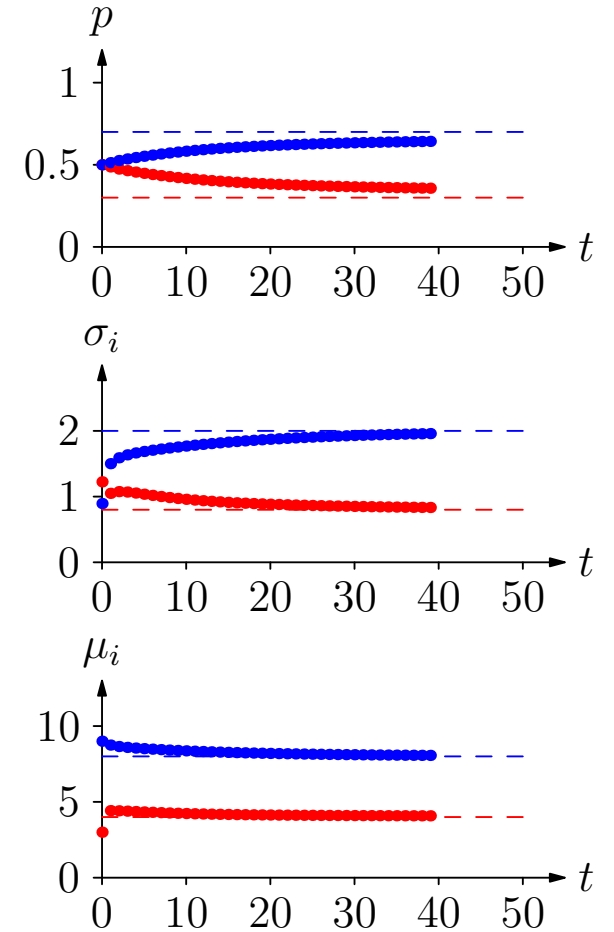
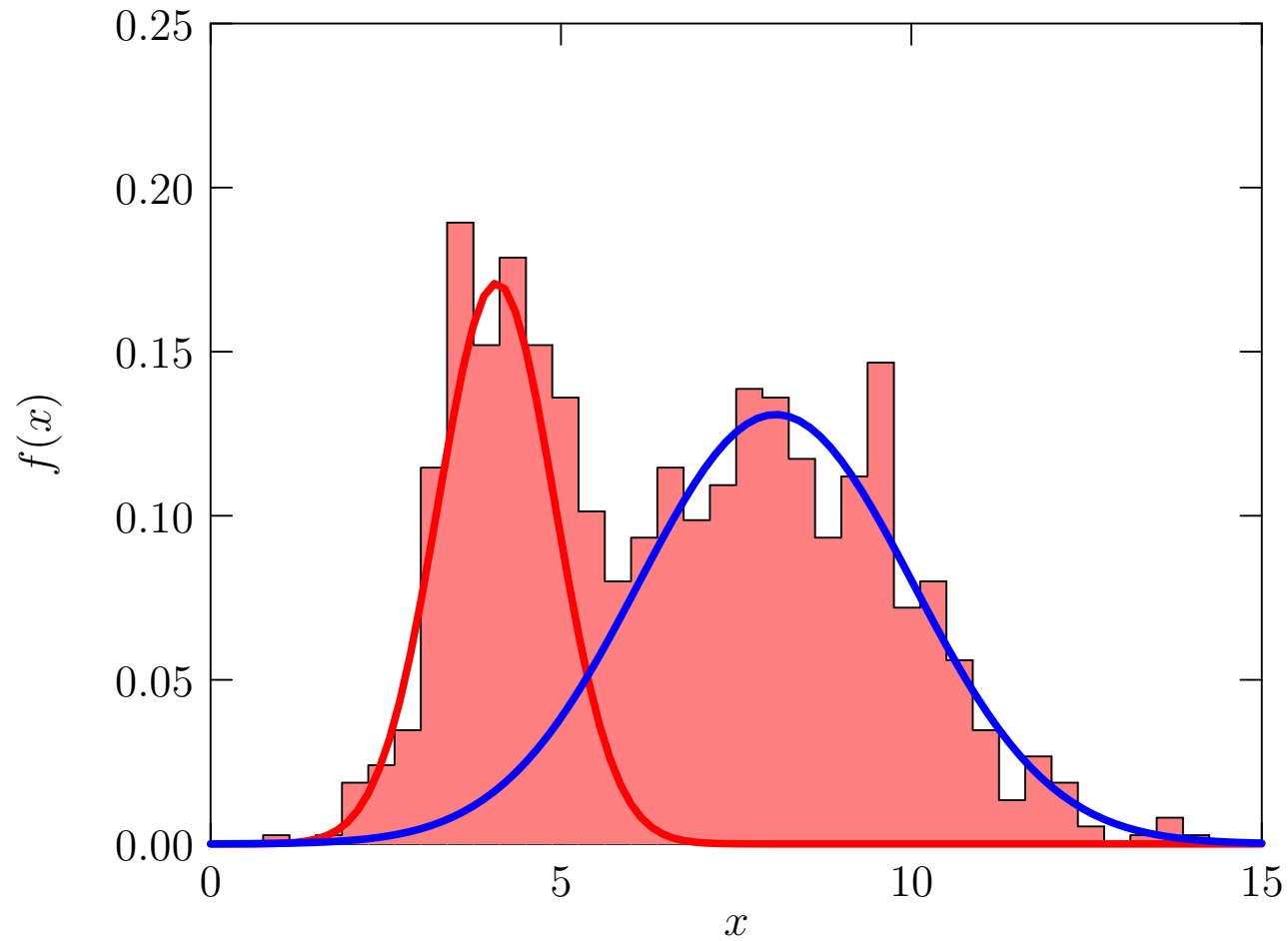
Example



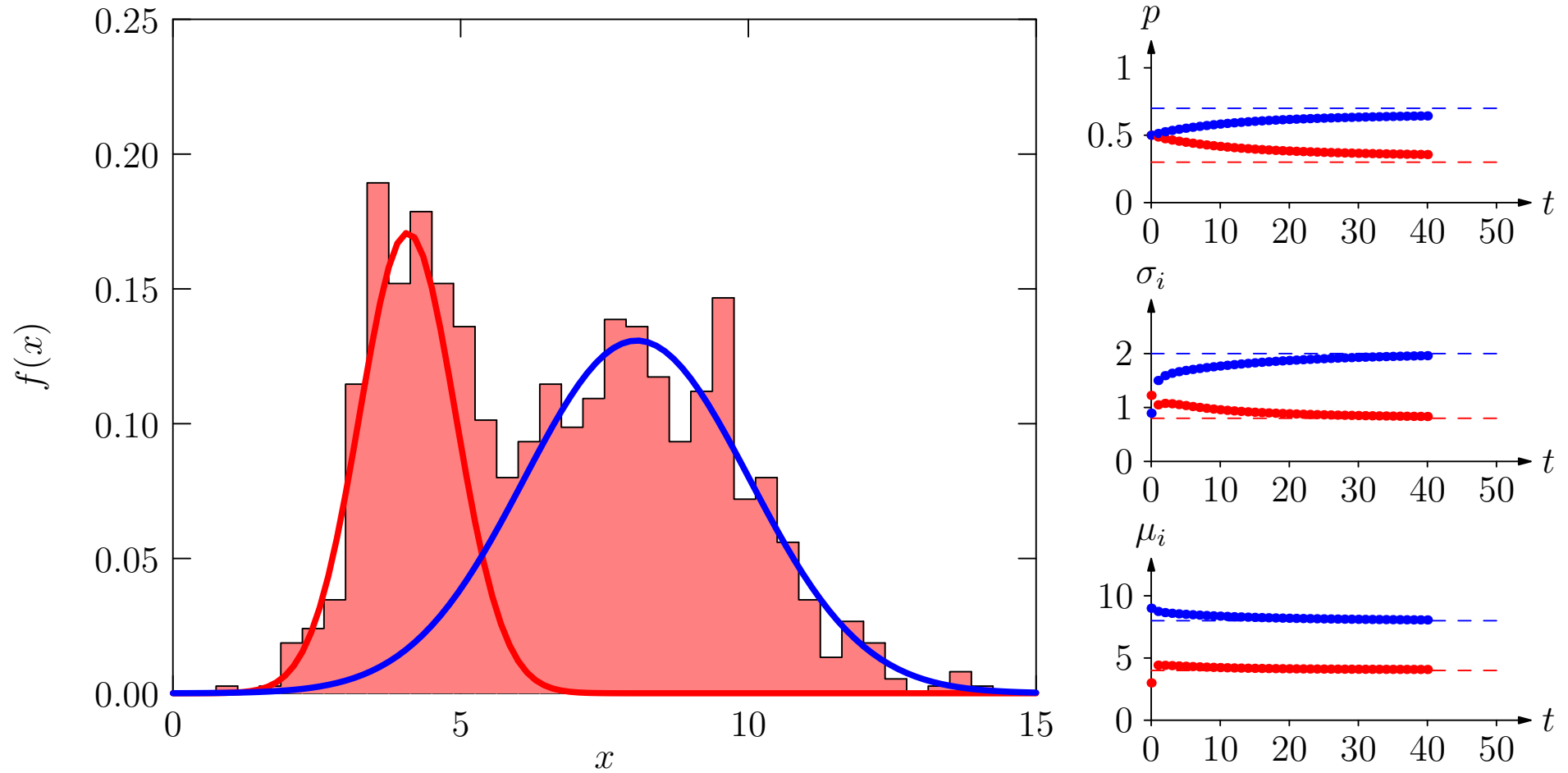
Example



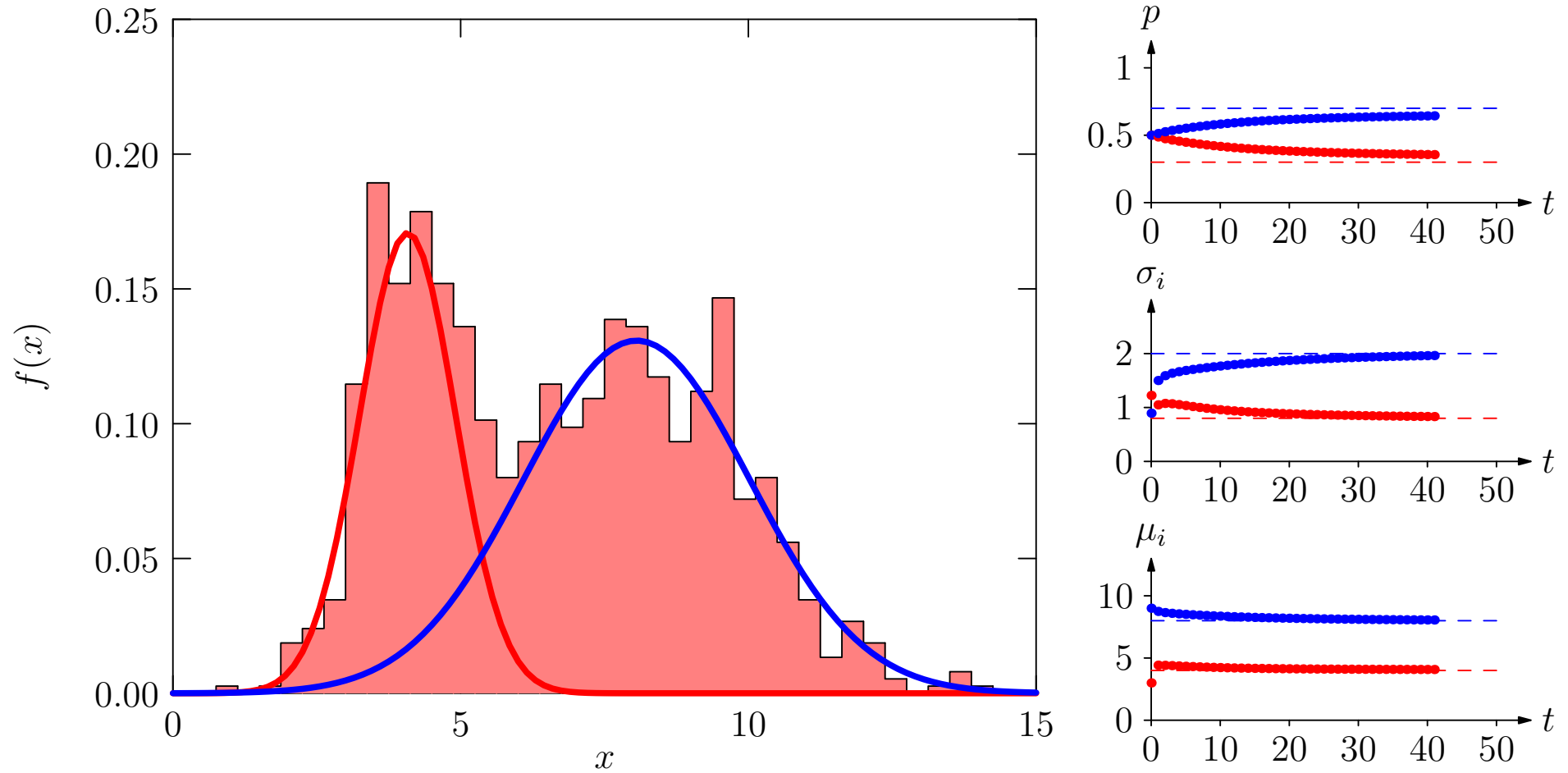
Example



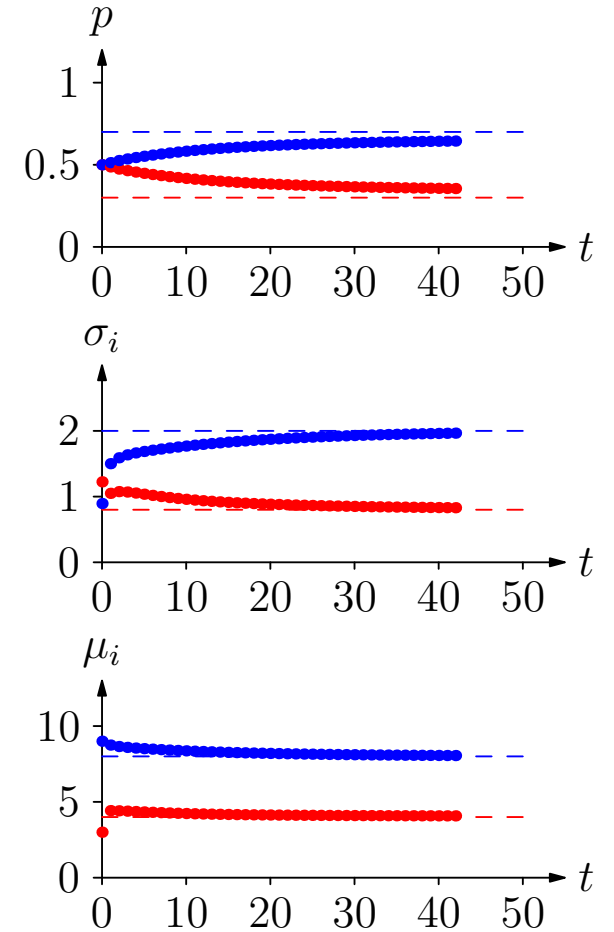
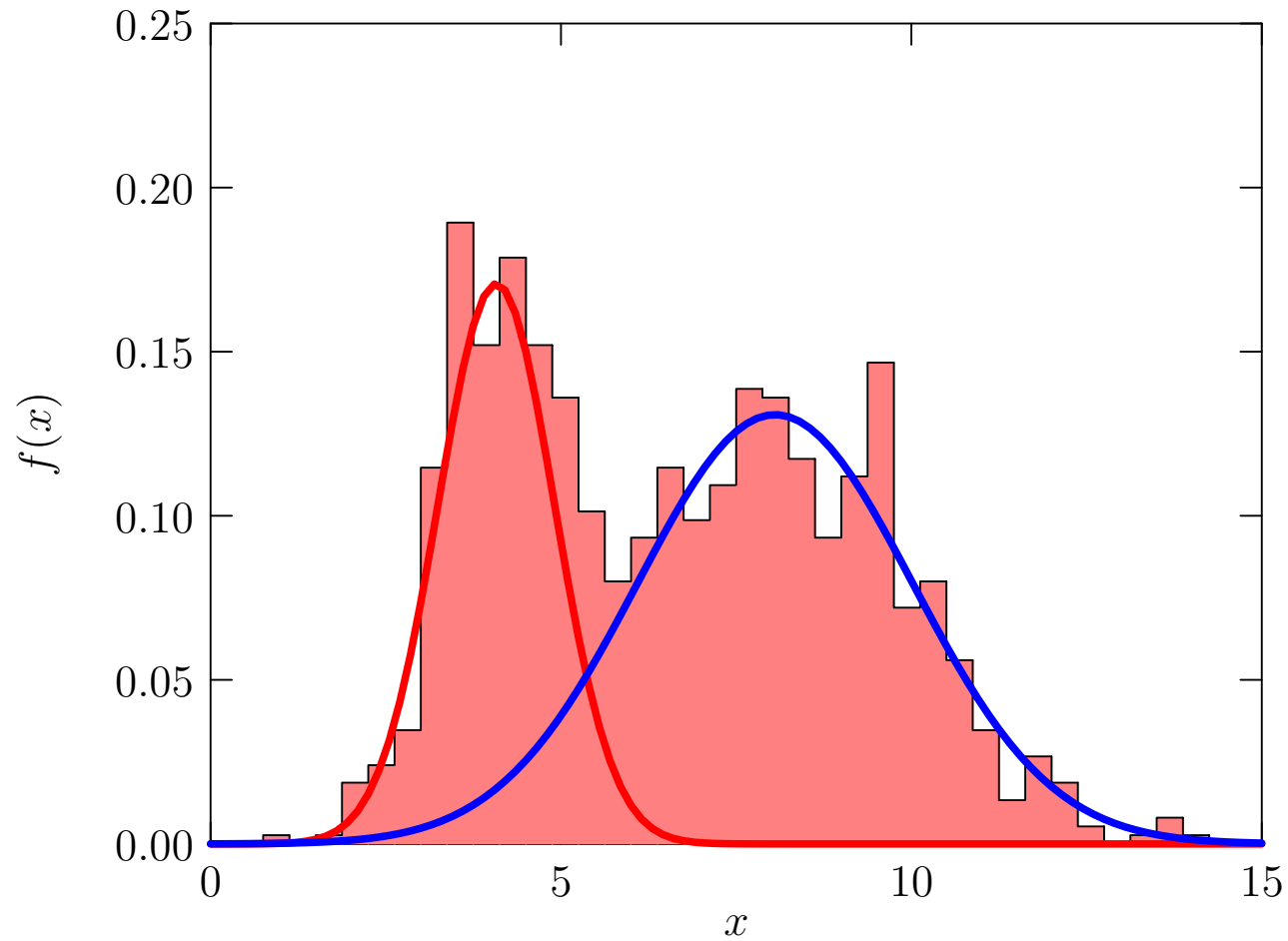
Example



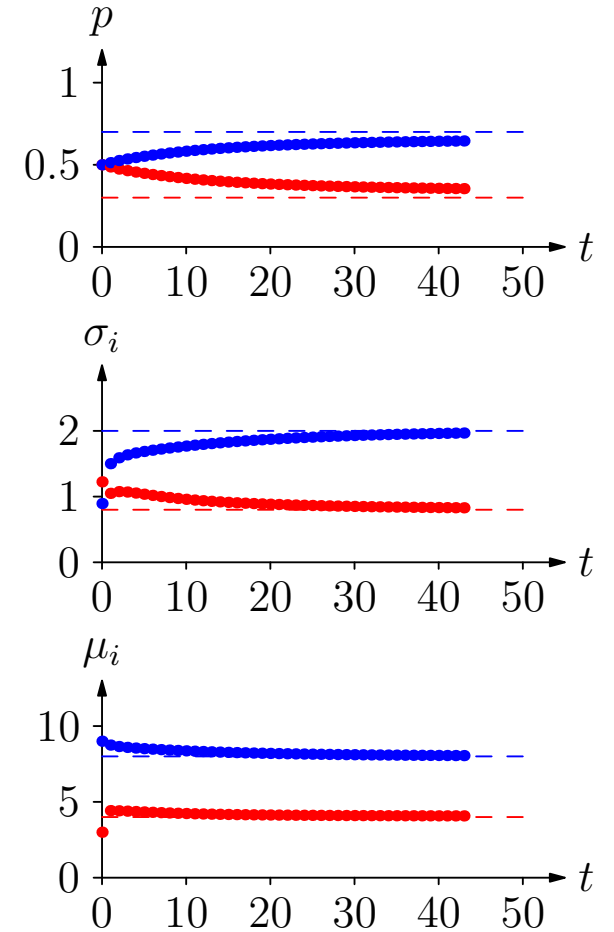
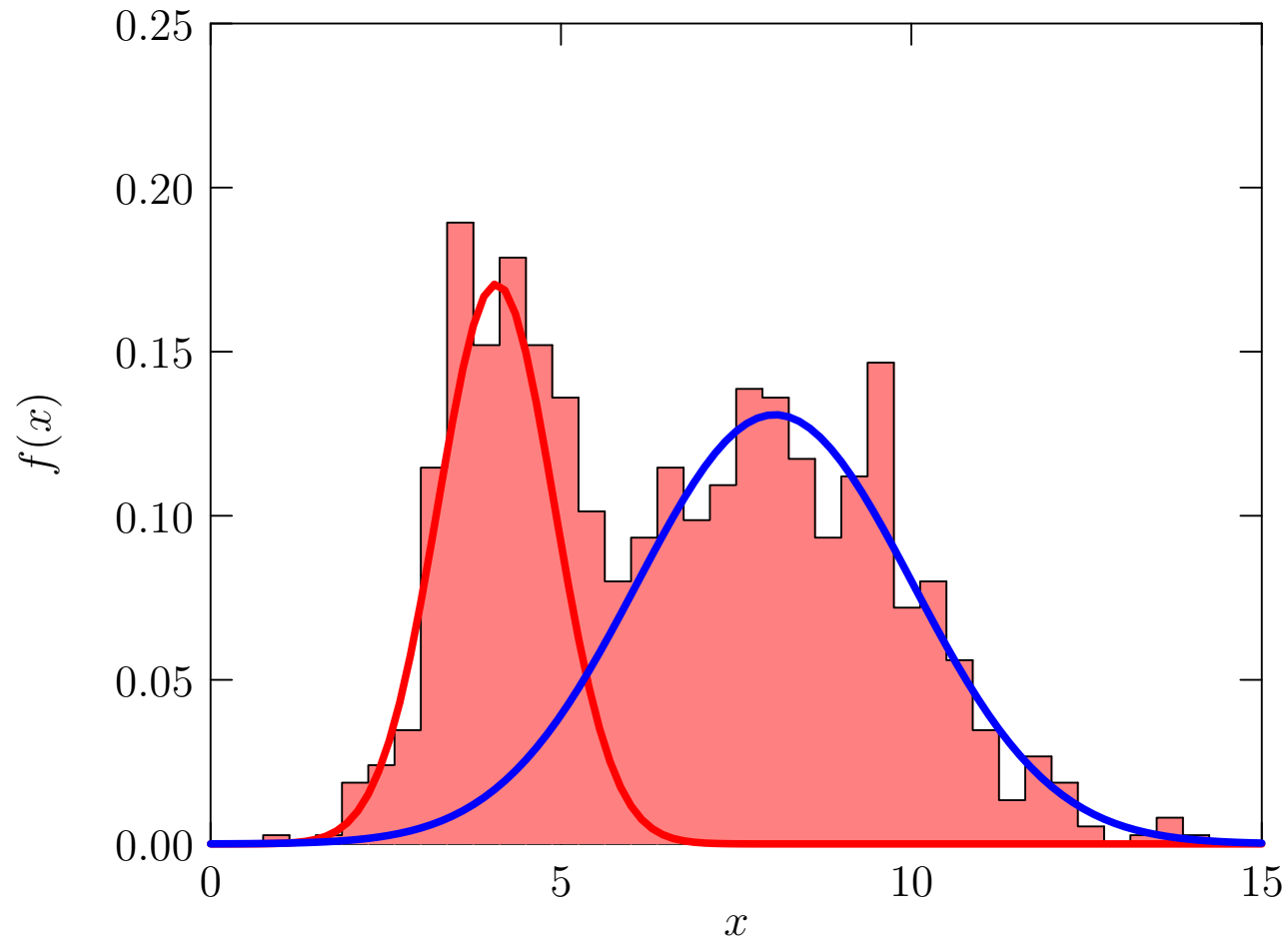
Example



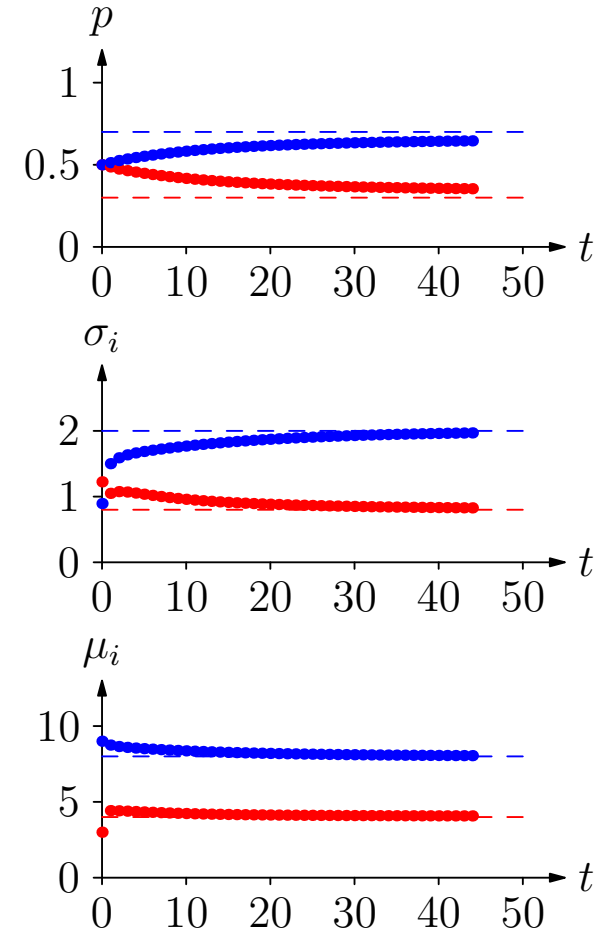
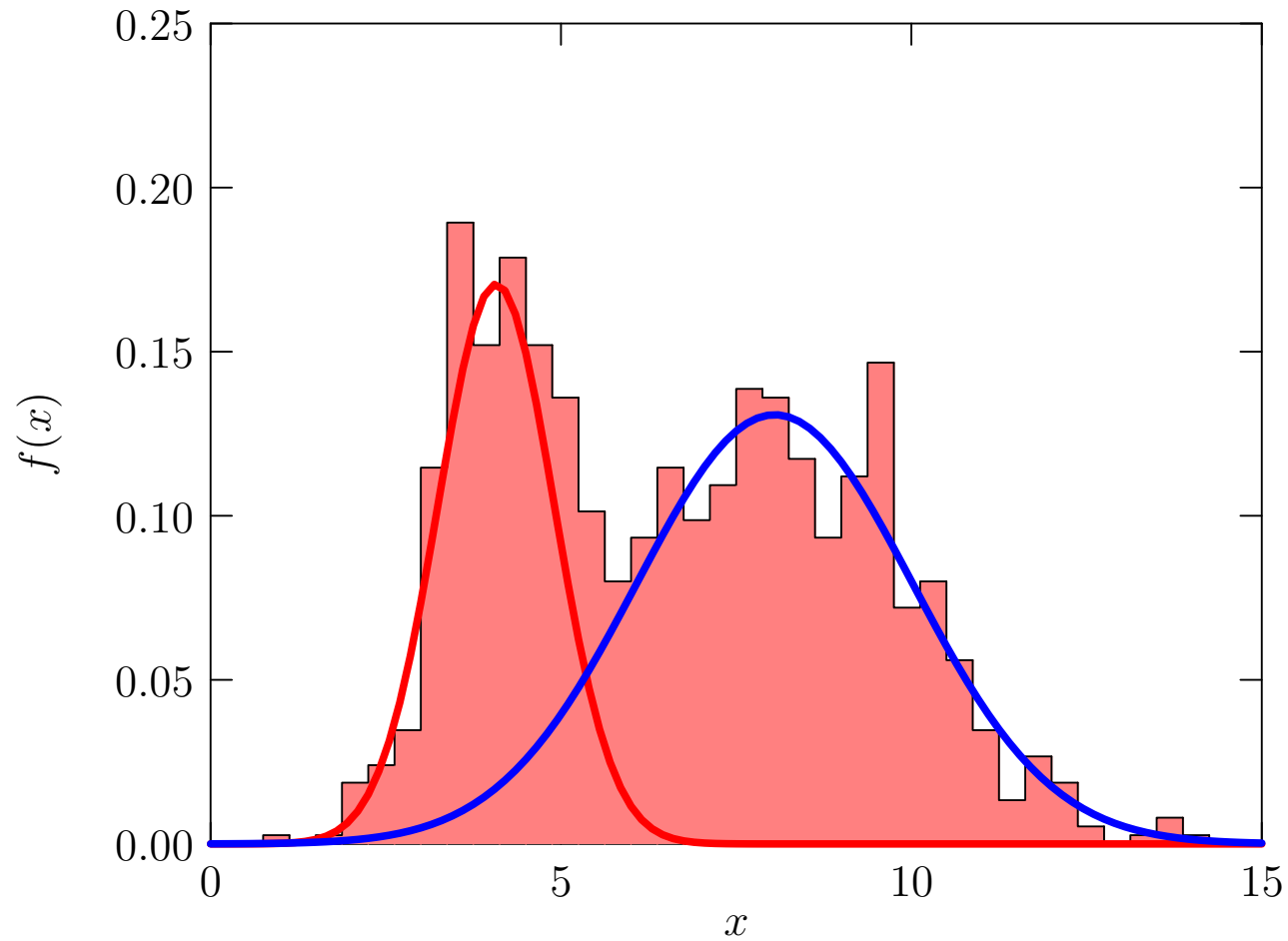
Example



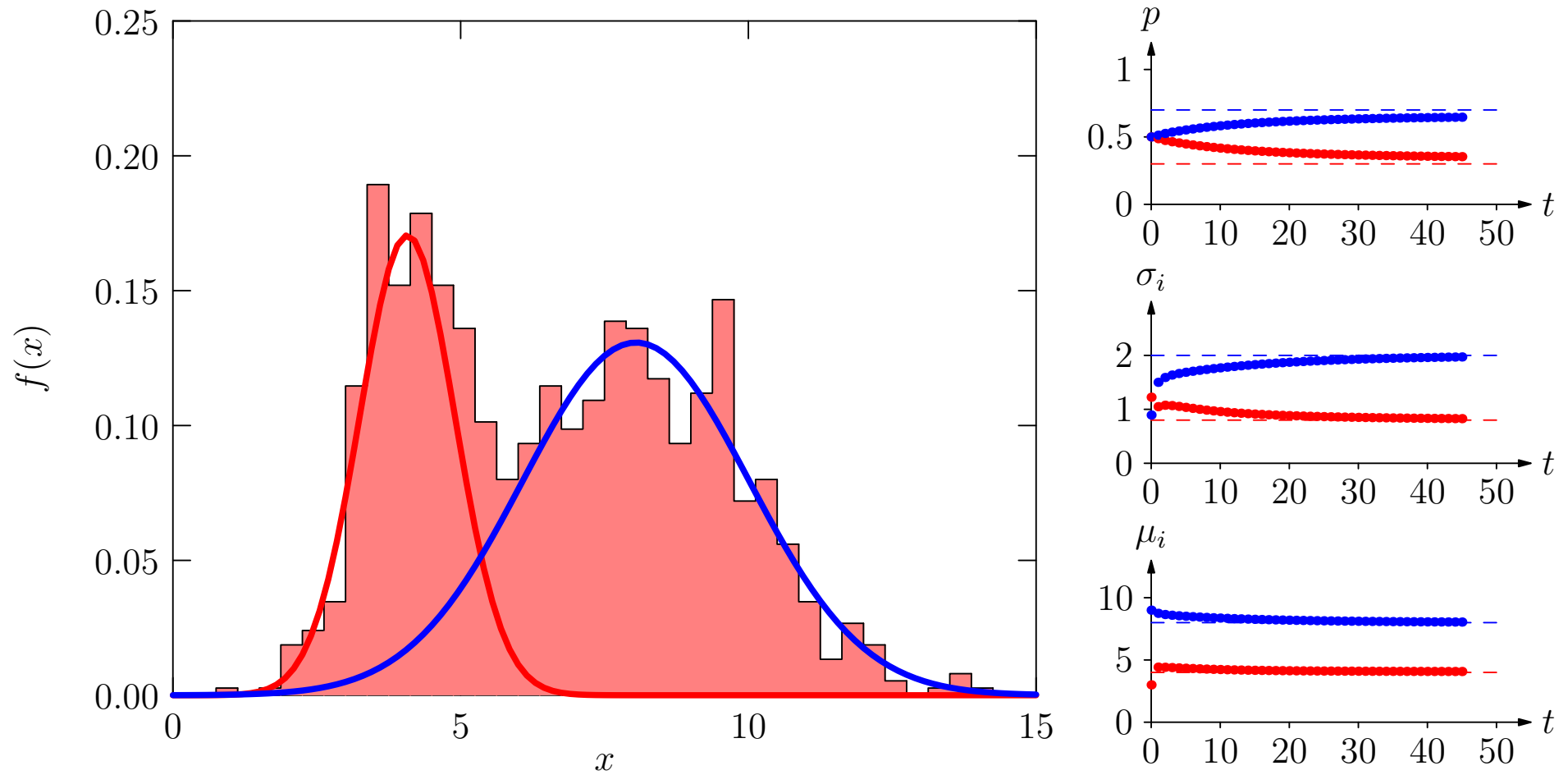
Example



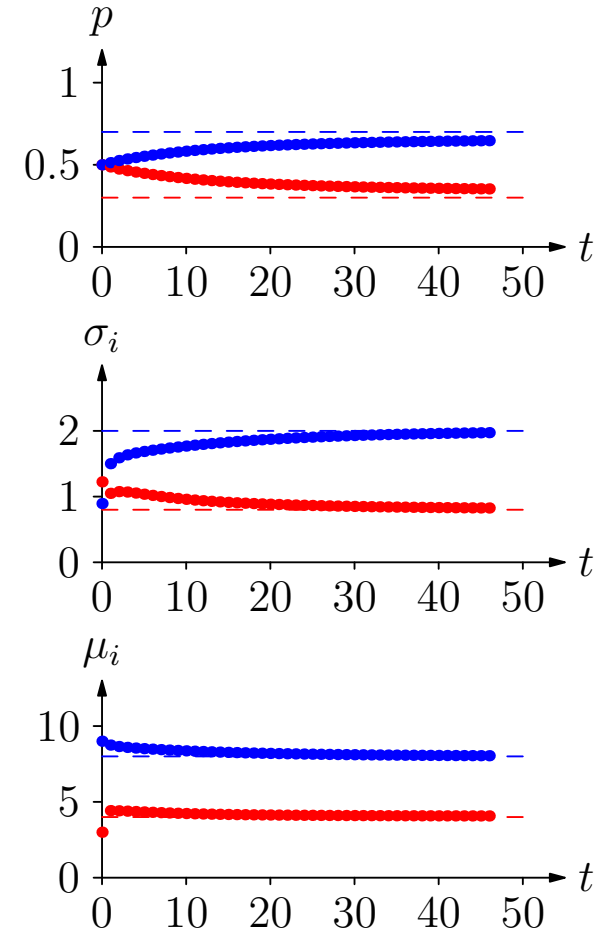
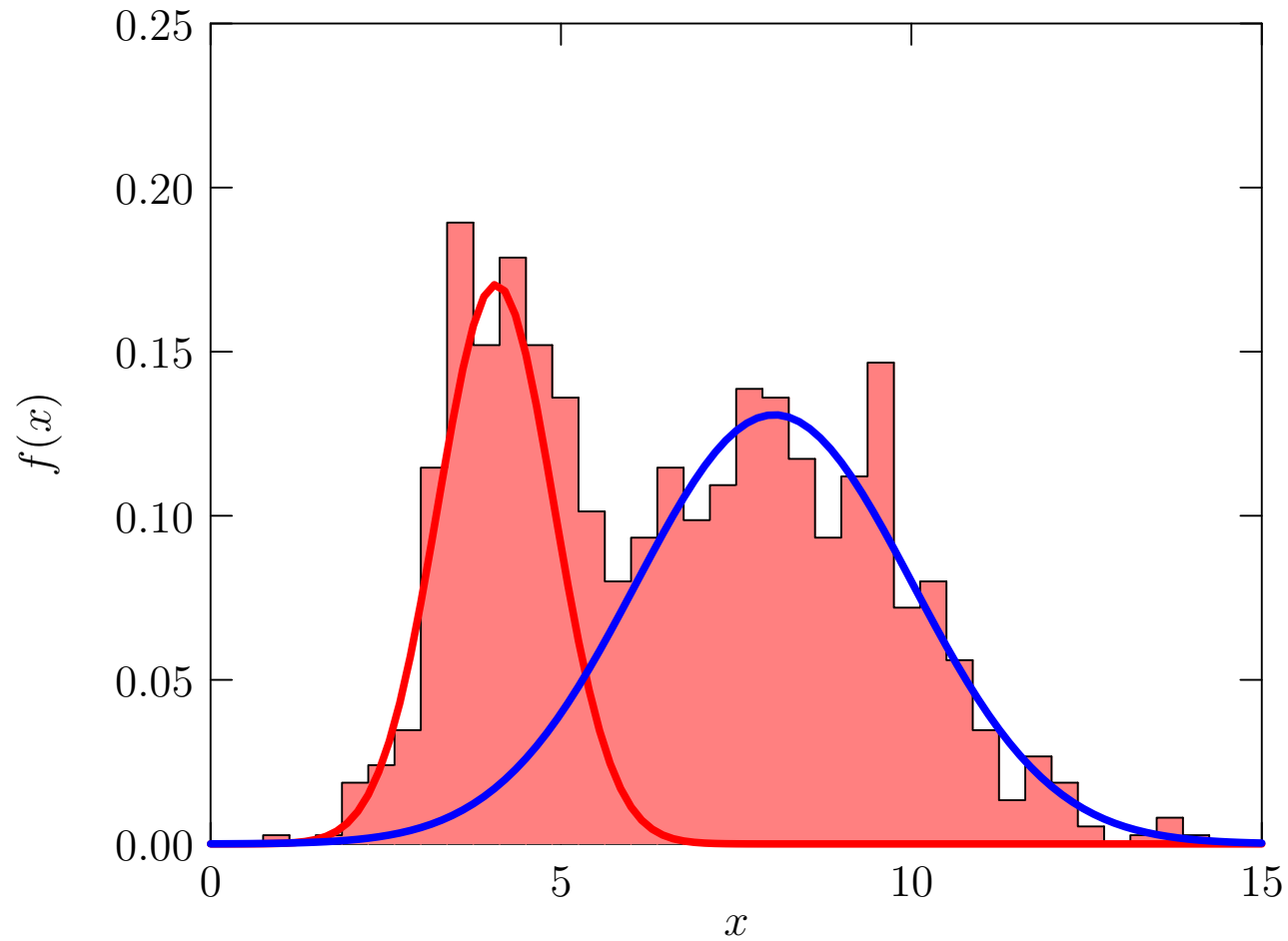
Example



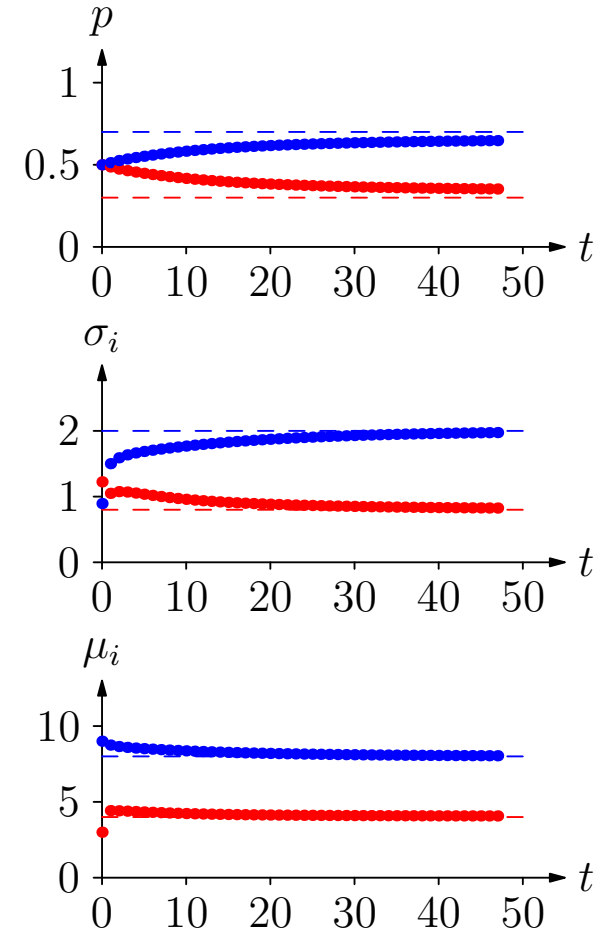
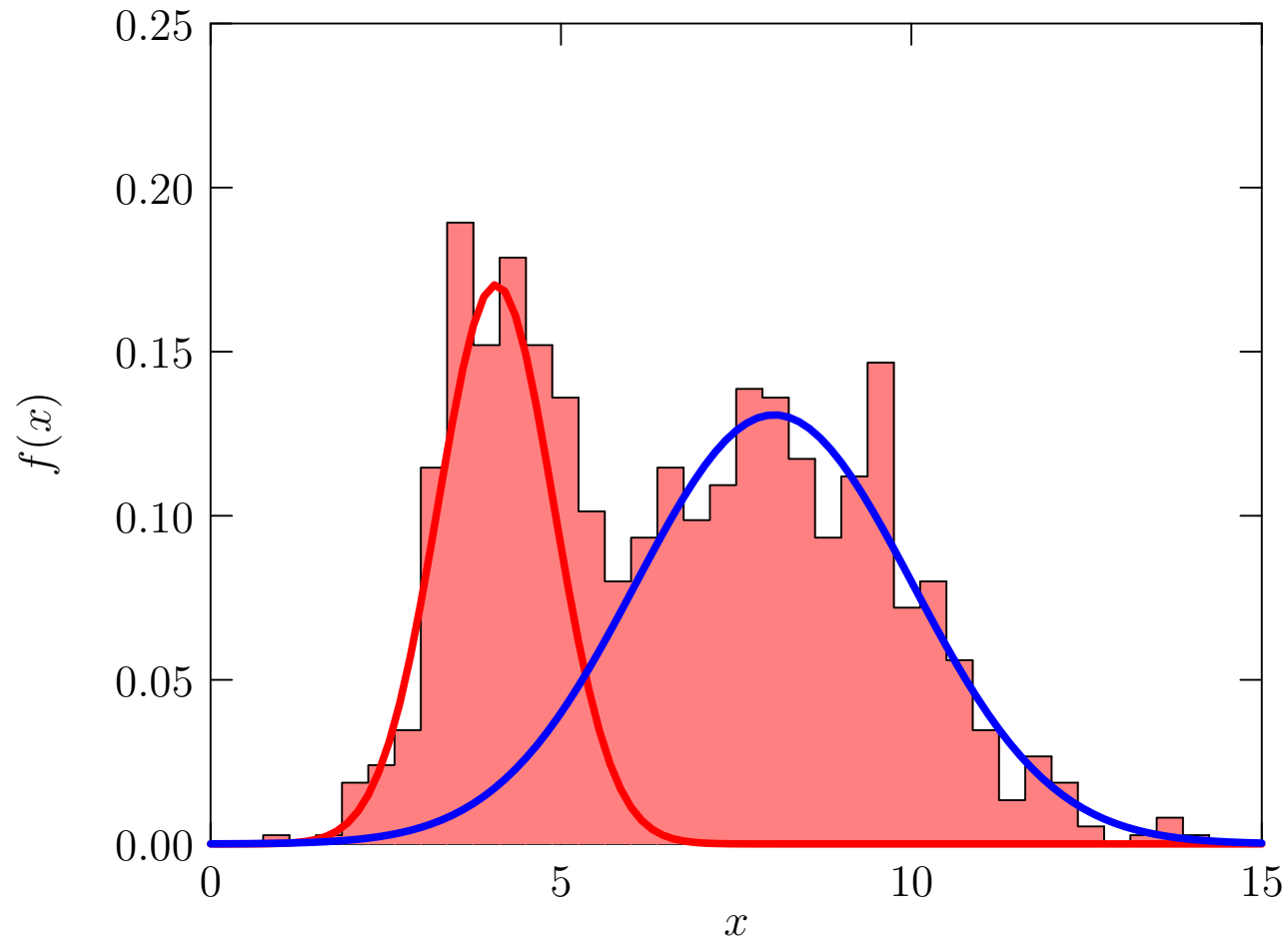
Example



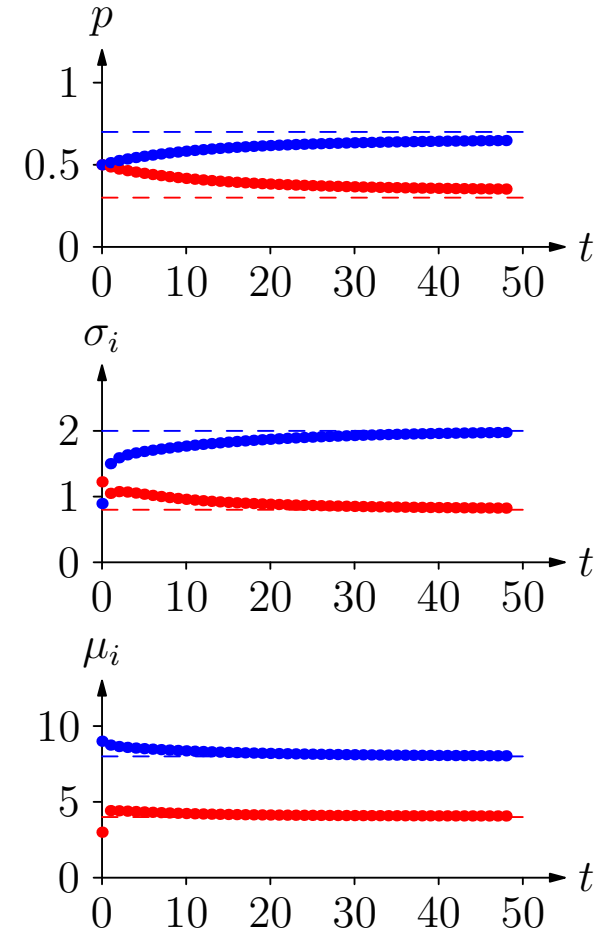
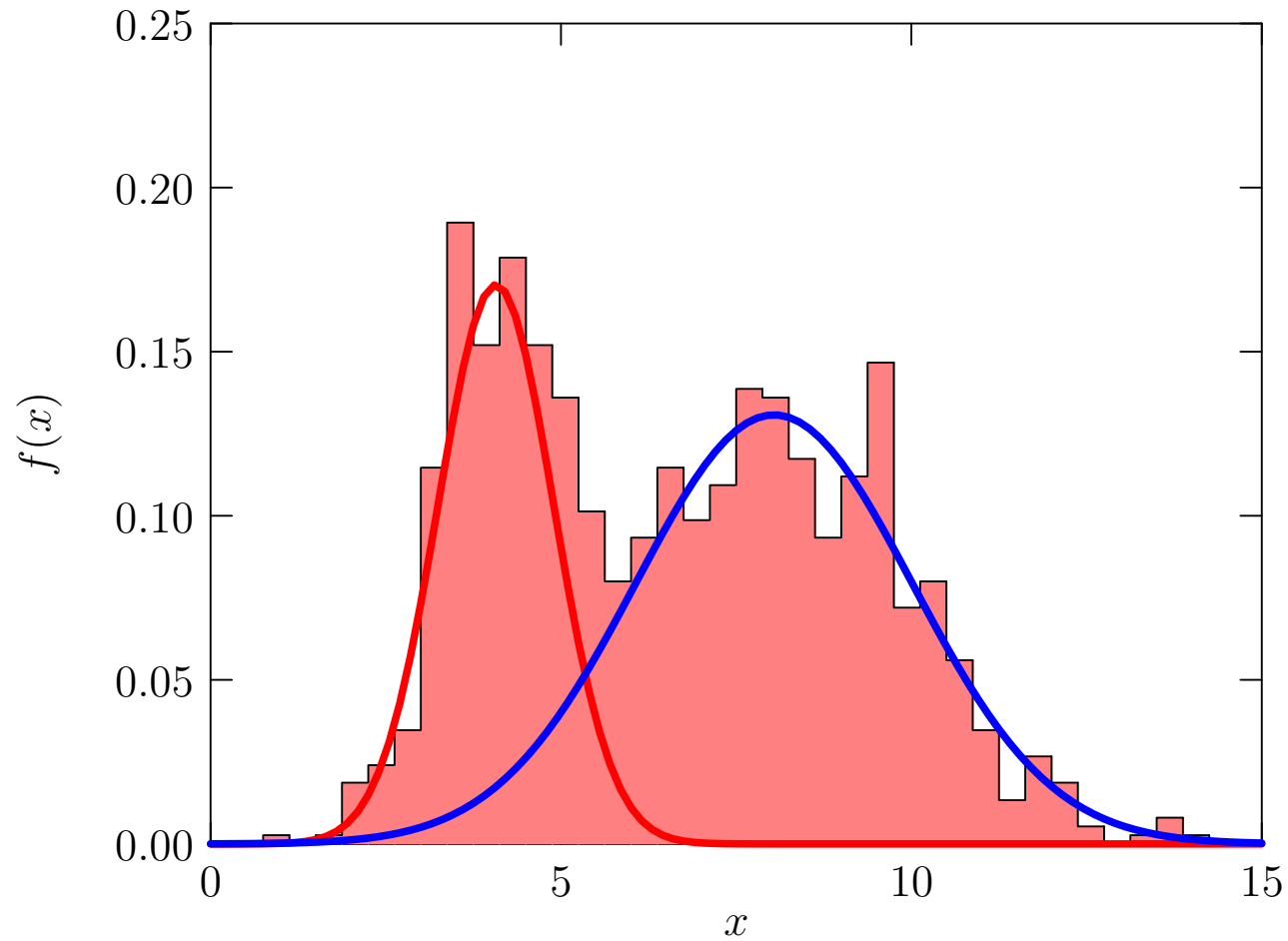
Example



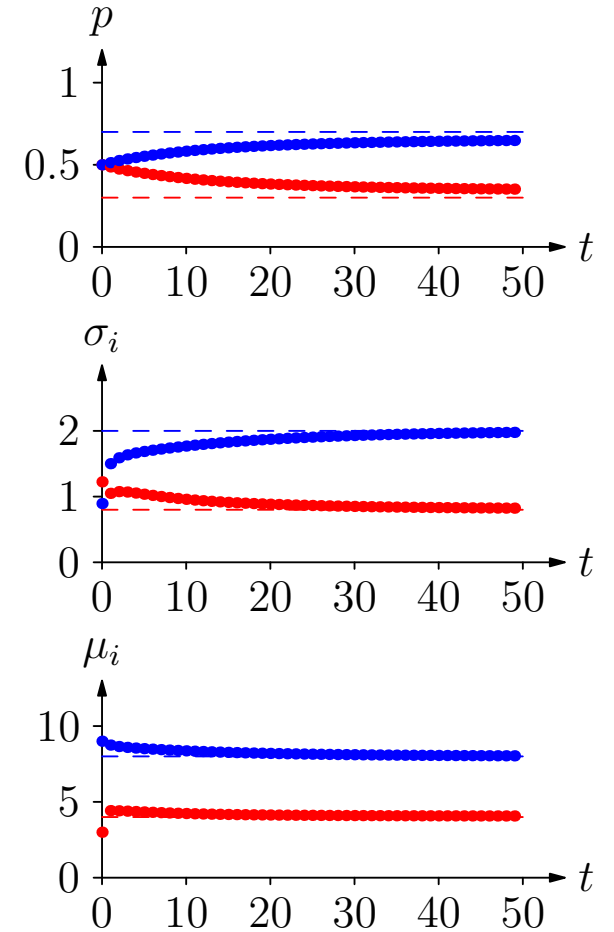
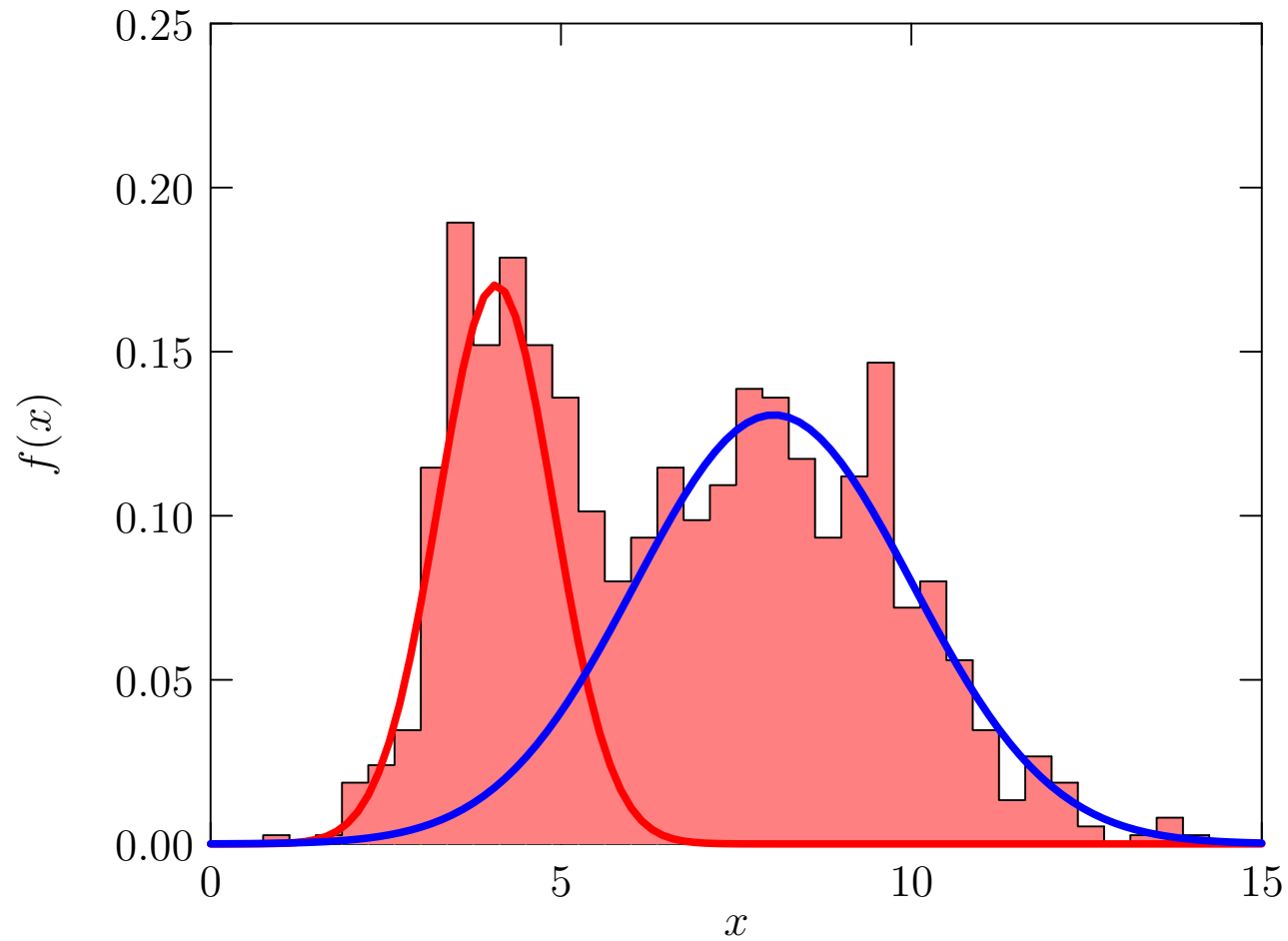
Example



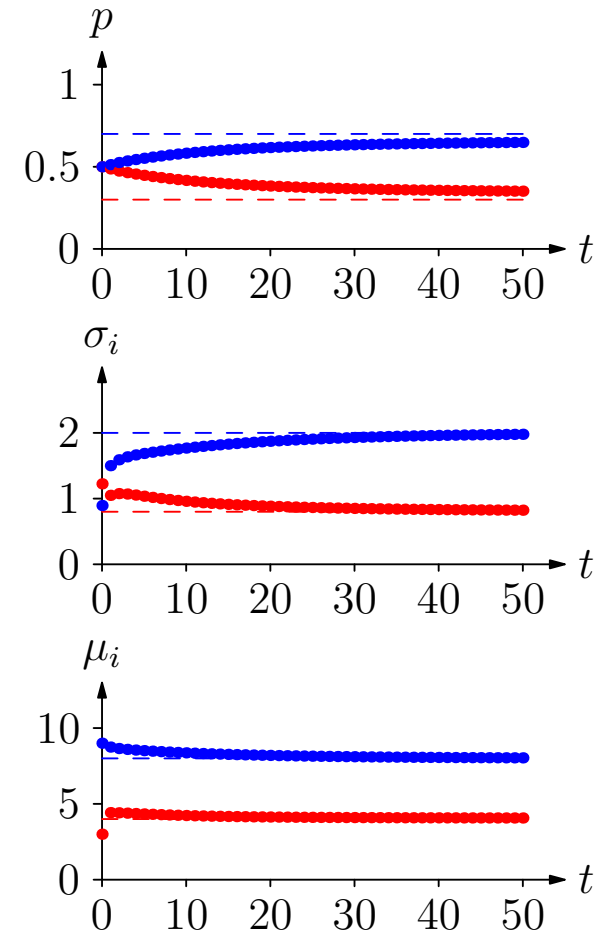
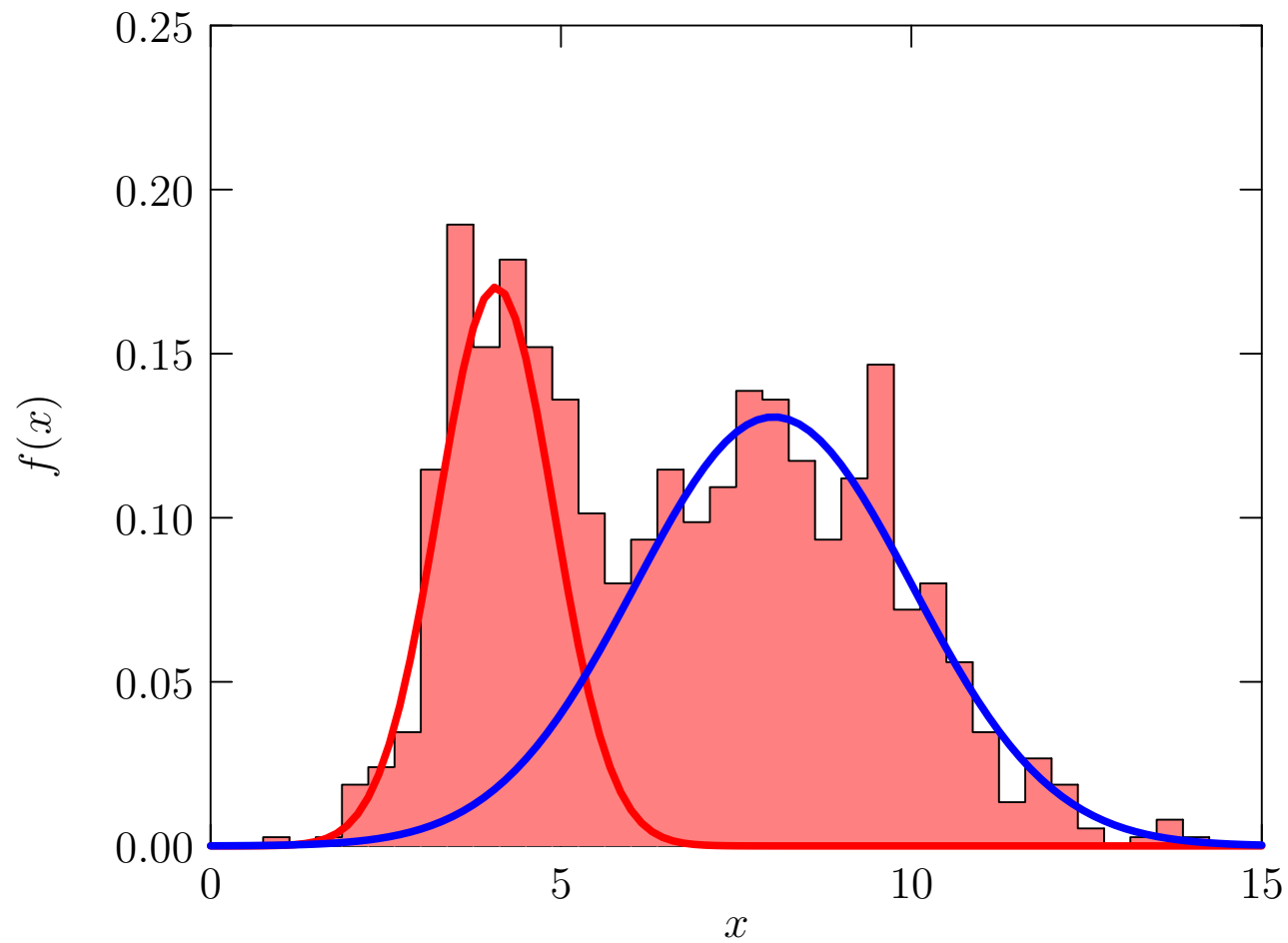
Example



Example



Example



Summary

- Building probabilistic models is an intricate process
- Identifying random variables that describe the system is the first step
- Often we need to introduce variables that we don't observe and need to be marginalised out
- The EM algorithm provide one approach to maximising likelihoods or MAP solutions when we have latent variables
- It often gives nice update equations, but convergence can be slow

Summary

- Building probabilistic models is an intricate process
- Identifying random variables that describe the system is the first step
- Often we need to introduce variables that we don't observe and need to be marginalised out
- The EM algorithm provide one approach to maximising likelihoods or MAP solutions when we have latent variables
- It often gives nice update equations, but convergence can be slow

Summary

- Building probabilistic models is an intricate process
- Identifying random variables that describe the system is the first step
- Often we need to introduce variables that we don't observe and need to be marginalised out
- The EM algorithm provide one approach to maximising likelihoods or MAP solutions when we have latent variables
- It often gives nice update equations, but convergence can be slow

Summary

- Building probabilistic models is an intricate process
- Identifying random variables that describe the system is the first step
- Often we need to introduce variables that we don't observe and need to be marginalised out
- The EM algorithm provide one approach to maximising likelihoods or MAP solutions when we have latent variables
- It often gives nice update equations, but convergence can be slow

Summary

- Building probabilistic models is an intricate process
- Identifying random variables that describe the system is the first step
- Often we need to introduce variables that we don't observe and need to be marginalised out
- The EM algorithm provide one approach to maximising likelihoods or MAP solutions when we have latent variables
- It often gives nice update equations, but convergence can be slow